

Psychology Book -1

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Course Coordinator

Vivek Singh

NATIONAL INSTITUTE OF OPEN SCHOOLING

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Chairman's Message

Dear learner,

As the needs of the society in general, and some groups in particular, keep on changing with time, the methods and techniques required for fulfilling those aspirations also have to be modified accordingly. Education is an instrument of change. The right type of education at right time can bring about positivity in the outlook of society, attitudinal changes to face the new/fresh challenges and the courage to face difficult situations. This can be very effectively achieved by regular periodic curriculum renewal. A static curriculum does not serve any purpose, as it does not cater to the current needs and aspirations of the individual and society.

For this purpose only, educationists from all over the country come together at regular intervals to deliberate on the issues of changes needed and required. As an outcome of such deliberations, the National Curriculum Framework (NCF 2005) came out, which spells out in detail the type of education desirable/needed at various levels of education- primary, elementary, secondary or senior secondary.

Keeping this framework and other national and societal concerns in mind, we have currently revised the curricula of all the subjects at the secondary level, making them current and need based. Textual material production is an integral and essential part of all NIOS programmes offered through open and distance learning system. Therefore, we have taken special care to make the learning material user friendly, interesting and attractive for you.

I would like to thank all the eminent persons involved in making this material interesting and relevant to your needs. I hope you find it appealing and absorbing.

On behalf of National Institute of Open Schooling, I wish you all a bright and successful future.

(Dr. S. S. Jena)

Chairman, NIOS

A Note From the Director

Dear Learner,

The Academic Department at the National Institute of Open Schooling tries to bring you new programmes every now and then in accordance with your needs and requirements. We are now revising the curriculum in all the subjects at the secondary level. In order to bring to you a curriculum which is at par with other boards in the country, we consulted the curriculum in different subjects at the CBSE and Several State Boards of Secondary Education. The National Curriculum Framework developed by the National Council for Educational Research and Training was kept as a reference point. After making a comprehensive comparative study, we developed the curriculum that was functional, related to life situations and simple. Leading educationists of the country were involved and under their guidance, we have been able to revise and update the curriculum.

At the same time, we also had a look at the learning material. We have removed old, outdated information and added new, relevant things and have tried to make it attractive and appealing for you.

I hope you will find the new material that is now in your hands interesting and exciting. Any suggestions for further improvement are welcome.

Let me wish you all a happy and successful future.

(Dr. Kuldeep Agarwal)

Director (Academic), NIOS

A Word With You

Dear learner,

Welcome to the secondary course in Psychology. Psychology is very easy and interesting subject. It is always with you and around you. The way you remember, you learn or you imagine, you think and reason, you feel you behave, all comes under psychology. Actually psychology is the scientific study of human behaviour. Behaviour is the expression of our inner world.

Everyone modifies his/her behaviour according to situations. One's behaviour and personality can always be modified for the better if he/she knows the psychological processes involved in it. Hence the study of psychology becomes very important for everybody. In the present course we have tried to introduce the basics of psychology. This course has been divided in seven modules and twenty seven lessons.

This book contains three modules. In the first module 'Foundations of Psychology; you will know what psychology is and how do we study the behaviour. Second module deals with the Basic Psychological process like learning, attention and perception, emotion and motivation and memory etc. Third Module tells us 'How does a human being develop'.

I hope the entire course material will create an interest in you for further study of psychology and help understanding yourself and persons around you better.

Yours,

(Vivek Singh)

Executive Officer (Academic)

How to use the Study Material

Congratulation! You have accepted the challenge to be a self-learner. NIOS is with you at every step and has developed the material in Psychology with the help of a team of experts, keeping you in mind. A format supporting independent learning has been followed. You can take the best out of this material if you follow the instructions given. The relevant icons used in

the material will guide you.

Title: will give a clear indication of the contents within. Do read it.

Introduction: This will introduce you to the lesson and also link it to the previous one.

Objectives: These are statements of outcomes of learning expected from you after studying the lesson. You are expected to achieve them, do read them and check if you have achieved.

Content: Total content has been divided into sections and subsections. Section leads you from one content element to another and subsections help in comprehension of the concepts in the content elements.

Notes: Each page carries empty space in the side margins, for you to write important points or make notes.

Activities: Certain activities have been suggested for better understanding of the concept.

Intext Questions: Very short answer self check questions are asked after every section, the answers to which are given at the end of the lesson. These will help you to check your progress. Do solve them. Successful completion will allow you to decide whether to proceed further or go back and learn again.

What You Have Learnt: This is the summary of the main points of the lesson. It will help in recapitulation and revision. You are welcome to add your own points to it also.

Terminal Questions: These are long and short questions that provide an opportunity to practice for a clear understanding of the whole topic.

Answers to Intext Questions: These will help you to know how correctly you have answered the intext questions.

Hints for Terminal Questions: This will help you to find correct answer.

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MODULE:1

1. AN INTRODUCTION TO PSYCHOLOGY

As human beings our curiosity drives us to know the reasons behind various events happening around us. Whenever we meet somebody or see someone doing something we immediately try to understand as to why this person is doing this kind of activity. Similarly we often try to understand our own experiences and behaviors in different situations. The eagerness and curiosity guide us to think as to how people are different from each other in terms of their intellect, aptitude and temperament: Why do they become happy or sad? How do they become friendly or develop antagonistic relationships? How do some people learn anything quickly while others take relatively more time? The answer to all these questions may be given by a novice as well as a person who has learned psychology. The answer of the novice can be guided by common sense but a psychologist can study the reasons behind such activities in a systematic way and can provide scientific answers to these questions which can be tested time and again. In this lesson we will try to understand the nature and scope of psychology in detail.

OBJECTIVES

After studying this lesson, you will be able to:

- explain the nature of psychology and its definition;
- describe the scope of psychology;
- elaborate the basic psychological processes; and
- explain the fields of psychology.

1.1 NATURE OF PSYCHOLOGY AND ITS DEFINITION

In simple words psychology is a systematic and scientific study of mental processes, experiences and behaviors - both overt and covert. The word 'psychology' has its

origin in two Greek words 'psyche' and 'logos'. The word 'psyche' in Greek language refers to 'spirit' or 'soul' and the word 'logos' refers to 'discourse' or 'study'. During earlier times it was considered as a discipline which deals with the study of soul. In India the study of such questions was the main concern during Vedic and Upanishadic period. Various aspects of mental processes were analyzed. Subsequently the schools of Yoga, Samkhya, Vedant, Nyaya, Buddhism, and Jainism provided detailed accounts of mind, mental processes and methods to control mind. In modern period it started at Calcutta University in 1916 with establishment of the Department of Psychology.

In the western world, the formal beginning of psychology as an independent discipline goes back to 1879 when Wilhelm Wundt established the first experimental laboratory at the University of Leipzig, in Germany. Since then the growth of psychology has covered a long journey. Today it is one of the very popular subjects among social sciences. It studies all the shades of experiences, mental processes and behaviours. A comprehensive analysis of all these aspects provides a scientific understanding of human nature. In the following sections we will try to understand all the components which collectively define psychology.

(A) Study of experience

Psychologists study a variety of human experiences which are mainly personal or private in nature. They may range from experiences of dream, conscious experiences at different stages of life and experiences when the consciousness is altered through meditation or use of psychedelic drugs. The study of such experiences helps the psychologist to understand the personal world of the individual.

(B) Study of mental processes

Psychology as the study of mental processes tries to investigate the activities

happening in the brain which are primarily non physiological in nature. These mental processes include perception, learning, remembering and thinking. These are internal mental activities which are not directly observed but inferred from the behavioural activities of the person. For example, we can say that somebody is thinking if he or she displays certain activities related to finding solution to a mathematical problem assigned to him or her.

(C) Study of behaviour

The range of behaviours studied in psychology is very broad. It includes simple

reflexes (e.g. eye blinking), common response patterns such as talking to friends, verbal reports about feelings and internal states and complex behaviours such as handling computers, playing piano and addressing a crowd. These behaviours are either observed directly through naked eyes or are measured through instruments. They are generally exhibited verbally or nonverbally (e.g. facial expression) when an individual reacts to a stimulus in a given situation.

Thus in psychology the main unit of investigation is the individual human being and his or her experiences, mental processes and behaviours.

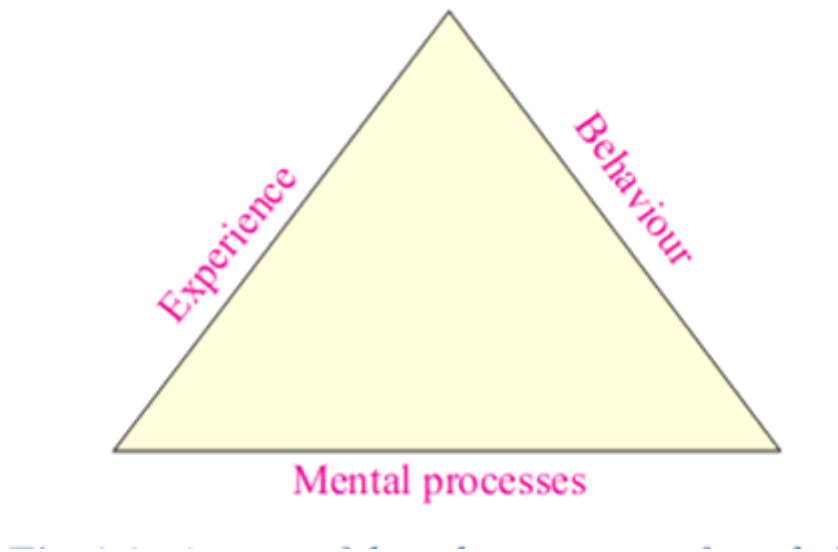


Fig. 1.1: Aspects of the subject matter of psychology

Figure shows a triangle depicting the three processes studied in Psychology

1.1 INTEXT QUESTIONS

1. Fill in the blanks:

a. The word 'psychology' has its origin in two Greek words _____ and _____ .

b. The main unit of investigation in psychology is the individual _____ .

c. Wilhelm Wundt established the first experimental _____
at the University of Leipzig, in Germany.

d. Psychology is the study of _____, _____
_____, and _____.

1.2 SCOPE OF PSYCHOLOGY

From the definition of psychology now it might be clear to you that psychology addresses a variety of issues related to mental and behavioural functioning of an individual. Such a study helps us to develop a basic understanding about human nature and facilitates us to deal with a number of personal and social problems.

The study of human beings starts with the functioning of biological systems especially the nervous system. Under central nervous system psychology studies the functions of various parts of brain which regulate our feelings, emotions and thinking. Within autonomic nervous system the role of hormones and neurotransmitters in determining our behavior is important. In addition psychology studies as to how a given sociocultural environment interacts with innate biological, intellectual and social attributes of the child and facilitates healthy development of the child.

As a living organism you encounter a lot of sensory input every moment. Your task is not only to process that information but to store and retrieve it when it is required to be used. Attention and perception help to process information. Memory helps us to register, retain and retrieve information, and thinking provides us the ability to manipulate and improve the stored information. Within psychology, all these are studied under the domain of higher mental processes or cognition.

In your life you must have come across many persons who have reached highest level of accomplishment. But such an accomplishment is not attained at once; it is the result of continuous learning in which the individual acquires the necessary skills and competencies through regular practice. Psychology helps us to understand the processes involved in reaching this high level of accomplishment.

In order to understand the purpose behind any kind of behaviour psychologists use the concept of motivation. It primarily focuses on the investment of mental energy and consistency of effort towards achieving the set goals. The various shades of feelings such as anger, fear, love, joy, and sadness which we experience during the course of our life are studied within the realm of emotion.

You will certainly agree that no two individuals are the same in terms of their physical attributes such as height, weight, skin colour or facial features as well as psychological attributes such as intelligence, personality, temperament and interest. An understanding of these and other aspects of individual differences facilitates the psychologist to select right person for the right job and to provide guidance and counseling for various matters of personal as well as professional concern. The understanding of individual differences also helps the psychologist to differentiate between normative (customary, accepted) and abnormal (deviant, unusual) behaviors.

Thus psychology as a discipline has a vast scope. It not only studies human beings across the life span but also tries to explore mental processes and potentials in order to facilitate achieving a better quality of life.

Activity 1: The Power of Positive Thinking: Health and Happiness

By thinking and saying positive thoughts to yourself and others, you can gradually create a positive change in yourself as well as in others. Try this exercise with your friends or family members. Ask your friends as to how many of them are in a 'positive' mood and how many in a 'negative' mood. Then instruct your friends to shake hands with others and genuinely say " I wish you happiness and good health" . Again ask your friends

about their mood, to see how many report 'positive' and how many 'negative'. Yes, you will be surprised - the number of reported 'positive' moods has gone up.

1.3 BASIC PSYCHOLOGICAL PROCESSES

While studying behaviour it is one of the most important tasks for psychologists to understand the processes which collectively influence a particular behaviour. These psychological processes are

(A) Sensation: It refers to our awareness about various stimuli which we encounter in different modalities such as vision, hearing, touch and taste.

(B) Attention: During attention we selectively focus on a particular stimulus among many stimuli available to us. For instance while listening to a lecture in the classroom we attend to the words pronounced by the teacher and, try to ignore the other stimuli present in the classroom, such as noise made by the fan.

(C) Perception: In the case of perception we process information and make out the meaning of the stimuli available to us. For example, we look at a pan and recognize it as an object used for writing.

(D) Learning: It helps us acquiring new knowledge and skills through experience and practice. The acquired knowledge and skills further bring a relative change in our behavior and facilitate our adjustment in varied settings. For example, we learn language, riding a bicycle and applying mathematical skills to solve various problems. .

(E) Memory: The information we process and learn is registered and stored in the memory system. Memory also helps us to easily retrieve the stored information when it is required for use. For example, writing the answers in the examination after studying for the paper.

(F) Thinking: In the case of thinking we use our stored knowledge to solve various tasks. We logically establish the relationships among various objects in our mind and take rational decision for a given problem. We also evaluate different events of the environment and accordingly form an opinion.

INTEXT QUESTIONS 1.2

1. State whether the following statements are True or False:

- a. In learning we acquire new skills through training and experience.
- b. In sensation stimuli are stored in the brain.
- c. In thinking we use stored knowledge to solve various tasks.

d. In attention we attend to all the stimuli present in the environment.

1.4 FIELDS OF PSYCHOLOGY

During its journey of more than a hundred years, psychology as a field of enquiry has grown in many directions. In contemporary times a number of specialized fields with focused area of application have developed. The emergence of different fields in psychology indicates the importance and relevance of these areas in our daily life.

In this section we will briefly learn about some of these fields.

Abnormal Psychology: This branch of psychology tries to describe, assess, predict and control those categories of behaviours which are considered as unusual and non-normative. It also deals with various categories of psychological disorders which affect the mental health of the individual. It assesses abnormality in the individual through standardized psycho-diagnostic tools and suggests problem specific treatment.

Cognitive Psychology: It deals with acquisition, storage, transformation and application of information the individual uses to understand and interpret events happening around him/her.

Clinical and Counseling Psychology: Clinical psychology is primarily concerned with diagnosis and treatment of various psychological disorders. A clinical psychologist is trained in psychotherapeutic techniques which are used to treat people suffering from psychological disorders. They are employed in hospitals and clinics dealing with mental health problems. The counseling psychologist deals with mild problems pertaining to adjustment in social and emotional life, and provides specialized services to deal with marital problems, stress oriented problems and career choice.

Educational Psychology: This field of psychology is mainly concerned with academic performance. It studies the role of various factors which influence learning processes in classroom setting. It focuses on assessing the aptitude, skills and intellectual potential of the students and evaluating their performance. An educational psychologist also helps students to overcome learning difficulties.

Environmental Psychology: This field is concerned with the study of interaction between physical environment and human behaviour. It studies the impact of noise, heat, humidity, pollution and crowding on human performance. It also focuses on the impact of physical environment on psychological health of individuals.

Health Psychology: This field focuses on the impact of various psychological factors (e.g. stress) on the onset progress and treatment of illness. It also deals with various life style diseases such as hypertension, coronary heart disease, cancer and diabetes.

Organizational Psychology: This field is concerned with application of psychological principles and models to study the selection and performance of

employees in organizational settings. It also studies the concepts of leadership, motivation, job satisfaction and performance appraisal.

Developmental Psychology: This field is concerned with the study of various physical and psychological factors which influence the systematic changes which the individual experiences across the different stages of life.

Emerging Fields: In addition to the fields mentioned above a number of new areas have emerged. Some of these fields are sports psychology, military psychology, aviation psychology, forensic psychology, peace psychology, neuropsychology, political psychology, feminist psychology and positive psychology.

INTEXT QUESTIONS 1.3

1. Describe any two fields of psychology and indicate their applications.

WHAT YOU HAVE LEARNT

- Psychology is a systematic and scientific study of mental processes, experiences and behaviors - both overt and covert – as they take place in a socio-cultural setting.
- In India the study of soul/consciousness was the main concern during Vedic and Upanishadic period.
- The formal beginning of psychology was in 1879 when Wilhelm Wundt

established the first experimental laboratory at the University of Leipzig, in Germany.

- The scope of psychology is wide as it addresses a variety of issues related to mental and behavioral functioning of the individuals.
- Study of psychology helps us to develop a basic understanding about human nature and facilitates dealing with a number of personal and social problems.
- The various psychological processes that are involved in human behavior are sensation, attention, perception, learning, memory, and thinking.
- Psychology has a number of specialized fields of study. Each field helps us understand human behavior in specific domain.

TERMINAL QUESTIONS

1. Define psychology. With the help of examples explain the different components of this definition.
2. Explain the various psychological processes which influence human behavior.
Support your answer with relevant examples.
3. Describe four different fields of psychology. Which field of psychology interests you the most? Give reasons.

ANSWERS TO INTEXT QUESTIONS

1.1

1. Psyche, logos
2. Human being
3. Laboratory
4. Experiences, mental processes, behaviors

1.2

1. True
2. False
3. True
4. False

1.3 Refer to section 1.4

Hints for Terminal Questions

1. Refer to section 1.3
2. Refer to section 1.5
3. Refer to section 1.6

2. METHODS OF PSYCHOLOGY

In the last lesson we noted that the interpretation of a psychological phenomenon by a psychologist may differ from that of a novice. A psychologist follows a systematic scientific procedure which has sound theoretical base in order to explain and interpret the phenomenon. Psychology has various methodological ways or approaches to understand and explain psychological phenomena. We will be studying about some of these approaches. In order to obtain responses from individuals a number of psychological tools or instruments are used. The responses taken on those tools constitute the basic data which are analyzed to study human experiences, mental processes and behaviours. In this lesson we will discuss these aspects in detail.

OBJECTIVES

After studying this lesson, you will be able to:

explain the different approaches to the study of psychological processes;
describe some important methods used in understanding human behaviour; and
describe various instruments used in understanding behaviour and psychological processes.

2.1 APPROACHES TO THE STUDY OF PSYCHOLOGICAL PROCESSES

As discussed in the previous lesson psychologists use a variety of approaches to describe, predict and control behaviour and mental processes. The main approaches are briefly described below.

Biological Approach: This approach focuses on biological structures and phenomena such as brain, genes, hormones, endocrine system and neurotransmitters in order to understand the dynamics of behaviour. Its main focus is on the role of

different parts of brain in regulating feelings, memories, emotions and other aspects of behaviour. Similarly the impact of **over-secretion or under-secretion** of different kinds of hormones in governing behaviour is studied. Behaviour genetics as one of the sub disciplines studies the genetic determinants of behaviour. Moreover, this approach looks for physiological basis of human behaviour.

Psychoanalytic Approach: The father of psychoanalytic approach Sigmund Freud focused on unconscious libidinal energy in describing the present state of the individual. He studied mind in terms of hierarchical arrangements of experiences in the form of different layers of consciousness (e.g. conscious, preconscious, and unconscious). Freud explored the nature and quality of unconscious through analysis of dreams, slips of the tongue, neuroses, psychoses, work of art, and rituals. He assumed that majority of human behaviours are triggered by unconscious motivation. Thus to understand the present human behaviour the analysis of unconscious mental contents is considered most important.

Humanistic Approach: Contrary to Freud, the father of humanistic approach Carl Rogers put greater emphasis on conscious experiences of the present situation, role of interpersonal experiences across the course of life, and people's capacity to grow toward psychological maturity. This approach basically assumes that a person is an active and self-actualizing agent and has a choice in deciding his behaviour. As a part of the self-actualizing process a person seeks to maintain a congruence between self and experience. However, because of past experiences with conditional positive regard, he may deny or distort the experiences that threaten one's self-system. Such a self-system can be changed in the therapeutic setting through genuineness, unconditional positive regard, and empathic understating of the client's problem by the therapist.

Behaviorist Approach: The unit of analysis for this approach is explicit, objective and overt behavior and its relationship with environmental stimulation. The father of behaviorism J. B. Watson emphasized on objective analysis of behaviour. He advocated that behaviour is largely governed by the association between stimulus and response and the behaviour can be shaped in a desired direction by manipulating this association.

Cognitive Approach: The cognitive approach emerged as an alternative to the mechanistic paradigm of behaviourism. This approach mainly focuses on the study of information processing capacity of the individual in terms of perception, remembering, thinking, language, reasoning, problem solving and decision making which are called higher mental processes. It proposes that we look out for information in the world and our behaviour depends upon the way we process this information. This approach

largely relies on computational models and assumes that behavior and mental processes can best be understood by treating them in terms of information processing.

The above discussed approaches indicate that mental processes, experiences and behaviours can be understood from various vantage points. In fact human behavior is complex and its varied aspects are to be appreciated in many ways.

INTEXT QUESTION 2.1

1. _____ focuses on the role of different parts of brain in regulating feelings, memories, emotions and other aspects of behaviour.
2. Focus of the _____ is on the information processing capacity of the individual.
3. _____ assumes that the person is active and self-actualizing agent and has a choice in deciding his behaviour.
4. According to the _____ majority of human behaviours are triggered by unconscious motivation.
5. The unit of analysis for the _____ is explicit, objective and overt behaviour and its relationship with environmental stimulation

2.2 METHODS TO UNDERSTAND PSYCHOLOGICAL PROCESSES

In order to understand human behaviour various scientific methods are used. The purpose of study or research is to develop principles and theories, test them and apply for solving different human problems. In this way we develop dependable understanding that helps us in guiding behaviour in various situations. Since human beings are complex living organisms their behaviours are shaped by many factors both intrinsic and extrinsic to him or her.

A psychological research carried out scientifically has the characteristics of *objectivity* which means that such researches are free from any kind of biases. It is *testable* time and again and can be open to all. One can verify its authenticity by following the same method in terms of getting the same result. It has scope for *self-correction*. In other words the researcher corrects his or her understanding if there is some error and goes for revision. The scientific studies have also the characteristic of *replication* which means that the results of the study are consistently verified by similar other studies across different settings.

Thus in psychology a number of methods are used to carry out scientific studies.

These methods are discussed below.

Observation: While shopping in the market you must have noticed various activities of the people. When you observe their activities you also think about as to why they

are doing those activities and probably you reach to a conclusion about the causes of such activities. Such a way of knowing about others is called **observation**. However, the meaning of observation goes a little further as compared to discussed above. Observation as a method of enquiry is often understood as a **systematic registering** of events without any deliberate attempt to interfere with variables operating in the event which is being studied.

Some Interesting Facts about Observational Method

Perhaps the most famous informal observations in the history of developmental

psychology are the observations made by Jean Piaget on his three children when they were infants. These observations went on to become the empirical foundation for Piaget's developmental theory. You can also observe the developmental patterns of your younger sibling or nephew, to understand the changes in sensori-motor development and other aspects of development. This method is used in natural as well as laboratory settings. When it is used to study the events happening in natural environment it is called **naturalistic observation** such as observing the behaviour of children on playground. In this case the observer (psychologist) has no control on the extraneous variables. He or she simply records the entire activities and then analyze them. On the contrary in the case of **laboratory observation** the event under study is controlled by the observer. For example, studying the effect of induced stress on task performance.

Observation is also divided into **participant and non-participant types** depending on the role of observer. In the case of **participant observation** the researcher mixes up with the event under study and conducts the study. Where as in the case of **nonparticipant observation** the researcher maintains an optimum distance and has little impact on the events under study.

One of the most important advantages of observation is that it studies the range of behaviours in the form in which they are happening. However, this method requires more time and effort. It often becomes victim of the biases of researcher.

Activity 1 Method of Observation

To develop an observation tool, you need to establish the indicators for the observation. Indicators are based on what you expect to find in the environment, or process. The second aspect is to consider each of the indicators and measure

them for their presence or absence.

A. Now try to use the observational method to collect data:

Make a video recording of any family function, or use a recording that is easily available and study it for non-verbal communication cues. Identify the indicators, such as smiling, shaking hands, the act of Namaste, use of hands

and the like. Count with the help of tallies how many times these cues are used by the people in the video recording. You can prepare a table with various aspects of observable behaviours; this will help you to understand which nonverbal cue is used more in the Indian setting. Also try to find out about gender difference, do men shake hands more than women, do women touch the feet of elders more than men? You can generate many more questions of your choice.

Experimentation: In the case of experiment the experimenter studies the effect of one variable on the other by **deliberately manipulating and controlling one**

variable. The variable which is controlled and manipulated by the experimenter is called **independent variable (IV)** and the variable on which the impact of independent variable is studied is known as **dependent variable(DV)**. In a simple experiment two groups are formed. One is experimental group in which participants receive the independent variable. The other is control group in which behaviour is observed without giving the independent variable. By manipulating independent variable the experimenter is in a position to state that change induced in one variable brings change in another variable. Apart from these variables the experimenter has to also simultaneously take care of other variables which are beyond his or her control. Such variables are called **relevant variables** and need to be controlled as they might confound the effect of independent variable.

In experimental studies three kinds of relevant variables are taken into account.

These are organismic variables, situational variables and sequential variables.

Organismic variables are related to personal characteristics of the participants such as age, sex, and personality features. Situational variables are concerned with the quality of physical environment during the conduct of experiment such as temperature, humidity and noise. Sequential variables are related to the very procedure of conducting the experiment when the participant is required to be tested across several conditions. Hence exposure of the participant to varied conditions may result either in attaining proficiency due to practice effects or in developing fatigue and monotony towards experiment.

Experimenters use following techniques to control the unwanted effect of relevant variables.

(1) Elimination: In this technique extraneous variables are eliminated from the experimental setting.

(2) Making Conditions Constant: In this technique the extraneous variables which cannot be eliminated are kept constant in order to make their effect same during the entire experiment.

(3) Matching: Through this technique the relevant variables are equated or held constant across all the conditions of experiment.

(4) Counter Balancing: This technique is used to minimize the effect of order or sequence. This is usually done by dividing the participants in two groups. On

first occasion half of the group (Group A) is given task 1 and the other half (Group B) is given task 2. On the second occasion Group A is given task 2 and Group B is given task 1.

Random assignment: In the case of random assignment all the participants have equal chance to be exposed to experimental and control conditions. It removes the systematic differences between groups.

In addition to experiments carried out in controlled setting (laboratory experiment) experiments are also conducted in natural life conditions. They are called **field experiments** and quasi experiments. Like laboratory experiment independent variable is manipulated and participants are assigned to different groups. In quasi experiments independent variable is manipulated in natural setting using naturally occurring groups to form experimental and control groups.

Case Study: You must have seen a doctor asking personal details in addition to the information about the medical problem of the patient or a media person asking so many questions about various aspects of life while taking interview of a popular person. The purpose behind asking these questions is to know more about the person in terms of his experiences, relationships and interaction with others so as to prepare profile of the person. In psychology this method is called case study.

In the field of psychological enquiry case study method has its own importance and relevance. In this method the main unit of analysis is the individual and his experiences across different contexts in life. It focuses on the individual's interactional patterns with significant others as well as his personal experiences across different real life situations. In order to prepare a case history of data are taken from many sources for example his or her family history, educational life, medical history and social life. This method is very popular in clinical psychology and life span developmental psychology.

In order to prepare the case history usually interview, observation and psychological tests are used to obtain information about the individual. The data collected through these techniques are analyzed in detail. A comprehensive profile of the individual is developed which reflects the description of events in his or her life. Case study helps to locate unique experiences of life as well as the various emotional and adjustment problems of the individual.

Though case study gives a detailed and in-depth description of individual's life we cannot make a very conclusive judgment about the individual without further establishing the reliability and validity of such information from various sources such as family members, friends and administration of some standardized psychological measures. Caution should be taken in planning data-collection from the individual and interpretation of the responses given by the individual.

Survey: You might be aware that television news channels or newspapers ask you to send your view through SMS on current issues of national or international importance. While doing this they try to seek the opinion of people on those issues

to communicate their view to the Government as well as to the society. For example they conduct opinion poll during the election as to which political party enjoys support of the majority of the people. Conducting such a study is called survey research. It is one of the popular research methods not only in psychology but also in other disciplines such as sociology, political science, economics and management.

In psychology survey method is generally used to study the pattern of opinions, attitudes, beliefs and values of the people. This method is also used to test the

hypothesis about the relationship of variables especially when some incident takes place. For example media tried to analyze the responses of the people across the country after the attack by terrorists on Mumbai. In order to collect the data from people a variety of sources are used such as directly contacting the participants with a set of questions and taking their interview, sending the questionnaire through email or through post and asking them to send SMS by their mobile phones. Thus in survey, research is generally conducted through questionnaire or interview. It can be conducted on a single individual as well as on a group.

INTEXT QUESTIONS 2.2

1. Observation is divided into _____ and _____
_____ observation depending on the role of observer.
2. In an experiment the experimenter studies the effect of one variable on the other by deliberately _____ and _____ one variable.
3. In the case study method the main unit of analysis is the _____
and his experiences across different contexts in life.
4. The variable which is controlled and manipulated by the experimenter is called _____ variable and the variable on which its impact is studied is known as _____ variable.
5. _____ method is generally used to study the pattern of opinions, attitudes, beliefs and values of the people.

2.3 PSYCHOLOGICAL TOOLS

While conducting psychological research a variety of tools are used to collect data and relevant information from the participants. These tools are in the form of paper, instruments or computer software. The administration of these tools helps the psychologist to obtain verbal, written, behavioural or physiological responses. In this section we will discuss some of the psychological tools which are frequently used in conducting research.

Psychological Tests: You must have heard about psychological tests which measure intelligence, aptitude and interest. Development of test is a major area of activity in psychological research. The tests are designed and developed to assess various psychological attributes. They are developed on the basis of a theoretical framework. For example a test of intelligence is developed following a theory of intelligence. These tests are administered to the individual alone or in a group setting. The obtained score of the individual on the test reveals his or her position in relation to others who also respond to the same test. Thus a psychological test provides an **objective assessment** of different qualities and limitations of the individual. A **standardized psychological test** has properties of reliability and validity. **Reliability** of a test refers to its consistency in terms yielding dependable scores. **Validity** of a test reveals the extent to which the test measures what it claims to measure.

Depending on the nature and administration a test can be either **verbal or nonverbal** (performance). In a verbal test the responses are taken in oral form. In nonverbal or performance test the responses are taken in the form of performance or certain behaviour. Psychological tests are also categorized as objective and projective. An **objective test** contains direct items about the psychological construct. The individual has limited freedom to respond to the items of the objective test. A **projective test** uses ambiguous, vague and unstructured stimuli such as pictures, inkblots, drawings, incomplete sentences etc. In this type of test the individual is free to give his/her responses.

Thus a number of tests are used to assess psychological attributes of the individual. The score on the test reveals the extent to which the individual possesses those attributes. Such scores help the psychologist to decide about the future course of action.

Questionnaire: A questionnaire consists of a set of questions to which the individual is required to respond. The items (questions) of the questionnaire can be either in **closed-ended** form or in **open-ended** form. In the case of closed-ended item the individual is provided with limited alternative and he or she has to choose only one alternative which reflects his or her view on the item. In open-ended items the individual is free to give his or her response the way he or she likes. The instruction as to how to respond to the items of the questionnaire is written on the first page. Data from a large number of individuals can be taken at a time as the questionnaire can be easily administered to a group of people. The items of questionnaire are written in simple and explicit language so that anyone can understand it. All the items tap various aspects of the construct which is measured. The items are often arranged in the sequence from general to specific.

Interview: It is a techniques of data collection in which a **face-to-face interaction occurs** between two persons with a set objectives. The person who conducts interview is called interviewer and the person who give responses is called interviewee. The interviews are also conducted through telephone, internet and video conferencing. The main purpose of interview is to understand various personal characteristics such as attitudes, values, interests and preferences.

Two types of interviews are often used to obtain information. These are **structured interview and unstructured interview**. In the case of *structured interview* the questions are already framed with the possible response options. The interviewee is required to respond to the set of framed questions by choosing one option. For example the attribute of friendliness can be measured by giving the option ranging from ‘highly friendly’, ‘often friendly’ to ‘least friendly’. *Unstructured interview* is a little flexible. It comprises of a variety of open-ended questions and the interviewee gives his or her responses as freely as possible. During the course of interview the interviewer also frames and reframes the questions and facilitates the entire process of interview. Conducting interview to recruit suitable candidates for a particular job is a good example of unstructured interview.

In order to conduct an interview the interviewer should possess certain skills which help him/her to elicit maximum responses from the interviewee. A skilled interviewer easily establishes rapport with the interviewee by reliving his/her anxiety and making him comfortable during the course of interview. He has command over his language which helps him to put even difficult questions in simple and lucid way in order to probe at a deeper level. He has control over his feelings and emotions which does not give any cue to the interviewee during the interview.

In the field of psychology interview is used for recruitment and selection, counseling, marketing and advertising, attitude survey etc. In this chapter you have learnt about the various approaches that help understand the nature and causes of human behaviour, and about the different methods that help us to gain more knowledge about mental processes.

INTEXT QUESTIONS 2.3

1. A _____ provides an objective assessment of different qualities and limitations of the individual.
2. _____ of a test refers to its consistency in terms yielding the scores from the representative sample for which it has been designed.
3. _____ of a test reveals the extent to which the test measures what it claims to measure.
4. A _____ uses ambiguous, vague and unstructured stimuli such as pictures, inkblots, drawings, incomplete sentences.

5. The items (questions) of the questionnaire can be either in _____
_____ form or in _____- form.

6. Interview as one of the techniques of data collection is often referred as a _____ between two persons with a set objective.

7. In the case of _____ the questions are already framed with the possible options/

8. _____ comprises of a variety of open-ended questions and the interviewee gives his or her responses as freely as possible.

WHAT YOU HAVE LEARNT

There are different approaches to explain, describe, predict and control behavior and mental processes. The main approaches are Biological, Psychoanalytic, Behaviouristic, Humanistic and Cognitive. Scientific method has the characteristics of being objective, testable, self-correcting and replicable.

Different methods are used to understand human behaviour. Observational method helps to describe a phenomenon in a laboratory or natural setting. It can be participant or non-participant.

Experimental method studies the effect of one variable on another variable by manipulating one and controlling other variables.

Case study focuses on one person. The person is studied in great detail to understand the underlying issues.

A questionnaire consists of a set of questions, which the respondent answers. It can be an open or closed ended.

Interview is a face-to-face interaction regarding a given topic. Interview can be structured or unstructured.

TERMINAL QUESTIONS

1. Describe three main approaches used by psychologists to understand mental

processes. Why do we need so many approaches to understand human behaviour?

2. Describe the characteristics of scientific method. Explain the use of observation for data collection.

3. Discuss the experimental method as a scientific method. Identify the techniques used to control relevant variables.

4. Discuss how psychological tools are used to understand human behaviour and psychological processes.

ANSWERS TO INEXT QUESTIONS

2.1

1. Biological approach
2. Cognitive approach
3. Humanistic approach
4. Psychoanalytic approach
5. Behaviouristic approach

2.2

1. Participant, non-participant
2. Manipulating, controlling
3. Individual
4. Independent, dependent
5. Survey

2.3

1. Psychological test
2. Reliability
3. Validity
4. Projective test
5. Close-ended, open-ended
6. Face –to-face interaction
7. Structured interview
8. Unstructured interview

Hints for Terminal Questions

1. Refer to section 2.1
2. Refer to section 2.2
3. Refer to section 2.2
4. Refer to section 2.3

3. INDIVIDUAL DIFFERENCES

It is often said that no two individuals are exact duplicates; they differ from each other in some way or the other. Hence the job of the psychologist is to identify and understand this uniqueness in individuals. Such a similarity or difference between persons reveals individual differences. It happens in our day-to-day life when we see people around us. A question comes to mind; how and why people appear similar or different to each other? For example when we think about their physical appearance, we often ask ourselves why some people have dark or fair complexion, why some people are tall and some are short, why some are thin and why some are very fat. When we think about their psychological characteristics we often come across people who are very talkative or less talkative, some laugh too much whereas others take much time even to smile, some are very friendly whereas some prefer to be alone. The present lesson tries to answer all such queries which can bother us in our everyday life. In psychology, these are called **individual differences** referring to the extent and kind of variations or similarities among people on some of the important psychological aspects such as intelligence, personality, interest, and aptitude. This lesson will also help us understand how to assess such similarities or variations among individuals.

OBJECTIVES

After studying this lesson, you will be able to:

describe the nature of individual differences;

understand how to assess individual differences;

discuss the nature and definition of intelligence and tools to measure it;

explain the nature and definition of aptitude and tools to measure it;

explain interest and tools to measure interest; and

discuss the nature of personality and describe the different techniques used in personality assessment.

3.1 NATURE OF INDIVIDUAL DIFFERENCES

It has been observed that if we collect information about people's characteristics

from a large sample and examine the pattern of distribution (as shown in Fig.-3.1) we find that a large majority of the people fall in the middle range while a small proportion lies in extreme categories. For example, most of the people fall in the category of average height and very few are very tall or very short. This holds true for many more characteristics including intelligence and other psychological attributes.

Fig. 3.1 Distribution of height of a large sample of persons

The fact that people are different from each other is a very common observation. The differences in psychological characteristics are often consistent and form a stable pattern. By 'consistent', we mean that people tend to show regularity in their behavior and their patterns of behaviour do not change very frequently. This consistency and stability in behaviour is unique to every person. People develop their unique traits/ characteristics and patterns of behaviour due to their genetic make up and the environment in which they are brought up. Once we know these differences systematically we can utilize the capabilities of people efficiently for their healthy development. Knowing about the specific characteristics of a person is necessary in order to extend support and utilize his or her potential to optimal level.

Individual differences occur due to interaction of genetic and environmental factors. We inherit certain characteristics from our parents through genetic codes. The phenotype or the expressed forms of our characteristics depend on contributions of the socio-cultural environment. This is the reason why we are not exactly like our parents and our parents not exactly like our grandparents. We do share similarities with our parents in respect of many physical attributes like height, colour of eyes, shape of nose etc. We also inherit certain cognitive, emotional and other characteristics from our parents like intellectual competence, love for sport, creativity etc. However, our own characteristics develop largely by the support from the environment which we inhabit.

The environment is responsible as how we are reared, the kind of atmosphere at

house, whether it is liberal or strict, the type of education that we get, what we learn from people, around us, books, cultural practices, peers, teachers and media All these aspects refer to 'environment' which help in developing our potentials.

Environment, by providing models and other opportunities, helps us develop many traits and skills. Our inheritance alone cannot decide what we become but our environment also contributes. We know the example of Dr. Babasaheb Ambedkar, who was born in a very poor family but, with the right education and environment, became a great lawyer and also designed the Constitution of India. The latest and most well known example would be of our former President Dr. A.P.J. Abdul Kalam. You can also remember many such examples, who did not become great because of inheritance but because of the environment. Now you know that our genetic codes vary. At the same time, surrounding environment also differs from person to person. It sets limits or defines a range by offering different opportunities. That is why the pattern of development of each of us is different from others. It makes us think why we are similar in certain ways and different in others.

INTEXT QUESTIONS 3.1

1. Explain the term 'individual differences'.

3.2 ASSESSING INDIVIDUAL DIFFERENCES

Psychological assessment refers to the use of specific procedures for evaluating personal qualities, behaviours and abilities of individuals. These procedures describe people by specifying how they are different from or similar to other individuals. Such assessments are frequently done by most of us when we make judgments such as 'nice', 'good', 'bad', 'attractive', 'ugly', 'genius' and 'idiot' etc. Such judgments can be erroneous many times. Scientific psychology tries to systematize these procedures so that assessment can be made with minimum error and maximum accuracy. Psychologists have developed 'tests' to assess these characteristics. A psychological test is a structured technique used to generate a carefully selected sample of behaviour.

In order to be useful for the purpose of drawing inferences about the person being tested, it is necessary that the test should be reliable, valid and standardized. A test is reliable if it measures a given characteristic consistently. For instance, if you assess something the scores on separate occasions should be more or less similar. Thus a person, if found to be of average intelligence on one occasion should also appear of average intelligence if tested after two weeks. If a test tells two different values while assessing the object on two occasions then it will be called unreliable. A test of intelligence can be called reliable only when a person scores high or low consistently on both the occasions. A good test is found to have high reliability.

The validity of a test refers to the degree to which it assesses what it intends to assess. A valid test of personality gives a measure of a person's personality and predicts behaviour in situations where that aspect of personality is pertinent.

Finally, in order to be useful, an assessment tool should be standardized.

Standardization involves establishing the procedure of administration of a test to all persons in the same way under the same conditions. It also involves establishing group norms so that an individual's score can be compared with those of others in a defined group. It is very important because a test score is only a relative score. It does not provide an absolute value as found in physical measurement. Standardization ensures uniformity and objectivity in administration and makes the results interpretable.

Psychologists have developed tests to measure different human characteristics. In schools, we use achievement tests which measure what people have learnt.

Psychologists frequently use tests of ability and personality. The tests of ability tell what an individual can do when he or she is at his/her best. Ability tests measure capacity as potential rather than achievement. Tests of intelligence and aptitude come under this category. Aptitude refers to the ability to learn a particular kind of skill required in a specific situation. Personality tests measure the characteristic ways of thinking, feeling or behaving.

INTEXT QUESTIONS 3.2

1. Define psychological assessment. Explain the need for psychological assessment.

2. Discuss in brief the main properties of a psychological test.

3.3 NATURE OF INTELLIGENCE AND ITS ASSESSMENT

You often come across the term intelligence in everyday life. We generally use this term whenever we find somebody doing something very good which goes beyond our expectation. Intelligence is one of the psychological terms used quite frequently in various settings (e.g. school). Who can be called 'Intelligent'? The one who gets highest marks in exams? That person who earns many educational degrees? Is the doctor more intelligent, or the engineer or the lawyer or the artist? One may answer these questions in different ways depending on the meaning of intelligence.

Intelligence is much more than getting degrees. Intelligence refers to "multifaceted abilities of people". It gets expressed in many ways. It comes in many forms. Some people are good in studies, some are good in repairing machines, some are good in acting and some are great in sports. People are very good in one subject and average in some other. The most important thing is that 'intelligence' is 'functional'. It is 'used' to do something and to achieve something.

In psychology, the term intelligence has been defined in many ways. One of the earliest definitions of intelligence was given by Binet and Simon in 1905 who defined it as the “ability to judge well, to understand well, and to reason well”. One of the most popular definitions of intelligence was given by Wechsler who defined it as “the aggregate or global capacity of the individual to act purposefully, to think rationally, and to deal effectively with the environment”.

Gardner defined intelligence as “the ability or skill to solve problems or to fashion products which are valued within one or more cultural settings”. He used the term ‘Multiple Intelligences’ and advocated that there are eight types of intelligences such as Linguistic, Logical-mathematical, Spatial, Musical, Bodily-kinaesthetic, Interpersonal, Intrapersonal, and Naturalistic.

Though the first attempt to measure intelligence was made by Sir Francis Galton a more systematic approach was developed by Alfred Binet, a French Psychologist. In 1905, Binet gave the concept of Mental Age (MA) which refers to an individual’s level of mental development relative to the environment in which he/she lives. The term Intelligence Quotient (IQ) was first devised by William Stern, a German psychologist, in 1912. IQ is defined as mental age divided by chronological age, and multiplied by 100: ($IQ = MA/CA \times 100$). For example if the mental of a child is 12 and his/her Chronological age is 8 then the IQ of the child would be 150. The intelligence test developed by Binet was revised subsequently and in 1916 the test was given the name of Stanford - Binet test of intelligence. One of the most popular and widely used tests of intelligence is Wechsler Scales of Intelligence. These scales have been designed for individuals of different age groups such as Wechsler Adult Intelligence Scale (WAIS) for adults and Wechsler Intelligence Scale for Children (WISC) for children between the age of 6 and 16 years.

Intelligence tests are of two kinds Individual test and Group test. An individual test of intelligence can be administered to a single individual at a given time whereas a group test is administered to more than one individual at a time. On the basis of nature of items, intelligence tests are Verbal, Non-verbal, and Performance Tests. A verbal test requires understanding of written words. Hence it can be administered to literate individuals only. In non-verbal test, pictures or illustrations are used as item of the test. Performance tests are made up of certain concrete tasks. Both non-verbal and performance tests can be administered to literate and illiterate individuals.

INTEXT QUESTIONS 3.3

1. What do you understand by the term intelligence?

2. What is IQ?

3.4 APTITUDES AND INTERESTS, AND THEIR ASSESSMENT

You may have come across people who are very proficient in one task but not that much in another task. Some of your friends may be very good in mathematics but very poor in social science. Such people possess special abilities or characteristics in a particular field but relatively low in other field. In psychology, this is called aptitude. An aptitude is a combination of characteristics that indicates an individual's capacity to acquire some specific knowledge or skill, after training. These qualities can be harnessed by appropriate training. In other words, if a person does not have the special abilities required to become a musician, such as discrimination between pitch, tone, rhythm, and other aspects of musical sensitivity, he/she would not be a good musician, even after sufficient training.

There are salient differences between intelligence, aptitude, and achievement.

Intelligence relates to the ability of a person to do certain thing at a given time.

Aptitude refers to the potential ability of an individual to perform a task, which

generally consists of a combination of abilities. Achievement involves performance at any given point of time in a particular subject (e.g. mathematics) with which you have been made familiar.

Aptitude tests are used for predicting success in a vocation such as clerical aptitude, mechanical aptitude, musical aptitude, typing aptitude, etc. Each of these tests usually contains a number of sub-tests. Several multiple aptitude test batteries have been developed to assess aptitude, such as Differential Aptitude Test (DAT), the General Aptitude Test Battery (GATB), and the Armed Services Vocational Aptitude Battery (ASVAB).

Psychologists have constructed a number of tests to measure interest for children and adults. There are vocational interest tests which help identify the interest of a person in particular vocational areas. There are also general interest inventories. Vocational interest tests are paper-pencil tests that assess a person's interest and match them with success in various occupations.

INTEXT QUESTIONS 3.4

1. Define aptitude.

2. What do you understand by aptitude batteries?

3.5 NATURE OF PERSONALITY

We all use the term 'personality' in day-to-day life. In psychology personality is thought as a person's unique and relatively stable behaviour pattern which remains consistent across situations and over a period of time.

The term personality has been defined in diverse ways. Allport in 1937 gave a very popular definition of personality which is still referred to by scholars. According to him “personality is the dynamic organization within the individual of those psychophysical systems that determine his unique adjustment to his environment”. Psychologists have approached personality from different perspectives. Each of these perspectives explains some aspects of personality. Now let us study in brief about some of these perspectives:

I) The trait perspective tries to describe personality in terms of various traits. Some times the traits are grouped into clusters. These clusters are called “types”. For instance introversion and extraversion are two types of personality traits.

II) The psychodynamic perspective calls attention to the unconscious needs and conflicts as well as the influence of earlier stages of development on our lives. Sigmund Freud, a noted psychologist, provided the core concepts for this perspective.

III) The socio-cultural perspective highlights the importance of the social and cultural environment. In view of this theory, our personality and behaviour patterns are acquired through interaction with others and adoption of social and cultural norms.

IV) The humanistic perspective emphasizes the enormous potential for freedom and growth present in each one of us. It is a view which is optimistic and emphasizes on positive aspects of life and potentialities.

INTEXT QUESTIONS 3.5

1. Define personality. Describe any two perspectives on personality?

3.6 TECHNIQUES OF PERSONALITY ASSESSMENT

In the field of personality assessment the major focus is on the study of the nature and intensity of various characteristics, which are generally reflected through the behaviour of the individual in a given situation. Personality assessment measures are classified into three categories.

- A. Objective or Self report measures of personality.
- B. Projective or Indirect measures of personality.
- C. Situational measures of personality.

A. Objective or Self-Report Measures of Personality

In measures of this kind, the person is asked to report about himself/herself on a set of statements organized in the form of a questionnaire or inventory. The nature of the statement can be either closed ended or open ended. In the case of closed ended statements, rating scales or optional answers are given whereas in the case of open-ended statements, the person is free to describe himself or herself. A number of tests have been developed to assess different trait domains of personality. Some of the self-report measures of personality, which are widely used to assess personality, are being discussed below.

(i) Neuroticism, Extraversion, Openness- Personality Inventory-Revised (NEOPI-R): This test is currently one of the most popular personality assessment tools being used across the world. This test has been developed by Costa and McCrae (1992) and is based on Five-Factor Model of Personality. The test has been developed based on longitudinal studies over 15 years. It provides scores on five major dimensions of personality. These five dimensions and facets are as follows: Neuroticism, Extraversion, Openness to Experience, Agreeableness and Conscientiousness.

(ii) Sixteen Personality Factor Questionnaire (16 PF): This test was originally

developed by Raymond B. Cattell, Karen Cattell and Heather E. P. Cattell in

1949. Since then, the test has undergone four revisions in 1956, 1962, 1967-69,

and 1988-93. Currently the fifth edition developed by Conn & Rieke (1994) is in practice. This test is popularly known as 16 PF and is used for age group of 16 and above. It yields scores on 16 primary traits such as social boldness, dominance, vigilance, emotional stability and rule consciousness. It also measures five global factors, which are Extraversion, Anxiety, Tough-Mindedness, Independence and Self-control.

(iii) In India a number of self- report measures of personality have been developed. Some of these modified versions are Bell Adjustment Inventory by Mohsim and Hussain, Maslow Security-Insecurity Inventory by Singh and Jamuar, Maudsley Personality Inventory of Eysenck in Hindi by Singh and Jodhpur Multiphasic Personality Inventory by Joshi and Malik, amongst others.

B. Projective or Indirect Measures of Personality

Projective or indirect measures of personality are widely used personality assessment tools. Unlike the self-report measures where the individual is provided with structured test stimuli, in projective tests the individual has to respond to unstructured or ambiguous stimuli. The basic assumption behind using unstructured test stimuli in projective tests is that the individual projects his/her latent or unconscious feelings, needs, emotions, motives etc. on to the ambiguous stimulus. The responses of the individual in the form of projection reflect the nature of his/her personality.

Some popular Projective Technique are Rorschach Ink Blot Test, Thematic Apperception Test and Rotter Incomplete Sentences Blank.

C. Situational or Observation Measures of Personality.

Another technique to assess personality is to observe the behaviour of the individual in purposefully designed situation. Such tests are more effective in the assessment of leadership, dominance, responsibility, extraversion-introversion etc. One of the first situational tests was developed by Hartshorne, May, and their associates (1928, 1929, 1930) for the Character Education Enquiry (CEE). This test measures traits like altruism, honesty and self-control. However, situational measures of personality came into light during World War II when the United State Office of Strategic Services (OSS) used situational tests for selection of military personnel for critical overseas assignments. One of the variants of this test is Situational Stress Test in which the individual has to react to various stressful, frustrating, or emotionally disruptive situations where his helpers are obstructive and uncooperative.

INTEXT QUESTIONS 3.6

1. What is a self-report measure of personality?

2. Discuss the use of projective technique in measuring personality.

3. What do you understand by situational or observational measure of personality?

WHAT YOU HAVE LEARNT

The study of individual differences focuses on the extent and kind of variations or similarities among people on some of the important psychological aspects such as intelligence, personality, interest, and aptitude.

Individual differences occur due to interaction of genetic and environmental factors. The differences in psychological characteristics are often consistent and

form a stable pattern. This consistency and stability in behaviour is unique to every person.

Psychological assessment refers to the use of specific procedures for evaluating personal qualities, behaviours and abilities of individuals. These procedures describe people by specifying how they are different from or similar to other individuals.

A test should be reliable, valid and standardized. A test is reliable if it measures a given characteristic consistently. The validity of a test refers to the degree to

which it assesses what it intends to assess. Standardization involves establishing the procedure of administration of a test to all persons in the same way under the same conditions.

Intelligence refers to “multifaceted abilities of people”. In psychology the term intelligence has been defined in many ways. Binet and Simon in 1905 defined it as the “ability to judge well, to understand well, and to reason well”.

One of the most popular definitions of intelligence was given by Wechsler in 1939. He defined intelligence as “the aggregate or global capacity of the individual to act purposefully, to think rationally, and to deal effectively with the environment”.

Gardner used the term ‘Multiple Intelligences and advocated that there are eight types of intelligences such as Linguistic, Logical-mathematical, Spatial, Musical, Bodily-kinaesthetic, Interpersonal, Intrapersonal, and Naturalistic.

The first systematic effort to assess intelligence was made in 1905 by Alfred Binet, a French psychologist, who gave the concept of Mental Age (MA).

The term Intelligence Quotient (IQ) was at first devised by William Stern, a German Psychologist, in 1912. IQ is defined as mental age divided by chronological age, and multiplied by 100.
$$IQ = \frac{MA}{CA} \times 100.$$

There are several Intelligence Scales developed by Wechsler such as Wechsler Adult Intelligence Scale (WAIS) for adults.

There are salient differences between intelligence, aptitude, and achievement.

Intelligence refers to the ability of a person to do a certain thing at a given time.

Aptitude refers to the potential ability of the individual to learn to perform a task. Achievement involves performance at any given point of time in a particular subject (e.g. mathematics) with which you have been made familiar.

Aptitude tests are used for prediction of success in various kinds of vocations

such as clerical aptitude, mechanical aptitude, musical aptitude, typing aptitude, etc. Several multiple aptitude test batteries have been developed to assess aptitude. The most widely used is Differential Aptitude Test (DAT).

Personality refers to a person's unique and relatively stable behaviour patterns which remain consistent across situations and over a period of time. Allport defined personality as the "dynamic organization within the individual of those psychophysical systems that determine his unique adjustment to his environment".

Psychologists have approached personality from different perspectives. These View points can be grouped in to four major types: Psychodynamic perspective, Socio-cultural perspective, Humanistic perspective, and Trait perspective.

Personality measures are classified into three categories: Self report measures, Projective measures, and Situational measures.

TERMINAL QUESTIONS

1. What do you understand by the term “individual differences”?
2. Define Intelligence. Name any two Intelligence Tests.
3. Explain the differences between Intelligence, Aptitude and Achievement.
4. Identify the major perspectives towards the understanding of personality.

ANSWERS TO INTEXT QUESTIONS

3.1

1. Extent and kind of variations or similarities among people on psychological characteristics. They occur due to interaction of genetic and environmental factors.

3.2

1. Use of specific procedures employed in evaluating personal qualities, behaviours and abilities of individuals.
2. Reliability, validity, standardization.

3.3

1. Aggregate or global capacity of the individual to act purposefully, to think rationally and deal effectively with the environment.
2. Intelligence quotient is mental age divided by chronological age multiplied by 100 $IQ = MA/CA \times 100$

3.4

1. Combination of characteristics that indicates an individual's capacity to acquire some specific skill or knowledge after training.
2. Are developed to assess aptitude eg. Differential Aptitude Test (DAT) and General Aptitude Test Battery (GATB)

3.5

1. Person's unique and relatively stable behaviour pattern which remains consistent across situations and over a period of time. Trait perspective, biological perspective, psychodynamic perspective, socio-cultural perspective, humanistic perspective (any two).

3.6

1. Person is asked to report about himself / herself on a set of statements.
2. Individual has to respond to unstructured or ambiguous stimuli. The responses in the form of projection reflect the nature of his/her personality.
3. Behaviour of individual is observed in purposefully designed situations.

Hints for Terminal Questions

1. Refer to section 3.1
2. Refer to sections 3.2 & 3.3
3. Refer to section 3.4
4. Refer to section 3.5

4. LEARNING

You must have observed a young human baby. If you wave your hands in front of the eyes of a new born you will see that the child automatically closes her eyes. A human baby is born with simple reflexes and when grown as an adult the individual becomes capable of performing many complex behaviours. This happens because of learning. The process of learning is continuous and goes on throughout life. From simple behaviours, like switching on the light, or picking up a book from the table to complex behaviours like driving a car or piloting a spaceship, all become reality only due to learning.

Learning empowers a person by making him/her what one is not and wants to become. All children are born with limited capacity of reflexes but it is only through learning that people become great scientists like Homi Bhabha, Vikram Sarabhai, leaders like Mahatma Gandhi and Pandit Nehru or any one whom you may admire. Without learning one is handicapped. People realize their potentials through learning and become what they dream. This lesson will help us understand the process of learning and its types.

OBJECTIVES

After studying this lesson, you will be able to:

explain the process of learning and its definition;

describe the nature of classical and operant types of conditioning;

explain learning by observation; and

explain concept learning, skill learning and transfer of training.

4.1 LEARNING

Most of our behaviours are acquired through the process of learning. Learning may be defined as a relatively durable change in behaviour due to experience. Thus, if

the change in behaviour is temporary, or due to instinct or maturation, it is not learning. The key feature of learning is experience. Any change in behavior in the absence of practice or experience does not qualify as learning. For example the maturational changes in the child, like crawling, standing and walking at certain ages are not included in learning. Similarly, illness and drug induced temporary changes are not due to learning. To qualify as learning, a change in behaviour must be relatively enduring.

Interestingly enough learning is not directly observable. It is often inferred from changes in the external behaviour. It is apparent in terms of improvement in the performance. You may recollect your own early childhood experiences when you were required to learn alphabets. In the beginning we make more errors. However, when we start practicing the errors decrease and a time comes when the behaviour becomes flawless. The decrement in the number of errors with increase in practice clearly indicates that learning is taking place.

How Do We Learn?

Learning helps us adapt to the surrounding environment. After living in a particular sociocultural environment for some time, we learn the norms of the society and all that is expected of us and become responsible citizen and members of family and work organization. All this is possible on account of learning. We use learning to acquire various types of skills. But the crucial question is how do we learn?

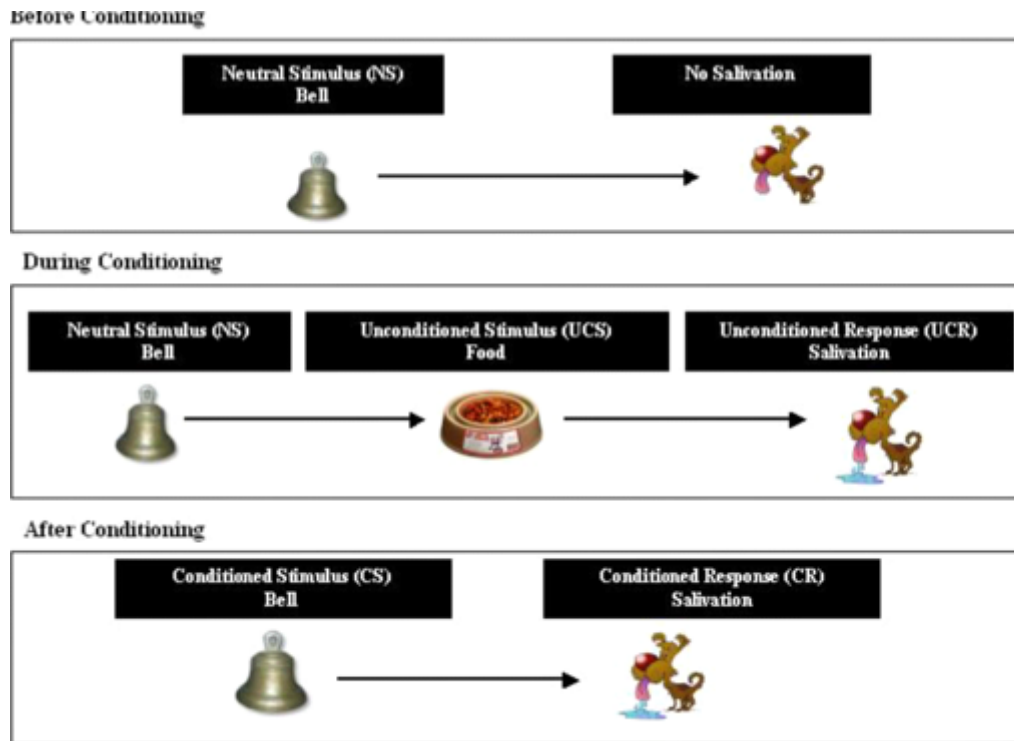
Psychologists, on the basis of studies on human beings and on animals, have tried to explain the process of learning. They have identified some procedures that are used in the acquisition of simple as well as complex responses. The two basic types of learning are classical conditioning and operant or instrumental conditioning. In addition, we have observational learning, verbal learning, concept learning, and skill learning. Let us briefly consider some of the important types of learning.

(i) Classical Conditioning : Learning through Association

Classical conditioning is also known as Pavlovian conditioning because it was discovered by a Russian scientist Ivan P. Pavlov who was interested in studying stimulus-response relationship. He worked with dogs. The animal was harnessed and food was presented to the dog by a laboratory assistant. Pavlov realized that the salivary response had been learned by the dog. He proceeded to study this type of learning in detail. (See Fig. 4.1) Pavlov presented to the dog a series of trials in which a tone (buzzer) was paired with food (biologically important stimulus). The learning trials consisted of pairing the tone, (Conditioned Stimulus or CS) with food

(Unconditioned Stimulus or UCS). The tone presented was rather short (e.g., 10 secs) and the time interval between the tone and presentation of food, was between 2 to 3 minutes.

Fig. 4.1: Classical Conditioning



During the initial trials the dog salivated when the food was presented. Secretion of saliva to food is a natural response and, therefore, the saliva is called Unconditioned Response(UR). Later, the buzzer was presented first. It was followed by food. After some pairing trials of buzzer and food, the dog started salivation at the ringing of buzzer alone. The buzzer is called Conditioned Stimulus (CS) because the dog has been conditioned to salivate at the presentation of the buzzer itself. The secretion of saliva on the presentation

of buzzer is called Conditioned Response (CR). The steps of conditioning process are shown in Fig. 4.2.

Procedure Response Response

Before conditioning

Food Salivation

(UCS) (UCR)

Buzzer Orienting Response Orienting Response

(CS)

During conditioning (Acquisition phase)

Food + Buzzer Salivation

(UCS) (CS) CR)

(Repeated pairing of the UCS and CS)

After conditioning

Buzzer Salivation

(CS) (CR)

UCS = Unconditioned Stimulus; CS = Conditioned Stimulus;

UCR = Unconditioned Response; CR = Conditioned Response

Figure 4.2: The process of classical conditioning.

It can be observed from Fig. 4.2 that originally the sound of the buzzer (CS) elicits only an orienting response (alerting response) and elicits salivation when the food is consistently presented in close association with the buzzes. Immediately after the buzzer is rung, the animal elicits salivation. This is called **acquisition**.

It has been found that if the buzzer (CS) is presented on each trial but the food (US) is not presented, **extinction** will take place. That is, the buzzer will no more elicit saliva (CR) and if it is continued for some time extinction will take place.

It has also been found that if a gap of some duration occurs after extinction and if the buzzer (CS) is again presented without food (US) the dog will salivate (produce CR) for a few trials. This recovery of CR after extinction is called **spontaneous recovery**.

(ii) Operant Conditioning: Modification of Behaviour through Reinforcement

If a child completes the home work she is praised (rewarded) by the parents and the child learns to perform the task. If the child breaks a plate, he/she is scolded (punished) and she will learn not to repeat the behaviour. This is called Operant Conditioning or instrumental conditioning. In other words, we learn to perform behaviours that produce positive outcomes and avoid behaviours that yield negative outcomes.

We may define operant conditioning as a process through which organisms learn to repeat behaviours that produce positive outcomes or avoid or escape from the negative outcomes. **B.F. Skinner** is considered as the most influential psychologist advocating the role of operant conditioning in learning. He developed an experimental chamber (called Skinner Box) to study learning process in rats. (see Fig. 4.3)

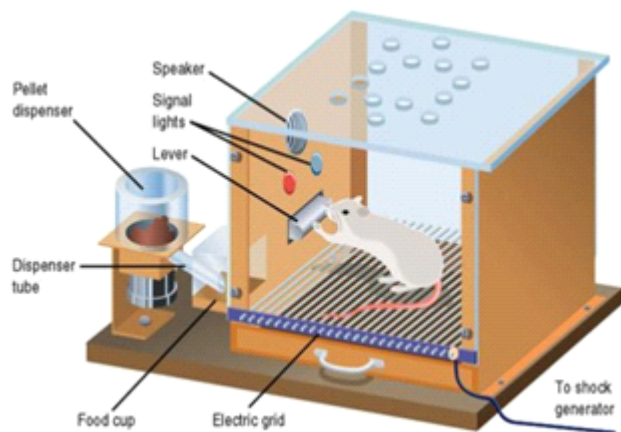


Fig. 4.3: Skinner Box

This figure shows a rat inside a cage. There is a lever on one side of the cage which is connected to a food box. When the rat presses the lever, the lid of the food box will open and a small amount of food will fall into the cage.

The chamber included a lever attached to the front wall. Pressing the lever is the response to be learned. The hungry rat is placed in the chamber and it starts doing random activity in it. After some time, the rat accidentally presses the lever and a pellet of food drops automatically in the plate and the rat eats it. After eating the pellet the rat again starts activity in the chamber. After some activity it again presses the lever and gets pellet (a reward). Gradually the random activity changes to more specific activity around the lever. Finally, the rat learns that pressing the lever results in dropping of the food, a satisfying outcome. In other words the pressing of lever by the rat is instrumental in providing food (reinforcement). The response (pressing the lever) is reinforced

and the behaviour is acquired or learned. The pressing of lever by the rat is instrumental in getting the food, a satisfying consequence (positive reinforcement) and that is why this type

of learning is also called instrumental learning. It is also called operant conditioning because the behaviour of rat or any organism is a kind of operation on the environment.

INTEXT QUESTION 4.1

Fill in the blanks with appropriate words.

1. Learning is a _____ permanent change in _____.
2. Pavlovian conditioning is known as _____ conditioning.
3. _____ advocated the role of operant conditioning.
4. Operant conditioning is also called _____ learning.

4.2 REINFORCEMENT AND OBSERVATIONAL LEARNING

Much of the human learning could be explained with the help of operant conditioning method. For example, parents and other authorities attempt to discourage aggressive behaviours by punishing them and reward for good behaviours. The role of reinforcement is very crucial in operant conditioning. It can be positive or negative. Let us understand these two types of reinforcement.

Positive Reinforcement :Reinforcement is any operation or action that increases the rate of response. In Skinner's experiment the rat responded by pressing the lever and obtained food. This is called positive reinforcement. So, a positive reinforcement or reward (e.g., food, sexual pleasure, etc.) is the operation that increases the strength of a specific behaviour. Positive reinforcement is any stimulus that strengthens a response that precedes it (e.g., lever pressing is reinforced by food).

Negative Reinforcement :Another quite different way of increasing the rate of response is through negative reinforcement. Suppose that in the Skinner Box the rat receives electric shock to the feet every second. When the rat presses the lever, the shock is removed for 10 secs. This increases the rate of response. This procedure is called negative reinforcement which involves application of an aversive stimulus (e.g. heat, electric shock, scudding etc.). The word "negative" refers to the nature of the reinforcer (aversive stimulus). It is a "reinforcement" because it increases the rate of response. This procedure is called "escape"

learning because the rat can escape the shock if it presses the lever. Another kind of negative reinforcement can result in conditioning called "avoidance" learning where the rat can avoid the shock by pressing the lever. In escape or avoidance learning the reinforcer is negative and the organism learns to escape or avoid its presence.

Schedules of Reinforcement

How do we reinforce the correct responses? It could be by using continuous or partial reinforcement. In the case of continuous reinforcement, every correct response is reinforced. For example, when the rat presses the lever every time it gets a food pellet (reinforcement). Alternatively, the responses are reinforced partially or intermittently (some times only). The continuous reinforcement is useful for establishing or strengthening new behaviours. The partial reinforcement, on the other hand, is more powerful in maintaining the acquired

behaviours.

Observational Learning : Modelling

Observational learning is the third major way we learn. Acquiring new skills by observing the behaviour of others is very common. It is a part of everyday life. Observational learning depends on the existence of appropriate model in one's environment. That is, the child picks up behaviour while the appropriate model is performing an activity. For example, young people learn aggression through watching the actions of others (models). Television programmes and movies provide much of the learning to the young people. When children watch violence on TV they tend to learn such behaviours. We learn various social roles through observational learning. However, observational learning is a complex process, far more complex than mere imitation. Children acquire information and learn skills through observational learning but do not put it into immediate use. People, particularly youngsters, can often be influenced in positive ways when they have appropriate role models.

INTEXT QUESTION 4.2

State whether the following statements are True or False:

1. Positive reinforcement is any action that reduces the rate of response.
2. The aversive response is called as escape learning.
3. Reinforcement can be partial or continuous.
4. Partial reinforcement does not help in maintaining behaviour.
5. Observational learning depends on the existence of appropriate model.

4.3 SOME OTHER TYPES OF LEARNING

Till now we have read about learning and conditioning. Now we will read briefly about the other types of learning.

Verbal Learning

You are reading this lesson and trying to understand the concept of learning. This is possible because of verbal learning. You have learned language. People in different parts of the world learn different languages. The process of learning language is called verbal learning.

If you recollect your younger days you will notice that you started with identifying alphabets, then moved to words and finally sentences. When you learn a foreign language you use pairs of words. Psychologists study how various procedures like serial learning and paired associates learning are used.

Concept Learning

This is about developing categories of objects and events. It is very important in our life that we should discriminate between things on the basis of some criterion. For example, the terms 'boys', 'girls', 'fruits' and 'furniture' refer to concepts. A concept involves a variety of objects clubbed together. Use of categories or class names helps us to communicate and perform different activities. The concepts may be natural or artificial. They may be as abstract and concrete. Love, freedom and democracy are examples of abstract concepts. Cow, table, boy, girl, orange and rose are examples of concrete concepts. While learning a concept we make one response to all the stimuli belonging to that category. Thus we refer to all kinds of tables as table or all boys as boy. In fact all kinds of higher learning necessarily involves concept learning. Concepts help us to reduce the complexity of our world. In lesson 7 you will read more about concept formation.

Skill Learning

An important area of learning involves acquiring various types of skills like riding a bicycle, writing, car driving, piloting an air craft, leading a group and motivating others etc. All of these involve skills. Those who are able to learn these and other skills get opportunities in life. Once the skill is acquired one may over learn it. It may become automatic and one is able to perform it with ease and comfort. As a result people perform tasks spontaneously and can perform more than one task at a time (e.g. talking to a person and car driving).

4.4 TRANSFER OF TRAINING

It is interesting to note that learning of one task does not remain confined to that specific task only. That learning is used in other situations also. A person's ability to utilize knowledge, skills and any kind of learning is highly appreciated. If a child learns to multiply or divide she can use the process not only in the class but also in the market place or at home if it is required. Transfer of training refers to the process of using earlier learning in a new situation.

It is positive if the first learning helps the second learning. It can be negative if the first learning interferes with the second learning. It can be zero if the first learning neither helps nor interferes with the second learning.

Learning transforms a person in many ways and occurs throughout one's life. Good learners utilize each and every opportunity as an occasion for learning. The methods or types of learning mentioned earlier provide only some basic ideas about learning. The changes that take place in personality, attitude and interest are also consequences of certain types of learning. They take place in complex ways. As learning progresses your ability to learn also develops. If you learn you become a better person, flexible in approach and competent enough to appreciate the reality.

Some Tips for Better Study Habits

Studying of a topic can be facilitated if you realize that effort and motivation both are important. In order to gain more while preparing a topic or studying some thing you need to attend to the following points.

Study at a Specific Place: You must study in a quiet, well-lighted area free of disturbance. If possible, you should also have at least one place where you only study. Do nothing else at the spot: Keep magazines, radios, friends, and other distractions out of that area. In this way, studying will become strongly linked with one specific place. Once there, you'll find it relatively easy to get started.

Organize Your Study through Spaced Sessions: It is quite reasonable to review intensely before an exam. However, if you are actually learning some topic for the first time you are asking for trouble. It has been found that spaced practice is a more efficient way to study. It involves a large number of relatively short study sessions, rather than one or two long ones (called massed practice).

Use Memory Aids: Psychologists have found many techniques useful for improving memory. Most of them link new information to certain ideas or images that are easy to remember.

Try to Test Yourself: When studying you can arrange to take several practice tests before the real one in examination. In other words, studying should indicate self-testing by use of flash cards, learning checks, a study guide or questions you ask yourself.

Over Learning: It has been found that many students *underprepare* for exams,

and most *overestimate* how well they will do in exams. A solution to both problems is overlearning. To overlearn, you should continue studying beyond bare mastery of a topic. This means that you need to give yourself time for extra study and review after you think you are prepared for a test.

Avoid procrastination: A tendency to procrastinate is almost universal.

Procrastinators put off work until the last moment. They work only under pressure, skip classes and give false reasons for late work. This has negative effect on learning and performance. Use of a formal time schedule and setting up specific goals may help to study more effectively.

INTEXT QUESTIONS 4.3

1. What are the main types of learning?

2. What is transfer of training?

WHAT YOU HAVE LEARNT

Learning may be defined as a relatively durable change in behaviour due to experience. If the change in behaviour is temporary, or due to instinct or maturation, it is not learning.

Learning helps us adapt to our environment.

The two basic types of learning are classical conditioning and operant conditioning or instrumental learning.

Classical conditioning involves pairing of a neutral stimulus and a natural stimulus (CS and UCS) leading to formation of association between neutral stimulus and response.

Operant conditioning is a process through which organisms learn to repeat behaviours that produce positive outcomes or avoid or escape from the negative outcomes.

Skinner defined reinforcement as any operation or action that increases the rate of response.

In escape or avoidance learning the reinforcer is negative and the organism learns to escape or avoid its presence.

In the case of continuous reinforcement, every correct response is reinforced.

The process of learning language is called verbal learning.

An important area of learning involves acquiring skills like riding a bicycle, writing, car driving, piloting an air craft, leading a group, motivating others etc.

Observational learning is also used in acquiring new skills by observing the behavior of others.

Transfer of training refers to the process of using earlier learning in a new situation. It is positive if the first learning helps the second learning.

Learning transforms a person in many ways and occurs throughout one's life. Good learners utilize each and every opportunity as an occasion for learning.

TERMINAL QUESTIONS

1. What is learning? Give an example?
2. Describe the process of classical conditioning.
3. What are the main types of reinforcement? What is the role of reinforcement in operant conditioning?
4. Give examples of observational learning and skill learning.
5. What is the significance of transfer of training?

ANSWERS TO INTEXT QUESTIONS

4.1

1. relatively, behaviour
2. classical
3. B.F. Skinner
4. Instrumental

4.2

1. False
2. True
3. True
4. False
5. True

4.3

1. Skill learning, verbal learning and concept learning.
2. Transfer of training refers to the process of using earlier learning in a new situation.

Hints for Terminal Questions

1. Refer to section 4.1
2. Refer to section 4.1
3. Refer to section 4.2
4. Refer to section 4.3
5. Refer to section 4.4

5. SENSORY PROCESSES: ATTENTION AND PERCEPTION

The purpose of describing sensory receptors, attention and perception is to familiarize you with the way our sense organs collect information and how it is processed by our brain. We have five sense organs through which we acquire information. These include eye, ear, skin, nose and tongue. We have mainly two functions of our senses: survival and sensuality. If we could not see any colors or the beauty of flowers or the pictures on our television or the traffic lights, our life would become dull and risky. Colors do not really exist “out there” in objects rather our world of colour is a product of sensory and perceptual processes of brain. We derive sensual pleasure in breathing fresh air enjoying tasty food, good music or feeling relaxed by gently touching a cat or dog. Our senses do more than just making contact with our external world. They add to happiness, variety and satisfaction in life. Sensation is a process by which neutral impulses are created by stimulation of sensory neurons that results in awareness of conditions inside or outside the body. Perception refers to the elaboration and interpretation of these sensory experiences. It is governed with our past and present experiences. With the help of this lesson you will learn how we derive a world of reality from the information that we receive from our sense organs.

OBJECTIVES

After studying this lesson, you will be able to:

1. understand various senses and the sense organs;
2. know the qualities that are common to all senses;
3. understand the factors in attention; and
4. identify some other senses beyond these five senses.

5.1 SENSATION

Sensation can be explained as the process by which one form of energy is converted into another form. For example light is converted into neural impulses by which we code sensory events in our system that can be processed by our brain. The sensory systems process information reaching to the brain. The motor systems process information going out of the brain to muscles and glands. Sometimes your parents switch on the T.V. and adjust the volume of sound that they can hear. Suppose you adjust it to a volume which you are able

to detect but your mother says that she is unable to hear and asks you to increase the volume. If your mother asks you to stop after some point that means that difference of adjustment has been noticed by your mother. This minimal amount of change of volume between two stimuli that is being recognized by your mother is called a “difference threshold”. Background of a stimulus also affects our sensation. For example stars are present in the sky in day time and at night but are visible only after sunset or at nighttime because they can not be detected due to intense background of the daylight sun.

We all are also guided by factors of expectations and experiences. Experience of sensation is not simply a yes/no, present/absent mechanism. For example you are expecting a very close friend of yours to visit your home at 4 p.m. At 4 p.m. that friend reaches your house and pushes the doorbell button. Other members of your house do not notice it but you are able to notice that sound. It is primarily because of your expectation that you notice this second clearly while others do not notice it. The minimum amount of physical energy needed to produce a sensory experience is called “absolute threshold”. We have several sense organs. The table below describes them

Table 5.1: Fundamental human senses

Sense	Stimulus	Sense organ	Sensation
Sight	Light waves	eye	Colour, pattern,textures
Hearing	Sound waves	ear	Noise, tones, music
Skin sensation	External touch	skin	Touch,cold,warmth,pain
smell	Volatile materials	nose	odors

taste

Soluble materials tongue

Sweet, salty, bitter

Let us look briefly at each of the senses.

Vision: Vision is extremely important for all of us. Humans and animals with good vision have advantage in each and everything in life. We experience vision with the help of our eyes which function like a camera. The eye gathers and focuses light like a camera. Sir Isaac Newton, who in the seventeenth century discovered the laws of

motion and gravity, also discovered that when white light passes through a prism it separates into a rainbow of color – the visible spectrum.

Color blindness: Not everyone sees colors in the same way. Some people are born with a color deficiency. Color blindness is the partial or total inability to distinguish colors. Most color blind people have trouble in distinguishing red from green.

Hearing: Hearing is equally important for our daily life. It is a principal sensory modality for human communication. Sounds are created when actions cause objects to vibrate. When vibrating objects push the molecules of medium back and forth we can experience sound. Frequency refers to the number of cycles a wave completes in a given amount of time. It is usually expressed in cycles per second (CPS) or hertz (Hz). Sound cannot travel in a true vacuum (such as outer-space) because there is no medium there to move or vibrate.

Pitch : Pitch is the highness or lowness of a sound determined by the sound's frequency. High frequencies produce high pitch and low frequencies produce low pitch.

Loudness: The loudness or physical intensity of a sound is determined by its amplitude. Sound waves with large amplitudes are experienced as loud and those with small amplitudes as soft. Loudness of sound is measured in units called decibels (dB).

Timbre: The quality of a sound wave's complexity is its timbre. The sounds that we call noise contain many frequencies that are not systematically related to each other.

Sense of bodily orientation (vestibular sense)

It is the sense of bodily orientation with respect to gravity, especially how our heads are positioned, whether straight leaning, reclining or upside down. The vestibular sense also tells us when we are moving or how our motion is changing.

Sense of bodily position and the movement of body parts (Kinesthetic sense)

Kinesthetic sense is the sense of body position and the movement of body parts relative to each other. It is a sense that provides sensory feedback about motor activities of our body, for example how the hand moves to pick up the telephone when it rings.

Sense of smell (olfaction)

The sense of smell involves a sequence of bio-chemical activities that triggers neural impulses. Once activated these neural impulses convey odor information to the brain.

Sense of taste (gustation)

The taste receptor cells are gathered in the taste buds on the upper side of the tongue. The experience of sweetness or saltiness is affected by these taste buds. There are only four true or primary taste qualities: sweet, sour, bitter and saline. Taste sensitivity develops in infancy but decreases in old age. Taste receptors can be damaged by

excessive use of alcohol, smoking, acids or hot foods but they are also replaced every few days and a permanent loss of taste is extremely rare.

Skin senses: Our skin contains nerve endings that are stimulated by contact with external objects and it produces sensations of cold, warmth or pressure. The sensitivity to pressure is maximum on face, tongue and hands and it is less on our backs. Touch plays an important role in human relations and emotions.

Sense of pain: Pain is the body's response to stimulation from noxious stimuli. Acute pain is reaction to sharp or sudden stimulation. The pain one feels in everyday life is also related to psycho-social and cultural habits. What "pain" a person experiences depend upon the meaning one attaches to pain and also on attention one receives from near and dear ones.

INTEXT QUESTIONS 5.1

1. Discuss the role of vision and hearing in our lives.

2. Briefly explain the concept of 'threshold'.

5.2 PERCEPTION

In the last section we dealt with sensation, the process of bringing information into the brain. This section is about perception, or how we use sensations into meaningful patterns. As we encounter a variety of events in our daily lives, the brain actively selects, organizes and integrates sensory information to create a picture of the world. Some of our perceptions are native or inborn and many other perceptions are a result of our past experiences.

Sensation is the stage where neural activity codes the information about nature of stimulation. Perception is the next stage in which an internal representation of an object is formed. This representation provides a working description of perceiver's external environment. Perception involves synthesis of simple sensory features into percept of an object that can be recognized.

This helps in identification and recognition, and meaning is assigned to the percepts. Perception and recognition are combined processes that do not act separately. For example a circular object may be a cricket ball or orange.

Stages of perception: The physical object in the world is called the distal stimulus

(distant from the observer) and the optical image on the retina is called the proximal stimulus (proximate or near to observer). The major task of perception is to determine the distal stimulus based on information of proximal stimulus – to know what the world out there is “really like” using one’s imagination of mind. There is more to perceiving which includes physical properties such as shape or size and past experiences.

Mental Processes

Expectations Knowledge Beliefs

Identification/Recognition

Analysis into parts

Perceptual Synthesis of Features

Organization □ □ □ □ Depth □ Consistency

Sensation

Sensory Processes

Environmental Stimulation of Objects

Fig. 5.1: The Process of Perception

INTEXT QUESTIONS 5.2

1. Explain the concept of perception.

2. Describe the process of perception.

5.3 PERCEPTUAL ORGANIZATION

Perception is an organized process. The most common form of perceptual organization is called figure ground organization in which sensations are grouped into objects or figures

that stand out on a plainer background. Look at Figure 5.2. Do you see a nose or two faces?

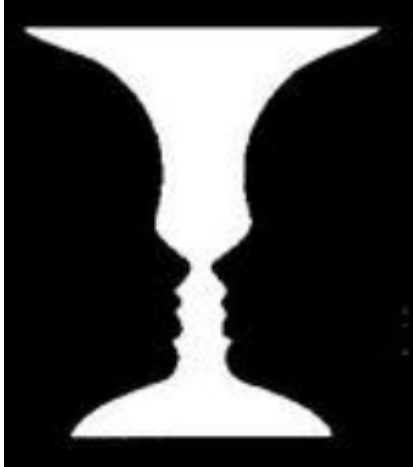


Fig. 5.2: Reversible Figure and Ground Organisation

Reversible goblet is a favourite demonstration of a figure-ground reversal. Note that either the light portion or the dark portion can be perceived as a figure against a background.

A Laws of Perceptual Grouping

The factors which determine perceptual grouping are:-

(a) Proximity: All other factors being equal, stimuli that are near one another tend to be grouped together. For example, if four stand near one another and a fifth 10 feet away, the adjacent four will be seen as a group and the distant fifth as an outsider. Events that are close in time and space are also perceived together.

(b) Similarity: Stimuli that are similar in size, shape or color tend to be grouped together.

(c) Continuity: Perception tends toward simplicity and continuity. Even if there are dots in a circular fashion, the person will see them as a complete circle.

(d) Closure: It is the tendency to complete a figure that is incomplete but has a consistent overall form.

(e) Common region: Stimuli that are found within a common area tend to be seen as a group.



Fig. 5.3: Laws of Perceptual Grouping

B. Perceptual Constancy

Perception of an object's shape, size or brightness remains the same even though its image on the retina has changed. This is called perceptual constancy and is found in all senses, though most examples given here are of vision.

If the perceived size of an object remains the same, even though the size of its image on the retina changes it is called size constancy.

In shape constancy the shape of an object remains stable even though the shape of its retina image changes.

Brightness constancy refers to the fact that the brightness of objects appears to stay the same as lighting conditions change.

C. Depth Perception

It is the ability to see three-dimensional space and to accurately judge distances. Without depth perception you can't ride on a motorcycle, or drive a car, catch a ball, thread a needle or simply walk around a room. The world would look like a flat surface. The ability of depth perception is partly innate and partly learned.

Depth cues are features of environment and messages from the body that supply information about distance and space. The cues which work with just one eye are called monocular cues and those which require two eyes are called binocular cues. Binocular cues are the most basic source of depth perception that is caused due to retina disparity (a discrepancy in the images that reach the right and left eyes). A person with one eye will have very limited depth perception.

Pictorial cues for depth are features found in paintings, drawings and photographs that impart information about space, depth and distance. This influence causes apparent perception of things which are not there. For example, if you stand between two railway tracks, they appear to meet at the horizon, even though they actually remain parallel.

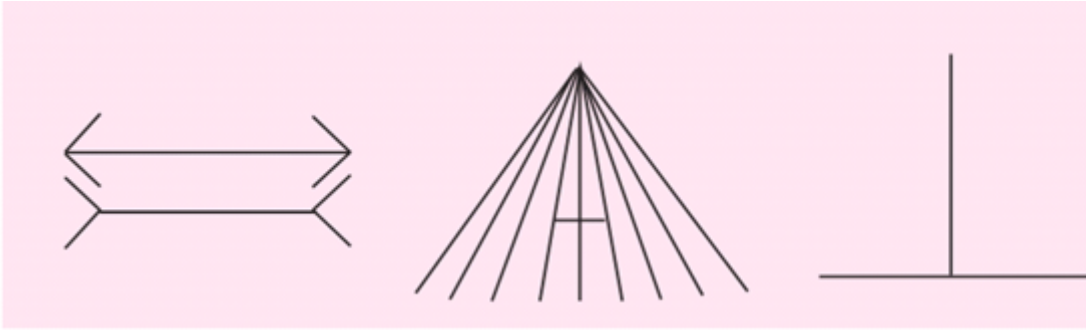


Figure 5.4: Three powerful illusions. In drawing A, known as the Muller-Lyer illusion, the bottom line looks longer than the top line. Actually, they are equal in length. In drawing B, the Ponzo illusion, the top horizontal line looks longer; again, both lines are equal. In drawing C, known as the horizontal-vertical illusion, the vertical line looks longer, check with a ruler to see that it's not!

Illusions are distorted perception of stimuli that exist, whereas hallucination is perception of objects or events that have no external reality. An example of illusion is the Muller-Lyer illusion in which horizontal line with outgoing arrow-heads appears shorter than the line with inward arrows.

INTEXT QUESTIONS 5.3

1. Describe various determinants of perceptual organization.

2. What do you understand by the term perceptual constancy?

3. Discuss depth perception with help of an example.

5.4 ATTENTION

Attention: Have you ever noticed that while doing both, driving and talking on your mobile you may not pay attention to driving. Or when you are studying and loud music is on in the neighbourhood you lose attention on your studies. ‘Divided attention’ can lead to accidents when people are driving and watching some other objects.

‘Selective attention’ is a process in which we give priority to a particular incoming sensory message. Attention has broadly three possible functions: (a) as a sensory filter (b) as a response selection and (c) as a gateway to consciousness.

Determinants of Attention

Physical factors: All other things being equal, physical factors like repetition, contrast, shape, size, brightness and contrast do affect our attention. A good packaging or bright light attracts us. That is why all big companies invest a lot on packaging their products in an attractive manner. Similarly an advertisement which is published on a regular basis in newspapers and electronic media catches the attention of customers more easily than non-advertised products.

Motives and attention: Motives also play a role in shaping our perception. For example, if you are hungry food-related words are more likely to gain your attention than non-food words. Advertisers and propagandists take advantage of two motives that are very common in our society: anxiety and sex. Many products such as toothpastes, food products even automobiles highlight the health concern to catch the attention of

customers. On the other hand items of fashionable products play on the desire to be attractive. The political parties cash on appeals to fear to make their voice heard by the public at large.

INTEXT QUESTIONS 5.4

1. Discuss the concept of Attention?

2. Identify the various determinants of attention.

5.5 EXTRA SENSORY PERCEPTION

Extra sensory perception is the ability to perceive objects or events in ways that cannot be explained by known sensory capacities.

Parapsychology is the study of extra sensory perception phenomena. Events that seem to lie outside the region of accepted scientific laws are called psi-phenomena. An example of extrasensory perception when a person intuitively 'feels' that his brother who is in another city is not well. Another example could be when you just 'know' that an earthquake is going to happen in your city.

Clairvoyance: It is the ability to perceive events or gain information in ways that appear unaffected by distance or normal barriers. Telepathy or perception of another person's thoughts or the ability to read someone else's mind.

Pre-cognition: The ability to predict or perceive accurately future events.

Psycho-kinesis: The ability to exert influence over inanimate objects by will-power (mind over matter). Some rare persons are able to move objects only through concentration but without touching them.

A few psychologists strongly believe in extrasensory perception but the majority does not subscribe to it.

5.6 APPLICATIONS OF PERCEPTION IN EVERYDAY LIFE

There are several ways in which knowledge of the process of sensation, perception and attention can be used in everyday life.

Eyewitness: You may be aware that eyewitness testimony is key to decisions in the judiciary. Not only do advocates and police officers lay strong emphasis on eye witness

testimony, but they have a strong belief that it is usually correct. But psychologists in large numbers are of the opinion that eyewitness errors are very common in perception. In fact impressions formed when a person is surprised, stressed or threatened are especially prone to distortion. Therefore it would be advisable for the investigative agencies and jurors to gather more evidence instead of solely relying on eyewitness while coming to a conclusion.

Perceptual awareness and positive psychology: Do some people perceive things more accurately than others? Humanistic psychologists believe that some people perceive themselves and others with unusual accuracy. Habituation is when we stop paying attention to familiar stimuli. When a stimulus is repeated without change our response to it habituates or decreases. It seems that creative people attend to stimuli, even those that are repeated.

The value of paying attention: We have this general tendency to generalize without paying attention to the diversity of possibilities. Perceptual clarity requires rigorous effort of paying more and more attention. Breaking perceptual habits and interrupting habituation can lead to good results. If you begin to question your own perceptions by bringing another interpretation to the same reality you can get marvelous outcomes in your activities.

WHAT YOU HAVE LEARNT

Sensation is the process by which neural impulses are created by stimulation or sensory neurons that results in awareness of conditions inside or outside our body.

There are five main senses: sight, hearing, touch, smell and taste. In addition we also have vestibular sense and kinesthetic sense.

Perception is the process through which an internal representation of an object is formed. It involves synthesis of simple sensations, and assigning meaning to the whole.

Principles of perceptual organization include nearness, similarity, continuity and closure and help identify the figure against the background.

Depth perception is the ability to judge distance and perceive three dimensional space. It involves monocular and binocular cues.

Perceptual constancy refers to the fact that we interpret a familiar object as having the same size, shape or color even if the sensations point otherwise.

Attention is the ability to focus our senses on a particular object. It involves selectivity and is influenced both by physical and psychological factors.

Extra-sensory perception is the ability to perceive events or objects in ways that cannot be explained by known sensory capacities.

Knowledge of perceptual psychology helps in everyday life and in learning how to improve our attention and perception.

TERMINAL QUESTIONS

1. Describe the importance of perception in our daily life.
2. Define perception and also discuss principles of perceptual organization.
3. Discuss the concept of depth perception and perceptual constancy.
4. What do you understand by Extra-sensory perception?
5. Describe the various applications of perception.

ANSWERS TO INTEXT QUESTIONS

5.1

1. Vision is extremely important. We experience vision with the help of our eyes which are like a camera.
2. The minimum amount of change between two stimuli that can be recognised is called “difference threshold”.

The minimum amount of physical energy needed to produce a sensory experience is called “absolute threshold”.

5.2

1. Perception is when an internal representation of an object that was earlier sensed is formed.
2. When we sense an object, we sensation is in the form of the organisation, depth and consistency of stimulus. Perceptual processes then analyse this into parts and mental process help us identify the stimulus.

5.3

1. Factors which determine perceptual organisation are figure ground organisation, perceptual consistency and depth perception.

2. Perception of an object's shape, size or brightness remains the same even though its image on the retina has changed. This is called perceptual consistency.
3. Depth perception is the ability to see three dimensional space and judge distances.

For example depth perception helps us drive a motorcycle or a car and helps us catch a ball.

5.4

1. Attention is the process by which we notice stimuli. Selective attention is a process in which we give priority to a particular sensory message.
2. The determinants of attention are physical factors and motives.

Hints for Terminal Questions

1. Refer to unit 5.1
2. Refer to unit 5.2
3. Refer to unit 5.3
4. Refer to unit 5.5
5. Refer to unit 5.6

6. MEMORY

If you are asked to tell the name of your favourite actor, actress or singer, it won't take more than a second to come out with the answer. In the same way we all remember our childhood friends, interesting incidents relating to them, our family members, relatives and so many other things. Have you ever wondered how we remember all these things, and do not forget them over long periods? All this is possible because of memory.

The human memory has immense potential. You must be knowing that before the Vedas were scripted, the oral tradition existed which means that the immense wealth of knowledge was passed on from one generation to another by the oral tradition. This was totally dependent on our memory. Memory is crucial in our life because it helps building bridges in the flow of experiences and allows us to have a sense of continuity. In this lesson we will study about this dynamic system which we call memory.

OBJECTIVES

After studying this lesson, you will be able to:

explain the nature of memory;

differentiate between the stages of memory;

describe forgetting; and

list some strategies for enhancing memory.

6.1 MEMORY AND FORGETTING

Psychologists consider memory and learning to be different processes, though both are closely related. While, learning refers to the acquisition of new behaviours through experience, memory refers to the process of storing the outcomes of learning in the form of information that can be retrieved when required. In this lesson you will learn

about memory and forgetting. You can very easily understand the significance of memory by visualizing a situation about a person who has lost his memory. He will lose his identity and cannot connect various experiences, events, and people. Such a person will lead a miserable life. This happens in brain damaged people or in old age when people develop some kind of memory loss, as in Alzheimer's disease.

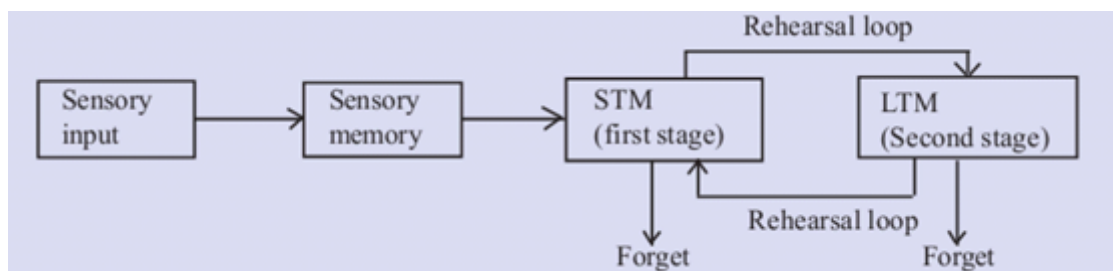
Memory refers to the set of processes involved in storing information. This specific process is termed as retention. Memory can be defined as a perceptually active mental system that receives, encodes, modifies, and retrieves information. We can not directly observe the process of memory. It can be studied indirectly by measuring retention. Three basic methods of measuring retention are : Recall, Recognition, and Relearning. Let us briefly discuss these methods.

Measuring Retention: Recognition, Recall and Relearning

When you are not able to remember someone's name but you know that you have seen his/her face before, this is the process of **recognition**. In this process you are checking the stimulus (face) with your memory content to find a match. In the same manner when one tries to remember a name without a face in front, it is called **recall**. Recall is a process of using a general stimulus and finding the information about it from memory. When you are given a question in an exam you use the process of recall. When you answer multiple choice items, you are asked to match the information with the provided pieces of information. You have to identify and match it with the information which is already stored in your memory. Recognition is easier than recall. **Relearning** is the method of learning the material another time. It usually takes less time than original learning.

6.2 STAGES OF MEMORY

Psychologists have discovered that memory is not a single or unitary system. It has more than one distinct system. In other words, there are more than one type of memory. According to the most acceptable model of memory, there are three major systems of memory : Sensory Memory; Short-Term Memory (STM), and Long-Term Memory (LTM). Information moves successively through these three systems if attention is given to the material. If attention (focused awareness) is not given, information does not move further into the system.



We get input from the sensory system, then this information goes into the Sensory memory and next into the short term memory. This is the first stage. Only when the information is rehearsed does this information go into the long term memory. Otherwise this input is FORGOTTEN Fig. 6.1: Stages of Memory explained

Sensory Memory : Hold any object about 12 inches in front of you. Look at it steadily for a while. Close your eyes and notice how long the clear image of that object lasts. A clear visual image of any object will last in sensory memory for about half a second after the stimulus is removed. Sensory memory holds representations of sensory input for very brief periods of time, depending upon the modality involved. There are different sensory registers for each of the senses.

Short-Term Memory (STM) : Look up for a moment and note what attracts your visual attention. Try to identify the sounds and sensations that you are experiencing now. What have you identified is the content of short-term memory or STM. It holds relatively small amounts of information for brief periods of time, usually 30 seconds or less. This is the memory system we use when we look up the phone number and dial it. If we are connected on the first instance the telephone number is forgotten. However, if we get the line engaged for some time, we keep on dialing the number and through repeated dialing rehearsal of the telephone number we push it to the long-term memory (LTM) storage.

However, it has been found that short-term storage is more than a passive “holding area” (e.g. holding a telephone number). On the contrary, it involves active processing of information. This finding has led psychologists to use the term working memory. It means that something active goes on during the short-term memory. Attempt Activity 1 to understand the working of STM.

Activity 1

Working of Short Term Memory (STM) Read out aloud the following sequence

Working of Short Term Memory (STM) Read out aloud the following sequence of numbers to a friend and ask him/her to immediately write in the order the

numbers that were spoken out aloud. After each set read the next set of numbers

and so on. Note that the series begins with three digits and increases by one digit each time.

4 2 8

1 4 6 3

5 7 4 9 2

6 0 8 1 3 5

1 8 4 7 9 2 6

3 5 0 1 7 4 2 9

2 7 3 9 6 8 1 4 5

How many digits can your friend recall?

Probably, your friend will recall up to $7 (\pm 2)$ digits.

STM has a limited capacity to hold information. In general it holds about 5 to 9 units (the “magic number” 7 ± 2) of information at one go. If the number of units goes higher, new information displaces or writes over the existing units. Thus in the example given earlier it is easy to remember a telephone number consisting of 7 to 9

digits. If more information is added we lose at least part of the phone number. However, this limitation in capacity can be effectively expanded by a process known as **chunking** i.e. several units of meaningful information are packed into one chunk. For example the number 194720021941 (12 digits) can easily be remembered if the 12 digits are chunked into three units 1947, 2002, 1941, all referring to calendar years.

Long – Term Memory (LTM) : It refers to the memory system for the retention of large amounts of information for long periods of time. It is the memory system that permits us to remember events that happened many years ago, yesterday, last year, and so on. It is the long-term memory that allows us to remember factual information making it possible for us to learn different subjects, appear for examinations and communicate with others. It brings continuity and meaning to our life.

When we pay attention to a piece of information and engage in active rehearsal the material is stored in the long-term memory (LTM). Information in the sensory memory enters short-term memory when it becomes the focus of our attention. If we do not pay attention to the incoming sensory information, the material fades and quickly disappears. We tend to pay attention to certain information and not to the other. Paying attention to certain aspects of our world is what we call. “selective attention”. The information from STM is often rehearsed by us. This rehearsal helps the transfer of that information from STM to LTM.

Activity 2: How Long Term Memory (LTM) works.

Ask the same friend, without looking at the written digits, to recall the four-digit number (1462). Even if you were to prompt your friend by saying that the number starts with ‘1’, it is highly doubtful that your friend will be able to recall even these four - digits.

Do you know as to why this happens? Because there was no rehearsal and therefore the four-digit number was not stored in the LTM.

Table 6.1: Nature of three systems of memory

Characteristics	Sensory Memory	Short-termMemory	Long-term Memory
Duration	Iconic about ½ sec	20 seconds	Life time

Echoic about 2 sec

Capacity	Large	7 ± 2	Unlimited	
Forgetting	Trace decay	Decay, interference and displacement	interference, Lack of Cues	Decay
Encoding	Sensory (auditory)	(visual) Phonological or visual-semantic	Semantic and sensory	
Retrieval	Parallel search	Serial exhaustive search	Parallel distributed search	

INTEXT QUESTIONS 6.1

1. What is recall?

2. List the different stages of memory.

6.3 TYPES OF MEMORY

In recent years psychologists have conceptualized memory into four types as given below:

Semantic :This deals with knowledge, meaning and generalized experiences. What ever we remember from books and information about world events and meanings of words are included in it.

Episodic :It refers to the experiences which are personal to an individual. You do so many things in a day. They are your unique experiences. Memory of such experiences is accessible by you only. They are part of your episodic memory.

Procedural :This deals with memory for actions or ways of doing certain things or performing certain activities.

Meta Memory :It is memory for your memory. We not only remember things but also remember that we can remember. People may be good or poor in understanding their own memories.

6.4 FORGETTING

It is a common experience that sometimes we are unable to remember information that we need at a particular time. We find that on some occasions this lets us down as we are unable to remember the information that we require desperately. During examinations we are unable to remember or retrieve what we had learned. This is what we call forgetting. Why does this occur? Many explanations have been offered to explain this kind of loss. Let us briefly consider some theories to explain the causes of forgetting.

(i) Decay :Some psychologists think that it is due to gradual fading of memory

traces or decay that happens with lapse of time. This does not explain why certain memories fade forever while others seem to be well preserved. Passing of time cannot, therefore, be considered as the main cause of forgetting. You can see that in Table 6.1

(ii) Interference : A more useful explanation is found in the view called **interference theory**. It argues that memories are lost because of what happens to them during the period when we retain them. Thus inhibition or interference by subsequently learned material becomes the major cause of forgetting. There are two types of such interference – **Retroactive interference** and **Proactive interference**. Retroactive interference (acting back wards) occurs, when the current information interferes with what has been retained earlier. For example, you learn Punjabi language on Monday and Bengali on Tuesday. During a Punjabi test taken on Wednesday interference is expected from the learning of Bengali words. Forgetting may also occur due to (acting forwards) proactive interference. This is a situation in which earlier information in the memory interferes with the current information. Suppose you have been driving a scooter with gears. Now, you have a new scooter which has no gears (automatic drive). You will tend to change the gears in the new scooter and this habit can persist for long. Proactive interference thus explains one's inability to acquire new knowledge or habit because of the powerful interference from old habits.

(iii) Level of Processing : While learning some material we may attend to it in detail and process at a deeper level or neglect it and attend at a surface level. This may be a cause of forgetting.

Activity 3: Effect of Delay and Interference on STM.

Ask a friend to remember a sequence of three consonant letters, such as, FJQ, while counting backwards from a given number, such as 462. After the three consonant letters have been said and your friend has counted backwards for about 15-20 seconds, ask him/her to recall the three consonant letters, that is, FJQ. Does your friend will find it difficult because the counting task prevents him/her from rehearsing the letters and allows the memory trace to decay.

Now give more three consonant letters (WCT, QRB) and ask the friend to count backwards (743, 947). You are likely to note that now your friend has greater difficulty in remembering the letters! Yes, this is due to proactive inhibition.

6.5 STRATEGIES FOR ENHANCING MEMORY

How good is your memory? We all would like to improve our memories so that we can retain more facts and information. This is possible with little effort and almost anyone can improve his or her memory. Here are some tips to enhance one's memory.

1. **Deep Processing** :If you want to learn something and wish to enter information into long-term memory, you have to think about it. You need to consider its meaning and examine its relationship to information you already have. Careful planning, considering the meaning of information, and relating it to existing knowledge is helpful in learning it and remembering it later. The retention of material is dependent directly upon the depth at which it has been processed. Deeper level in terms of meaning is very important. Second, the depth of

processing also refers to the rehearsal of material to be remembered. Greater the rehearsal, more the chances of recalling the material later.

2. Attending carefully :If you want to learn something well and ensure its retention in long-term memory you have to put in conscious effort to attend to it carefully.

3. Minimize interference :You have learned that interference is a major cause of forgetting and you should try to reduce it as much as possible. In general, the more similar the materials to be learned, the more likely they will produce interference. Thus, you should arrange your studies so that you don't study similar subjects one right after the other. For example, if you have to study two languages, study them on different days.

4. Distributed practice :While learning some material it is beneficial to learn it using distributed practice in which there is a gap between trials. Take rest after a period of study. If the chapter is lengthy, divide it into two or three parts and learn a part in one go. After mastering one part then go to the second and so on. Do not try to cram all the information you want to memorize at once.

5. Using memory aids :People use various cues, indicators and signs to connect events. This often facilitates remembering. You can use visual imagery to remember objects and places. For example, if you visit a new place, you can remember the location by remembering several associated objects and places. You may remember the place to be close to a cinema hall, in front of a signal post and so on.

6. Shorthand codes :You can develop your own shorthand codes to memorize long list of items. You can use the first letter of each word or item and construct a unique "word". For example, to remember the seven constituents of light we use VIBGYOR; where V stands for Violet, I for Indigo, B for Blue, G for Green, Y for Yellow, O for Orange and R for Red. During the school days. We used to forget the

order of Mughal emperors in history subject. We developed a shorthand code : BHAJSA” where B stands for Babar, H for Humayun, A for Akbar, J for Jahangir, S for Shahjahan and A for Aurangzeb. It helped remembering the names in order.

You can use the tips in Figure 6.2 to help you memorize your course material

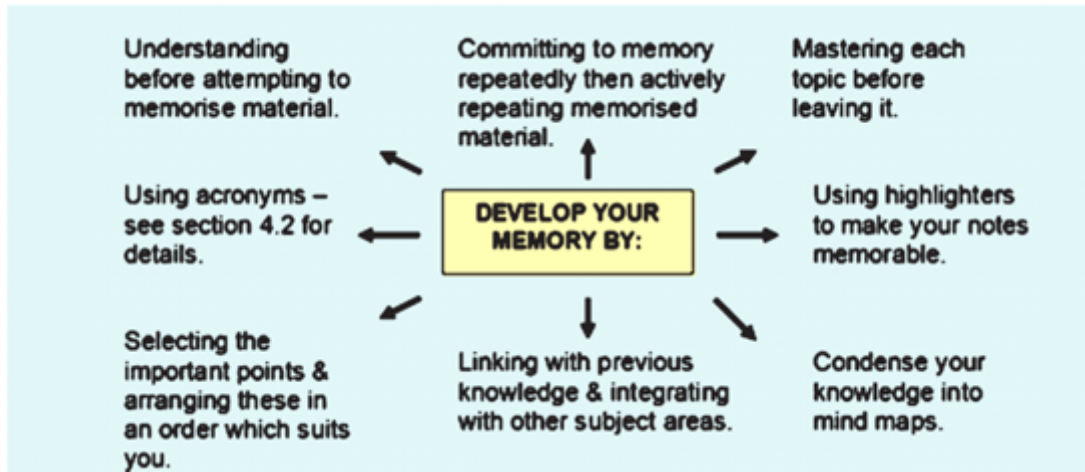


Fig. 6.2: Tips to memorize course material

6.6 REPRESSION: FORGETTING PAINFUL EVENTS

The events and experiences that are threatening or painful are eliminated from our consciousness. This is called repression. Freud believed that the repressed memories are pushed into hidden recesses of the unconscious mind. Such repressed memories may cause many psychological problems for the individual. Amnesia refers to a loss of memory due to illness, injury, drug abuse or other causes.

INTEXT QUESTIONS 6.2

1. Make a list of the major types of memory.

2. State any two causes of forgetting.

WHAT YOU HAVE LEARNT

Memory refers to the set of processes involved in storing information and the specific process is termed as retention.

Three basic methods of measuring retention are : Recall, Recognition, and Relearning.

Sensory memory holds representations of sensory input for brief periods of time, depending upon the modality involved.

Short – term memory or STM holds relatively small amounts of information for brief periods of time, usually 30 seconds or less.

Long – Term Memory is a memory system for the retention of large amount of information for long periods of time.

Forgetting is loss of memory or failure to recall. It is caused by several factors including decay, level of processing and depression.

Memory can be enhanced with the help of mnemonic strategies

TERMINAL QUESTIONS

1. Describe briefly the three stage model of memory

2. Describe the properties of short term memory.
3. List the main causes of forgetting.

ANSWERS TO INTEXT QUESTIONS

6.1

1. Recall is a process of using a general stimulus and finding the information that is in the memory.
2. Sensory memory, short term memory, long term memory

6.2

1. Semantic, Episodic, Procedural and Meta memory.
2. Decay, Interference, Level of processing

Hints for Terminal Questions

1. Refer to section 6.2
2. Refer to section 6.2
3. Refer to section 6.4

7.THINKING AND PROBLEM SOLVING

Have you ever thought about how you have acquired knowledge about all that you know today? It may instantly come to your mind that your teachers have taught you about this, your parents made you learn all these things, or you read about all these in books. In all these situations you have learnt something to build your knowledge base. Thus the process of knowing or of acquiring knowledge is called cognition. The process of acquiring knowledge is facilitated by cognitive processes such as attention, thinking, remembering, and reasoning. These cognitive processes are controlled and regulated by higher center of the brain, the cerebral cortex. The cognitive processes are very much specific to human beings and are guided by concepts, facts, propositions, rules, and memories. All these cognitive activities are governed by thinking. Hence, in this chapter you will learn about what is thinking, stages of problem solving, stages of creative thinking, and processes involved in making decisions.

OBJECTIVES

After studying this lesson, you will be able to:

understand the nature of thinking;

describe various stages of problem solving;

explain creativity and its role in life; and

explain the process of decision making.

7.1 THE NATURE OF THINKING

Thinking is perhaps one aspect of our mental activity which continues even when we are asleep. The difference between what is thinking and what is not thinking is just our awareness about the particular thinking process. Hence thinking is a complex mental process which involves manipulation of information. Such information is

collected through our senses (such as vision, hearing, smelling etc) from the environment, as well as the information which is stored in our memory because of our encounter with many events and situations in the past. Thinking is a **constructive process** in the sense that it helps us to form a new representation of any object or event by transforming available information. It involves a number of mental activities, such as inferring, abstracting, reasoning, imagining, judging, problem solving, and creative thinking. Such activities take place in our mind and can be inferred from our behaviours. Thinking is usually initiated by a problem and goes through a sequence of steps such as judging, abstracting, inferring, reasoning, imagining, and remembering. These steps are often directed towards solution of the problem. The example given below will help you to understand this in a better way.

In order to reach your new school on time suppose you are trying to find out the shortest route from your home to your new school. Your choice will be guided by many factors such as condition of the road, the density of traffic during your school time, safety while walking on the road etc. Finally you take a decision about the best possible shortest route after considering all these factors. Thus, a simple problem like this also requires thinking. The solution to this problem emerges after processing information that is available to us from the environment and our past experience. Thinking relies on a variety of mental structures such as **concepts and reasoning**. We will briefly learn about these mental structures.

Concepts: Concepts are one of the key elements of thinking. Concepts represent objects, activities, ideas, or living organisms. They also represent **properties** (such as “sour” or “brave”), **abstractions** (such as “anger” or “fear”), and **relations** (such as “smaller than” or “more intelligent than”). Concepts are mental structures which allow us to **organize knowledge** in systematic ways. We cannot observe them directly, but we can infer them from behaviour.

We as human beings have the capacity to abstract the essential characteristics of objects, events or whatever we perceive. For example, when we see a Potato we categorize it as ‘vegetable’, and when we see a towel we categorize it as ‘cloth’.

Whenever we encounter a new stimulus we tend to treat it as a member of a familiar or remembered category and take the same action toward it and give it the same label.

Reasoning: Reasoning is also one of the key aspects of thinking. It is a process that involves inference. Reasoning is used in logical thinking and problem solving. It is goal directed, and the conclusions or judgments are drawn from a set of facts. In reasoning,

information from the environment and the stored information in the mind are used following certain rules. There are two types of reasoning: deductive and inductive. In **deductive reasoning** we try to deduce or draw conclusion from a set of initial assertions or premises; where as in **inductive reasoning** we start from available evidence to generate a conclusion about the likelihood of something. Most cases of scientific reasoning are inductive in nature. Scientists and even lay people consider a number of instances and try to determine what general rule covers them all. For

example, the person is a priest, because he is wearing plain cloth, prays and eats simple food.

INTEXT QUESTIONS 7.1

1. What is thinking?

2. What are the different mental components of thinking?

7.2 PROBLEM SOLVING

Problem solving is part and parcel of our daily life. Every day we solve a number of problems ranging from simple to complex. Some problems take little time where as some take much time to solve. We look for alternative solutions if do not get the right kind of resources to solve the problem in hand. In the case of solving any type of problem our thinking becomes directed and focused and we try to use all the resources, both internal (mind) and external (support and help of others) to arrive at the right and appropriate decision. For example if you want to score good marks in an exam, you study hard, take the help of teachers, friends, and parents and finally you score good marks. Thus problem solving is directed thinking focused towards dealing with a specific problem. This thinking has three elements: **the problem, the goal, and the steps to reach the goal**. There are two methods which are used prominently in problem solving. These are- “**Means-end-analysis**” and “**Algorithms**”. In the case of Means-end-analysis a specific step-by-step procedure is followed for solving certain types of problems. In the case of ‘**heuristics**’ the individual is free to go for any kind of possible rules or ideas to reach the solution. It is also called rule of thumb.

Problem Solving and Mental Set: Sometimes we use a particular strategy/technique to solve a problem but we may or may not succeed in our effort to solve the problem. This creates a set to approach future problems that are encountered by a person. The set continues even if the problem is different. Despite this, we use the same strategy/technique whenever we come across the same problem and again fail to reach the solution. Such phenomenon in problem solving is called mental set. A mental set is a tendency on the part of an individual to respond to a new problem in the same manner that he or she has used earlier to solve a problem. Previous success with a particular rule produces a kind of mental rigidity/fixedness/set, which hinders the process of generating new ideas to solve a new problem. A mental set inhibits or affects the quality of our mental activities. However, in solving our real life problems we often rely on past learning and experience with similar or related problems.

In Activity 1 You may not be able to solve the problem, because of the mental set that one has to keep the lines within the grid of nine dots. By going outside the boundaries, you will succeed.

INTEXT QUESTIONS 7.2

1. Define problem solving. Discuss the two types of problem solving.

2. Discuss the role of mental set in problem solving.

7.3 CREATIVITY AND ITS ROLE IN LIFE

Do you know whatever you see around you, the things which you use for work in everyday life, the transport you use for commuting from one place to another etc are all the products of human thinking? Creativity is a particular kind of thinking which involves reaching out to the solution of a problem in a unique and novel way which was nonexistent earlier. Creativity is the mother of all inventions and discoveries in the world. Unlike routine solutions to the problems, creative solutions are novel, original, and unique, that others have not thought of before. The creative solutions or productions are sudden or spontaneous and are the outcome of a lot of work and preparation already done consciously and unconsciously. The sudden appearance of new ideas is called insight. The creative thinker can be any more such as an artist, musician, writer, scientist or sports person.

Stages of Creative Thinking: Graham Wallas , one of the leading psychologists of early twentieth century stated that there are five stages of creative thinking. These are Preparation, incubation, illumination, evaluation, and revision. We briefly discuss these five steps of creative thinking in the subsequent section.

1. Preparation: This is the first stage in which the thinker formulates the problem and collects facts and materials necessary for the solution. He/she finds that the problem cannot be solved after days, weeks, or months of concentrated effort. Unable to solve the problem the thinker deliberately or involuntarily turns away from the problem, initiating stage two i.e. incubation. At this stage of problem solving, it is important to overcome negative consequences of mental set and any kind of mental set or bias.

2. Incubation: This is a stage of no solution and involves a number of emotional and cognitive complexities. However, the negative effects of mental set, functional fixedness, and other ideas that interfere with the solution tend to fade. Perhaps, fatigue and too much of concern with the problem also mount up during this

period. Further, the unconscious thought processes involved in creative thinking are at work during this stage.

3. Illumination: In this stage a potential solution to the problem seems to be realized as if from nowhere. It is about having the insight about the possible solution. Illumination occurs with its “**aha**” experience when a sudden idea or solution appears into consciousness.

4. Evaluation: In this stage the obtained solution is verified or tested to see if it works. Frequently, the insight may turn out to be unsatisfactory, and may need some modification in the strategy of approaching the problem.

5. Revision: Revision is required in the case a solution which is not satisfactory.

It has been found that creative people are generally talented (e.g. artists, musicians, mathematicians etc.), and have specific abilities. Creative people have been found to have some specific personality characteristics such as they are independent in their judgments, self-assertive, dominant, impulsive, prefer complexity, etc.

INTEXT QUESTIONS 7.3

1. What is creativity? What are the possible characteristics of a creative person?

2. Discuss in brief the stages of creative thinking.

7.4 DECISION MAKING

We make several decisions in our day-to-day life, such as decisions pertaining to our personal life, social life, education, career etc. When we take a decision which gives us success where as our faulty decisions do not yield the desired result. Decision making is also related with another term ‘judgment’. Let us discuss these two aspects of thinking separately.

Decision Making: Decision- making is a kind of problem solving in which we select an appropriate alternative out of a number of alternatives available to us. For example, you have the option to choose between History and Psychology courses in your eleventh grade. You attend classes in both the subjects to decide upon the course to choose. Suppose you find that the contents of psychology are relevant, interesting and new and the teacher is intelligent, friendly, knowledgeable, and having good verbal ability; all qualities that you value in a teacher. So, on the basis of judgment about the subject and qualities of the teacher you decide to choose the psychology course.

Judgment: Judgment is a process of forming opinions, arriving at conclusions, and making critical evaluations about objects, events and people on the basis of available information. The process of judgment is often automatic and spontaneous. It does not require any prompting. Some judgmental choices are habitual like need for going for a morning walk before getting ready. Judgments involve evaluating information about the world (objects, events, persons, etc.), while decisions require making choices.

INTEXT QUESTIONS 7.4

1. What is the difference between decision making and judgment? Discuss in brief.

WHAT YOU HAVE LEARNT

The process of knowing or acquiring knowledge is called cognition. The process of acquiring knowledge is facilitated by processes such as attention, thinking, remembering, and reasoning. These cognitive processes are controlled and regulated by higher center of the brain, the cerebral cortex.

Thinking is a complex mental process which involves manipulation of information. Such information is collect through our senses (such as vision,

hearing, smelling etc) from the environment, or is based on information which is stored in our memory because of our encounter with many events and situations in the past.

Thinking is a constructive process in the sense that it helps us to form a new representation of any object or event by transforming available information. It involves a number of mental activities, such as inferring, abstracting, reasoning, imagining, judging, problem solving, and creative thinking.

Concepts are one of the key elements of thinking. Concepts are mental structures which allow us to organize knowledge in systematic ways. We cannot observe them directly, but we can infer them from behaviour.

Reasoning is another key aspects of thinking. It is a process that involves inference. Reasoning is used in logical thinking and problem solving. It is goal directed, and the conclusions or judgments are drawn from a set of facts.

There are two types of reasoning: deductive and inductive. In deductive reasoning we try to deduce or draw conclusion from a set of initial assertions or premises; where as in inductive reasoning we start from available evidence to generate a conclusion about the likelihood of something.

Problem solving is directed thinking focused towards dealing with a specific problem. It has three elements: the problem, the goal, and the steps to reach the goal.

A mental set is a tendency on the part of an individual to respond to a new problem in the same manner that he or she has used earlier to solve a problem.

Previous success with a particular rule produces a kind of mental rigidity/fixedness/set, which hinders the process of generating new ideas to solve a new problem.

Creativity is a particular kind of thinking which involves reaching out to the solution of a problem in a unique and novel way which was nonexistent earlier.

Creativity is the mother of all inventions and discoveries in the world. Creative solutions are novel, original, and unique that others have not thought of before.

There are five stages of creative thinking. These are *Preparation, incubation, illumination, evaluation, and revision*.

Decision-making is a kind of problem solving in which we select an appropriate alternative out of a number of alternatives available to us. Judgment is a process of forming opinions, arriving at conclusion, and making critical evaluations about objects, events and people on the basis of available information.

TERMINAL QUESTIONS

1. Give any 2 examples each for (a) Concepts (b) Reasoning (c) Problem solving.
2. Identify any 5 creative persons who are famous in India.
3. Explain the concept of decision-making and its importance in daily life.

ANSWERS TO INTEXT QUESTIONS

7.1

1. Complex mental process involving manipulation of information.
2. Concepts and reasoning

7.2

1. Directed thinking focused towards dealing with a specific problem Means-and - analysis and Algorithms
2. Mental set inhibits the quality of mental activities.

7.3

1. Thinking which involves reaching out to solutions in a unique and novel way

which was nonexistent earlier. Creative persons can be self-assertive, dominant, impulsive, may prefer complexity etc.

2. Preparation, incubation, illumination, evaluation revision.

7.4

1. Decision making is a kind of problem solving. Judgement is a process of forming opinions, arriving at conclusions and making critical evaluations.

Hints for Terminal Questions

1. Refer to section 7.1 & 7.2

2. Refer to section 7.3

3. Refer to section 7.4

8. MOTIVATION AND EMOTION

Have you noticed how each one of us is busy in one or the other activities? We try our best to achieve certain goals; and, if some difficulty comes in the way we feel frustrated. These statements show that our daily life is influenced by our needs such as hunger, thirst, achievement, affiliation etc. Not only that, we also feel happy, sad, angry or violent. The former aspects refer to motivation and the latter refer to different types of emotions. This lesson explains the relevance of motivation and emotions in human life. Motivation helps in answering the question: why do people seek to do certain things?. This helps us understand why people differ in terms of their behaviour. Emotions play an equally important role in everything we do, affecting our relationships with others and health. Both concepts are important because they help us understand the reasons for the various activities we indulge in and help determine the unique personality of each person.

OBJECTIVES

After studying this lesson, you will be able to:

- define the concepts of motives and needs;
- understand goal directed behaviour;
- describe the different aspects of emotions; and
- explain the role of emotions in organizing behaviour.

8.1 NATURE OF MOTIVATION

Have you wondered as to – ‘Why does your father/mother go to work everyday ?’ ‘Why do you study for long hours?’ ‘Why do you like to play with your friends?’ or ‘Why does your mother stay up the whole night and care for you when you are feeling unwell?’

Each of these questions has an answer as there is a reason or motive behind

each type of activity mentioned. Working, studying, playing and caring are some important activities which are considered purposeful. Motives, wants, needs, interests and desires help explain our movement towards the chosen goals. A need or motive directs us to act or behave in a particular way towards a certain goal. Thus you study for long hours to achieve good results and your mother looks after you because of the affection and concern she has for you as her child.

A. The Concept of Motivation

Motivation plays an important role in directing human behaviour. There are several characteristics of motivation. First, it pushes us to reach the goals we have set for ourselves. In order to reach our goals we need to be **aroused or feel active**. Too little arousal will make us feel dull and relaxed, whereas, too much arousal may make us withdraw from our goal. Hence we must be motivated to maintain a balanced or **optimum level of arousal**. Second, people are motivated to behave in certain ways because of the pleasing and pleasant properties of external stimuli, such as, money, good grades, food, which motivate the person to act in a purposeful/focused manner.

Third, motivation is the result of the person's thoughts and expectations. It can be of two types: **intrinsic motivation** that comes from within the person, based on personal enjoyment of any task, and **extrinsic motivation** which is based on external rewards such as money, pay and grades. You will find that sometimes we work hard and produce high quality work when motivation is intrinsic rather than extrinsic. On the other hand, we are also influenced by the external rewards. In life, both types of motivation are important. Finally, motivational needs are ordered such that basic needs such as hunger and thirst are satisfied first and then other higher-needs, such as need for achievement and power are fulfilled. Let us now learn about the hierarchy of needs.

B. The Hierarchy of Needs

According to psychologist **Abraham Maslow**, there are several levels of needs that each person must strive to meet before he or she can achieve the highest level of personal fulfillment. Thus, a person must be able to fulfil the **primary (biological) needs** at the lowest level, as indicated in Figure 1. Once these needs are met then **safety** becomes important, followed by **need to belong** and **to love and be loved**. The desire to belong to groups such as family, friends, and religious groups makes us feel loved and accepted by others. If we succeed in satisfying the mentioned needs then we feel the need to have **self esteem, confidence, and feeling of self-worth**. Next come **cognitive needs** which include need for **knowledge and understanding**, followed by

need for **order** and **beauty**. Finally a person moves towards reaching the full potential called **self-actualization**. Such a person has the qualities of **self awareness**, is **socially responsive**, and is open to the **challenges of life**. This type of organization of needs is called a **hierarchy**, or ranking in a series.



Figure 8.1: Maslow's hierarchy of needs

As life goes on people gain wisdom and knowledge of how to handle situations, thus they move up the ladder or hierarchy. A person may move up and down the

hierarchy depending upon the situation he or she finds himself in. This hierarchy may not hold true for different cultures. It has been found that in countries like Sweden and Norway quality of life is very important and social needs are more sought after than self-actualization. In some cultures security needs are more dominant than self-actualization making job security more important than job satisfaction.

C. Primary Needs

‘Need’ refers to a lack of something that is necessary for us. We feel hungry when our blood sugar level is low, thus we need to eat food to satisfy our hunger. Motives or

needs are characterized in terms of primary and secondary needs or motives.

Important primary needs are biological needs such as hunger, thirst, oxygen, rest, sleep, temperature regulation, warmth, shelter, excretion, and sex. These needs have to be satisfied for us to stay alive. Needs such as hunger and sleep are cyclical in nature or are repetitive. Thus we need to eat and sleep regularly. Attempt Activity 1 to understand the effects of not eating or sleeping for a long time.

Activity 1: Primary Needs

- a) Effect of lack of sleep on performance. Try to keep awake for a long period of time, for example for about 16 hours. Now list your experience on two aspects: i) how long you take to learn a lesson and ii) your feelings at this point of time.
- b) Effect of not eating for a long period of time. Try not to eat for a long period, for example, 8 hours. List your experience in terms of how you feel. Rate your feelings on a five point scale in terms of: i) extent of low feeling, ii) level of irritation and annoyance. Also try to concentrate on reading a new lesson. Are you able to read the lesson? If not, why is this so?

The starting point of many biological needs is an imbalance in the physiological conditions of our body. Our body tries to maintain a state of balance called

homeostasis. This balance is crucial. There must be enough water content in our body, body temperature must not be high or too low, glucose levels must be adequate and so on. Without maintaining this balance we would not be able to function properly in our daily life activities. Can you recall when you or a family member suffered from high fever and had to rest in bed for a few days? A state of balance must be maintained within our body to carry out our daily activities.

Why do we eat or feel hungry? There are several factors involved in the hunger need. Stomach contractions or ‘hunger pangs’ indicate that the stomach is empty and low levels of glucose and protein in the blood make us hungry. In addition, the smell, taste or appearance of food may lead to a desire to eat. When you reach home the smell of your favourite dish makes you feel hungry! Many factors work together to make us feel hungry. It is interesting that you can go hungry for long periods but not without water. We feel thirsty because we lose water content due to perspiration, breathing and urination.

Influence of Culture and Environment on Primary Needs

Do you know that our biological needs, such as hunger, are influenced by environmental factors? Often we are likely to feel hungry because of a number of reasons rather than due to low glucose levels in the blood. Environmental factors that have an effect on hunger and eating are:

Learned liking and habits, for example, a person who enjoys having pizza will want to eat it even if not hungry.

Stress can lead to eating, for example, examination stress makes some students eat more than is required.

External cues, such as when it is dinner time or your mother calls out for dinner you are likely to feel hungry. The smell and attractive appearance of food may also make you feel hungry.

Dietary restraints, if the doctor has advised a patient to avoid sweet foods there may be an increased desire to eat sweets.

D. Secondary Needs

Secondary needs lead to psychological or social motives. They are called social motives because they are learned in social groups, particularly the family. As children grow up and interact with others they acquire certain needs which can be fulfilled in a group setting. For example, achievement motivation is acquired by children through their parents, role models and socio-cultural influences. Some common social motives are:

Achievement: To strive for success and excellence, carry out difficult tasks.

Affiliation: To seek and enjoy the company of others, to make friends.

Power: To strive for impact on others, to influence, control and persuade others.

Nurturance: To help and take care of others such as children, elderly or sick persons.

Aggression: To fight, forcefully overcome opposition, to hurt and ridicule.

Exploration: To seek a variety of stimulation, to explore novel things.

The type and strength of social motives differ from person to person. For example, you may have high needs for achievement and affiliation, whereas your friend could have a moderate need for achievement and a low need for affiliation. Or, you could have a high need for affiliation and your friend a high need for exploration.

Mother Teresa's concern for the poor, sick and helpless made her open the Missionaries of Charity in



1950. It provides food for the needy and runs schools, hospitals, orphanages, youth counselling centres, and, homes for lepers and the dying. Mother Teresa's work has touched the lives of thousands of people. Her concern for others makes her high on need for nurturance. Do read the poem written by Mother Teresa as shown on www.youtube.com.



Nelson Mandela dedicated himself to the struggle of African people and was imprisoned for 27 years before he achieved freedom for his people. In fact he did not want personal freedom but fought for the setting free of common people. He was influenced by Mahatma Gandhi's way of thinking. For him it is important to live in peace and harmony and have equal opportunities for all people. Nelson Mandela was able to achieve difficult goals and his commitment helped him accomplish his purpose in life. Watch the video on www.youtube.com 'Making Poverty History – Nelson Mandela' and 'First Nelson Mandela Interview'

We can also classify needs into three basic types: **existence needs** that include all primary needs that are essential for survival; **relatedness needs** include safety, belongingness, esteem and other social relationships needs; and, **growth needs** include those aspects that help people to develop their full potential. It is important to understand that biological and psychological needs do not work in isolation. They influence each other and determine human behaviour.

INTEXT QUESTIONS 8.1

1. Explain the concept of motivation.

2. Differentiate between primary and secondary motives.

3. Describe Maslow's hierarchy of needs.

4. Discuss the role of culture and environment in our needs and motives.

Quiz 1

1. The process by which a person moves towards fulfilment of wants is called:

a. need b. incentive c. motivation d. goal

2. Secondary needs are acquired through

a. biological needs b. social interaction c. intrinsic procession d. innate needs

3. According to Maslow the most basic needs are:

a. safety b. cognitive c. love d. biological

4. Reema gets Rs. 20 for every good grade she obtains in school. Reema is receiving:

a. intrinsic motivation b. extrinsic motivation c. primary need d. growth need

5. Ashok has a high need for because he likes to study and investigate new things.

love b. nurturance c. exploration d. achievement

8.2 NATURE OF EMOTIONS

Each one of us has experienced strong feelings that are associated with pleasant or negative experiences. Examples of emotion are feeling happy when enjoying a movie, or getting embarrassed when we break a neighbour's beautiful vase, or sadness after an argument with our best friend. Emotions influence our everyday behaviour.

A. Functions of Emotions

Emotions play several important functions in our lives.

Emotions prepare us for action: When you see a person with a gun coming towards you, the emotion of fear and the accompanying bodily changes will help you run away from the situation.

Emotions help arrange our activities and future behaviour in some order:

Emotions help us to gather information and increase our chances of making the right response in the future. When a person threatens us we will try to avoid meeting that person out of fear. Positive emotions can act as reinforcement such as, wanting to repeat an action which brought us praise and joy.

Emotions help us interact effectively with others: When we show our emotions other people watching them may come to be familiar with our way of behaving and know how we are likely to act again in the future. Similarly, others' emotional expressions help us understand how to behave with them.

Activity 2

In column A are some situations which we all go through. In column B are emotions we might experience in these situations. Try and match column A to column B.

A B

- (i) You had fight with your best friend (i) Surprise, excitement
- (ii) You are about to go for an interview / exam (ii) Fear, nervousness
- (iii) You and your family are going for a picnic (iii) Sorrow, sadness
- (iv) You are about to receive the results of (iv) Disgust

an exam that you do not think you did well on.

(v) You have received an unexpected gift (v) Scared, nervousness

(vi) You see a man chewing pan/tobacco and (vi) Excitement, happiness
spitting on the road



Fig. 8.2: Emotional Expressions

Figure 8.2 showing different facial expressions

Ease Happy or amused Laughter

Anger Hate Aggressive

Sad Bored Mild

B. Components of Emotions

Emotion has three basic components or parts:

1. Physiological changes: when a person feels an emotion certain changes take place in our body such as increase in heart rate, blood pressure, pupil size is

increased, perspiration takes place, breathing is faster and the mouth may become dry. Think about the last time when you went to the examination centre to write your exam or when you were angry at your younger brother. You may have experienced these bodily changes.

2. Behavioural changes and emotional expression refer to the outward and noticeable signs of what a person is feeling. These include facial expressions, bodily postures, and gestures with the hand, running away, smiling, frowning, and slumping in a chair. The six primary emotions in facial expressions are: fear, anger, sadness, surprise, disgust and happiness. This means that these emotions are easily recognized in people from all over the world.

3. Emotional Feelings: Emotion also includes the personal/individual feeling. We can label the emotion as: happy, sad, angry, disgusted and so on. Labelling

of a feeling is shaped by past experiences and the culture one belongs to. On seeing a person with a stick in his/her hand we may run away or prepare ourselves to fight, whereas, if a well known singer is in your neighbourhood you will go to the place to hear him sing your favourite songs.

Cognition or understanding of the situation is the most important part of emotional experiences. Some psychologists feel that we think and then feel, while others believe that our emotional reactions are so fast that thinking about the situation does not arise. Attempt Activity 3 to check which explanation is correct according to you? Human emotions involve so many things, past experience, culture, individual differences therefore at times we may first act and then think about it, whereas in other situations we can first think before reacting.

Activity 3 Emotion and Thinking

Think and answer: when you suddenly see a dog snarling near by do you first run away or do you first think ‘What should I do?’ and then decide to run?

Make a list of other situations where you are likely to first react and other situations where you are likely to first think and assess the situation and then you will respond.

Most of the times we feel a mixture of emotions. For example, when a person wins a lottery she may show the emotions of shock, disbelief, surprise and joy. We are also familiar with each other’s feelings and know what it means to him or her. Hence you know when your friend is happy to receive a gift and sad when you have had an argument with her. However your experience of emotions of joy and fear is different from your friend’s. Going for a walk may provide you with pleasure whereas for your friend a hug from her mother gives the feeling of pleasure. The meaning a feeling has for you or your friend is influenced by past experiences and memories.

Activity 4

There are many ways by which we describe our feelings. Prepare a list of the various emotions people feel. Make your own list of emotions. Also ask your family members and friends. You will realize that the list is very long for human emotions have a wide range.

C. Emotions and Culture

Do people in all cultures express their emotions similarly? Yes, research indicates that the six basic emotions are expressed universally. Therefore, Indians, Americans, Germans or Japanese express joy and fear in the same basic way. Each primary emotion is associated with a unique set of facial and muscular movements.

Yet you must understand that the behaviour of people from various countries is different from your behaviour. Japanese tend to show very little emotional display as compared to Italians. Why does this difference in display exist if there are basic universal emotions? The answer is that as we grow up we come into contact with family members and others. The social experiences lead to the development of **display rules** which are approved way of showing of emotions. These rules are learnt during childhood and this learning becomes noticeable in the expression of emotions. Can you recall any instance when you have received a gift that you do not like? Have you not thanked profusely with a smile 'pasted on your face'? This is known as 'making of emotion'. Interestingly there may have been instances when you have been able to note the difference between a made up smile and a true smile.

INTEXT QUESTIONS 8.2

1. State the functions of emotions in our lives.

2. With the help of examples identify the components of emotions.

3. State how display rules are influenced by culture.

Quiz 2

State whether the following statements are true or false:

1. Emotions prepare us for action. T/F
2. When a person feels an emotion no bodily changes take place. T/F
3. Emotional expression is only shown through facial expressions. T/F
4. Labelling of emotions is influenced by past experience and the culture one belongs to. T/F
5. Display rules are approved ways that are noticeable in the expression of emotion. T/F

8.3 ROLE OF EMOTION AND MOTIVATION IN HUMANBEHAVIOUR

Can you imagine a situation when you are about to appear for an important examination or are going to receive a prize for securing the first position in class and you do not feel anything? Or, you are walking alone down a dark street at midnight and someone is following you and again you are not feeling anything? Absence of not being motivated to do something and absence of feelings is likely to make us act in a robotic fashion.

We have tried to understand motivation and emotions as two distinct concepts. In real life both are closely related and connected and separating the two concepts is a difficult task. In this unit you have learned that along with motivation we also feel our experiences. In the examples provided above the act of studying long hours and the experience of success provides you with a sense of achievement and happiness. The emotional feelings of love and affection your parents have towards you also motivate them to care for you. The death of a loved one brings on the emotion of sadness whereas we experience anger at being treated unfairly. Emotions not only motivate us, they also indicate the reason for motivation. They play an important role in our lives.

Hence your motivation is also accompanied by emotions. Any situation that brings about a strong emotion is likely to motivate you to either repeat it or to avoid it. Thus both motivation and emotion are forces that push us towards action. Primary needs like hunger, thirst and sleep and secondary needs like achievement, power or approval are accompanied by emotional reactions of satisfaction, joy and pride. It can be said that:

Both motivation and emotion move, activate and help control and plan human behaviour.

The feeling part of emotion has a motivational component. It helps us to get what we want and to avoid what we do not want in life.

Emotions can serve as motives. For example, a frightened child may cry out for help and seek comfort and security from a loved one, such as, a mother or father.

WHAT YOU HAVE LEARNT

Motivation is a want or need that helps us move towards our goal. It moves us to act or behave in a particular way.

Motives and needs are characterized in terms of primary motives (biological) and secondary (psychological) motives. Primary needs are due physiological imbalance while secondary needs are influenced by social experiences of the individual.

According to the hierarchy model, biological needs must be fulfilled before

psychological needs.

Emotions play three important functions: prepare us for action, organize our behaviour and future, and, interact effectively with others.

Emotions have three interrelated components: physiological changes, behavioural and emotional expression and subjective feelings.

Basic emotions and their expressions are universal in nature yet culture provides guidelines towards how and when to show emotions.

Motivation and emotion together activate and organize our life. Motivation leads to emotional experience and this further motivates (reinforces) a person to act.

TERMINAL QUESTIONS

1. Write three examples of motivated action. Now identify the important characteristics of motivation.
2. Identify the important components of Maslow's hierarchy of needs. Which motives are satisfied first? Provide reasons as to why this is so.
3. With suitable examples describe the role of motivation and emotion in organizing our behaviour.

Activities/Exercises

1. Interview a person who has achieved prominence in his or her field. For example, the president of a company, a sports person, a dancer or a physically challenged person. Ask the person about his/her life – starting from childhood to the current position, incidents that influenced them, the various challenges and how they were overcome. Prepare a case study.
2. Collect a number of magazines, and newspapers. Cut photographs which show different facial expressions. Make a collage of these photographs to understand the different emotions in human life.
3. Interview a dancer or theatre person. Ask the person about different emotional expressions and how they are used to communicate specific emotions. Try to note how the body posture, gestures, and facial expressions communicate emotions.

ANSWERS TO INTEXT QUESTIONS

8.1 Refer to Section 8.1

Quiz 1

1. c
2. b
3. d

4. b

5. c

8.2 Refer to Section 8.2

Quiz 2

1. T

2. F

3. F

4. T

5. T

Hints for Terminal Questions

1. Refer to section 8.1

2. Refer to section 8.1

3. Refer to section 8.3

Web Sites

The following websites will help you to further understand motivation and emotions.

1. <http://thesituationist.files.wordpress.com/2007/10/emotions.jpg>

Types of emotions: (a) In this site robots have been used to depict different types of emotions. Study in detail how each part of the face (forehead, eyebrows, size of eye pupil, etc.)

(b) https://sfari.org/image/image_gallery?img_id=136061&t=1241787803666

Whereas in this site humans have been used to show six different emotions. Can you identify them?

(c) [http://www.scribd.com/doc/12910868/Paul-Ekman-Emotions-Revealed-](http://www.scribd.com/doc/12910868/Paul-Ekman-Emotions-Revealed-Recognizing-Faces-and-Feelings-to-Improve-Communication-and-Emotional-Life)

[Recognizing-Faces-and-Feelings-to-Improve-Communication-and-Emotional-Life](http://www.scribd.com/doc/12910868/Paul-Ekman-Emotions-Revealed-Recognizing-Faces-and-Feelings-to-Improve-Communication-and-Emotional-Life)

This E-book by Paul Ekman has a number of photographs which you can observe to understand different types and display of emotions. On page 236 in the appendix there is a test 'Reading Faces' consisting of 14 photographs that you can attempt . At the end of the e-book answers with explanations are also provided .

9. NATURE AND DETERMINANTS OF DEVELOPMENT

One of the most important features of all living beings is change. The changes are most striking when we see the transition in human life spanning from conception to death. The journey from an unborn baby to an adult and then on to an old person is very fascinating. Even a casual observation of the human lives around you will make it clear that several changes happen each day in our bodies and psychological functioning. Some changes are noticeable while some can not be immediately or clearly observed. Some of these changes are determined more by hereditary or genetic factors, while some depend more on environmental and cultural factors. Different cultures hold different goals for development and use different strategies to bring up children. In order to help individuals develop in the best way possible, it is important to understand the nature and process of development throughout the life-span.

OBJECTIVES

After studying this lesson, you will be able to:

describe the nature of development and distinguish it from other related concepts;

describe the important domains and stages of human development;

identify major genetic and environmental influences on human development;

and

learn about the effects of environmental factors on pre-natal as well as postnatal development.

9.1 NATURE OF DEVELOPMENT

In this section, we will try to understand what is meant by the term “development”, distinguish it from other related concepts, and identify its main characteristics.

A. The Concept of “Development

The term “development” is generally used to refer to the dynamic process by which an individual grows and changes throughout its life-span. It is often thought of as the process of qualitative change taking place from conception to death. In this way development is a broad term and deals with all areas including physical, motor, cognitive, physiological, social, emotional and personality. It should be noted that developments in all these areas are inter-related. For example, a 13 year-old girl undergoes physical and biological changes in her body and such changes are in turn related to her mental, social and emotional development.

Life begins at conception when mother’s ovum fertilized by the father’s sperm and a new organism is created. From that point till death an individual keeps on changing. Such changes are not random but orderly and generally follow a pattern. It may be noted that the developmental changes are not always incremental or evolutionary. They may also involve a decline in the functioning called “involution”. A child loses milk teeth in the process of development while an old person may show decay in memory and physical functioning. Development, therefore, is best viewed as a gain-loss relationship in which newer and different kinds of changes take place. Older behavior patterns may lose their salience while new ones may emerge.

Development is shaped by the joint influences of both nature and nurture in a cultural context. Nature refers to the hereditary contribution a child receives from parents at the time of conception. Genetics determines several aspects of a person’s physical structure and functioning as well as some psychological characteristics to a certain extent. Nurture refers to the influences of the complex physical and social ecology in which we develop and grow. Various aspects of the child’s ecology (e.g. physical facilities, social institutions and rituals, and school) influence the developmental outcomes in important ways.

B. Key Developmental Concepts

The term “**development**” is often used interchangeably with “**growth**” and “**maturation**”, but these terms need to be carefully distinguished.

“**Growth**” generally refers to the quantitative additions or changes in the organic structure. For instance as we become older, the body size, height, weight, proportion of parts of our body change in measurable ways. Also, the vocabulary increases. “Development” on the other hand, is a broader term which often includes growth, but is

used more to refer to functional and qualitative changes in cognitive ability, perceptual ability, personality and emotional development and so on.

“Maturation” is a term that refers to the natural unfolding of changes with increasing age, e.g. hormonal changes as the individual reaches puberty. An example of this

can be seen in breast development which is influenced by release of estrogen when a girl reaches adolescence. Maturation refers to the changes which are primarily biological in nature and occur due to our genetic program. Our biological structure follows a predetermined course of changes with time. This can be seen in the development of teeth during childhood. Changes in body proportions with age provide an example of such predetermined universal trends. The size of the head is roughly half of the whole body at birth, but the proportion keeps decreasing until adulthood, when it is less than one-fourth of the whole body. Therefore, maturational changes in our body are primarily due to the ageing process rather than learning or other factors such as illness or injury.

It must be noted that changes in behavior also occur due to “**learning**”. Learning takes place as a result of a person’s interaction with the environment. Maturation provides the raw material and sets the stage for learning to occur. Taking the case of learning to read, the child needs to be biologically ready. The eyes need to develop proper focusing ability before a child can learn to read. Therefore, maturation and learning jointly bring about changes in a person’s behaviour.

“**Evolution**” is a term that refers to species-specific changes. Evolutionary changes happen very gradually and are passed on from one generation to the other so that the species is better equipped for survival. The evolution from apes to human beings took place over a period of about 14 million years. Changes that occur at the level of species are called **phylogenetic** and those that occur at the level of individual are called **ontogenetic**. The term evolution is also used to describe the incremental changes that take place in the course of development.

C. Characteristics of Development

We can now summarize the characteristic features of development to help us distinguish it from other kinds of changes.

Development is a **life-long process**, spanning from conception till death.

Developmental changes are often **systematic, progressive and orderly**. They usually follow a pattern, proceeding from general to specific, and from simple to complex and integrated levels of functioning.

Development is **multi-directional**, i.e. some areas may show a sharp increase

while other areas may show a decline. Developmental changes usually involve an increase in maturity towards higher levels of functioning, e.g. increase in vocabulary's size and complexity. But it may also involve a decrease or loss, such as in bone density or memory in old age.

Moreover the rate of growth and development is **not always constant**. 'Plateaus' are often seen in the pattern of development, indicating periods of no apparent improvement.

Developmental changes can be **quantitative**, e.g. increase in height with age, or **qualitative**, e.g. formation of moral values.

Development is both **continuous as well as discontinuous** process. Some changes may occur very rapidly and be overtly visible, such as appearance of the first tooth, while some changes may not be sharp enough to be observed easily in day-to-day functioning, such as understanding of grammar.

Developmental changes are **relatively stable**. The temporary changes due to weather, fatigue or other chance factors will not qualify as development.

Developmental changes are **multi-dimensional** and inter-related. They may happen together in many areas at a time, or can occur one at a time. During adolescence there are rapid changes in the body as well as in emotional, social and cognitive functioning.

Development is highly **plastic or flexible**. This means that the same person may suddenly show greater improvement in a particular area than expected from the past rate of development. An enriched environment can produce unexpected changes in the physical strength, or in memory and intelligence levels.

Development is **contextual**. It is influenced by historical, environmental and socio-cultural factors. The loss of a parent, an accident, a war, an earthquake and child-rearing customs are examples of factors which may influence development.

There are considerable **individual differences** in the rate or tempo of developmental changes. These differences could be due to hereditary factors or environmental influences. Some children can be very precocious for their age, while some children show developmental delays. For example, though an average child begins to speak 3 word sentences around 3 years of age, there will be some children who have mastered this much before 2 years, while there will be some who are not able to speak full sentences even by 4 years. Moreover, there will be some children who are not able to speak even beyond the upper limit of the range.

INTEXT QUESTIONS 9.1

1. Write True/False against each statent:

- a. Development refers to both quantitative and qualitative changes. T/F
- b. Development follows a constant rate. T/F
- c. The study of development does not cover old age. T/F
- d. A person may show increase in some aspects and show decline in other aspects at the same time T/F
- e. Enrichment of the environment can produce dramatic changes. T/F

2. Write short answers to the following questions:

a. Explain the concept of development.

b. State any 3 major characteristics of human development.

9.2 DOMAINS OF DEVELOPMENT

As we have noted development is an inclusive term that incorporates changes in several areas. These areas or domains deal with 3 broad categories:

1. Physical and motor development: It refers to changes in body-size and structure, functioning of various body systems, brain development, perceptual and motor development.

2. Cognitive development: It refers to the development of cognitive and intellectual processes, including memory, attention, intelligence , academic knowledge, problem solving, imagination and creativity. It also includes development of language.

3. Socio-emotional development: It refers to how we develop relationships with other people, and how our emotions emerge and change as we grow older. It includes emotional communication and self-control, understanding of self and

others, interpersonal skills, personality, and emergence of friendship and moral reasoning. Physical domain

Socio-emotional domain Cognitive domain

Fig. 9.1: Inter-relatedness of the domains of development

These domains or areas are inter-related and combine in a holistic manner to make a person's developmental pattern unique. Each domain influences and is influenced by others. As a baby develops physically, several motor skills are acquired. As the

baby becomes capable of grasping, reaching, sitting, crawling, standing and walking, he/she is able to better explore the environment, resulting in further cognitive development. Improved thinking and understanding helps formation of better social relations as well as emotional expression and comprehension. On the whole, each domain helps the child move towards enriched experiences, learning and overall development.

INTEXT QUESTIONS 9.2

1. Fill in the blanks in the following statements:

- a. Improvement in memory and language indicates development in the _____ domain.
- b. Interpersonal relationships fall in the _____ domain.
- c. Changes in size and structure of the body fall in the _____ domain of development.
- d. _____ broad domains of development can be categorized for the sake of convenience.

9.3 STAGES OF DEVELOPMENT

Even though development is a continuous process, some theorists believe that various stages can be identified for the sake of locating major shifts and determining the developmental tasks. This helps in monitoring the pace of developmental changes. It must be pointed out that there is no sharp dividing line between them. Each stage has certain characteristic features and prepares the ground for the next stage. Some theorists have suggested stages in specific areas of development. For instance Piaget who identified stages of cognitive development and Freud who suggested stages of psycho-sexual development. These theories are explained in other lessons.

A. Developmental Stages: A Life Span Perspective

Most psychologists identify the following stages of development:

Prenatal period (from conception to birth) : In this period, the single-celled organism changes into a human baby within the womb.

Infancy and toddlerhood (birth-2 years): Rapid changes in the body and brain help several sensory, motor, social and cognitive capacities to emerge.

Early childhood: (2-6 years): Motor skills are refined, language develops, ties are formed with peers, and the child learns through play.

Middle childhood (6-11 years): These are the school years when the child acquires literacy skills, thought processes are refined, friendships emerge and self-concept is formed.

Adolescence (11-20 years): This period is marked by puberty which signals the onset of rapid physical and hormonal changes, emergence of abstract thinking, sexual maturity, stronger peer ties, sense of self and autonomy from parental control.

Early adulthood (20-40 years): This is the stage of life when the youngster leaves home for the sake of education, or to find a career, and to form intimate relationships leading to marriage and having children.

Middle adulthood (40-60 years): At this stage the person is at the peak of his/her career. There is a need to help children begin independent lives, and to look after own parents who are aging.

Late adulthood (60 years till death): This period is marked by retirement from work, decrease in stamina and physical health, bonding with grand-children, and dealing with impending old age and death of self and spouse.



Fig. 9.2: Above figure shows stages of human development

Though the stages of development are universally recognized, the exact age range is arbitrary and depends on cultural factors. In some cultures, for example, there is no distinct phase of adolescence with the stress associated with it. One simply moves from childhood to adulthood. The concept of old age is undergoing a change with enhanced medical, health and cosmetic facilities. Life-span expectancy has also increased and reached to 65 years in India.

B. Life Stages : An Indigenous View

Ancient texts also view the human life-span in terms of stages. Life is seen as a process of evolution in four stages known as Ashramas, which literally means a dwelling place. There are four main Ashramas:

Brahmacharya: During this period the major task is learning while living as a student under the guidance of a Guru, with emphasis on discipline and simple living.

Grihastha: In this phase of life an individual has to work and take on the responsibility of raising and maintaining a family within the social context.

Vanaprastha: It is a period of seclusion after one has raised the family and completed the duties towards it. It requires severe discipline and austerity and sharing responsibilities with the younger people.

Sanyasa: It is the final stage of life when one needs to achieve complete detachment from worldly objects, freedom from desire, and move towards ultimate self-knowledge and renunciation from the world.

These stages or ashramas are based on the principle that a human being should grow, participate and discharge worldly duties and contribute to the progress and welfare of society and move towards liberation and spiritual growth. It is desired that the older persons should give way to the younger generation. In this scheme one lives in a web of obligations to society as well as environment including all living beings such as animals and trees. It emphasizes co-existence with this whole living world rather than being a consumer and exploiting the environment for personal gains.

C. Developmental Tasks

Each developmental stage is characterized by a dominant feature or a leading characteristic which determines its uniqueness. For example, a child is expected to go to school and study while an adult is expected to work and raise a family. Certain characteristics stand out more prominently than others and each period is called a stage. People learn certain behavior patterns and skills more easily and successfully at certain stages and this becomes a social expectation. For example, a child is supposed to be able to go independently to school in middle childhood. Such social expectations of a particular age common to all persons constitute “developmental tasks”. If a person is

able to master the developmental task of a particular stage, he or she is considered to have successfully moved on to the next stage of development.

INTEXT QUESTIONS 9.3

1. Write True or False after each statement:

- a. Developmental stages have a fixed age range.
- b. Prenatal period extends from birth to conception.
- c. Adolescence is marked by rapid physical and psychological changes.

- d. There are four stages in the Hindu view of development.
- e. The Hindu concept of development encourages staying away from the family.
- f. Developmental tasks are social expectations of a particular age group.

2. Write short answers to the following questions:

- a. State the major stages of development as described in ancient Hindu texts.

- b. What do you understand by the term “developmental tasks”?

9.4 INFLUENCES ON DEVELOPMENT

It is important to understand the factors that determine the course of development. Each one of us is a product of genetic factors and environmental influences. The role of maturation and learning has already been described briefly in an earlier section. Let us now look at the contributions of heredity and environment in some detail.

A. Genetic Influences

At the time of conception, the ovum of the mother and the sperm cell of the father unite to form a new cell. The small particles in the nucleus of the cell are called **chromosomes**. The chromosomes exist in pairs. The human cell has 46 chromosomes arranged in 23 pairs. One member from each pair comes from the mother and the other one from the father. Chromosomes store and transmit genetic information. The genes, which are the actual trait carriers, are found in very large numbers in each chromosome.

The fertilized zygote brings together various combinations of chromosomes. In this way, different genes are transferred from each child of the same set of parents. Due to this reason each child bears greater similarity to his or her blood relatives than to anyone else. At the same time there are also many differences amongst blood relatives.

B. Genotypes and Phenotypes

Genetic transmission is a complex process. Most characteristics that we observe in human beings are combinations of a large number of genes. Innumerable permutations and combinations of genes are responsible for the large differences in physical and psychological characteristics.

Only identical or monozygotic twins have exactly the same set of chromosomes and genes as they are formed by duplication of a single zygote. Most twins are fraternal or di-zygotic, who develop from two separate zygotes. These fraternal twins may resemble each other like brother and sister, but they will also be different from one another in many ways.

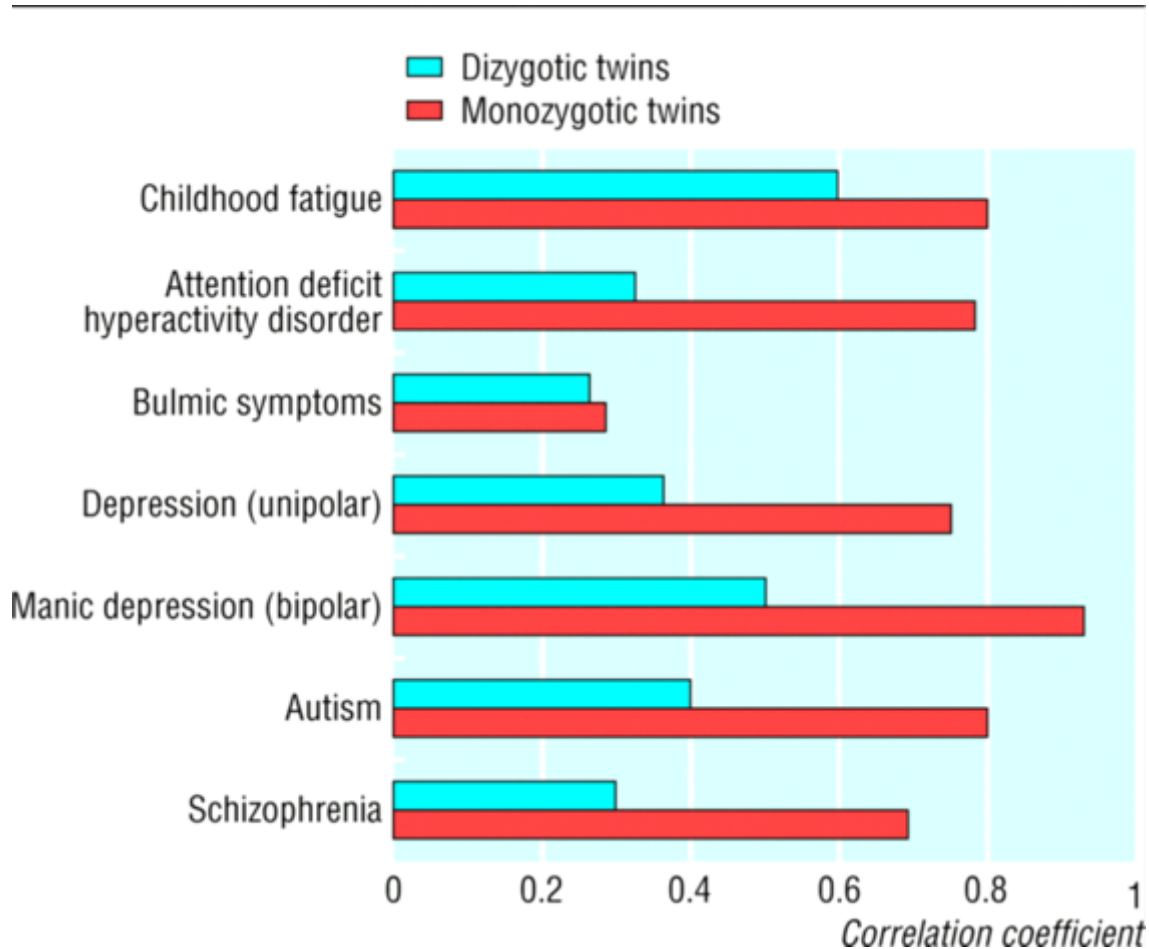


Fig. 3: Differences between Mono-zygotic and Di-zygotic twins

Genes can be **dominant or recessive**. It is a known fact that there is more color blindness or lack of sensitivity to certain colors among males than females. A

grandmother and mother can transmit this condition to the male child without being color blind themselves. This is because in the male this disorder is dominant, whereas in female it is recessive. The genes form pairs. If both genes in a pair are dominant, the individual will display the specific trait (e.g. color blindness). If one gene is dominant and the other recessive, the dominant will prevail. The recessive gene will be passed on and may show up in a later generation.

The dominant gene, therefore, is the one responsible for a particular trait to show up in a person. The characteristics which show up and are displayed e.g. eye color, are called **phenotypes**. The recessive gene does not show up as a trait, unless paired with

another gene just like it. The characteristics that are carried genetically as recessive genes but are not displayed are called **genotypes**.

Genotype, therefore, refers to the actual genetic material or a person's genetic heritage while **phenotype** refers to the individual's physical and behavioral characteristics which are determined by both genetic and environmental factors.

C. Environmental Influences

Nature refers to what a child has inherited genetically from her parents, while the influence of environment on the development of the child is referred to as nurture.

To understand the development of a person, we have to study the complex interaction between nature and nurture or heredity and environment.

Environmental influences are important both at the prenatal and postnatal stages of human development. At the prenatal stage, when a fetus is in the mother's womb, internal or external harmful agents, such as certain legal or illegal drugs, alcohol, lead and pollutants can harm the unborn baby's development. The mother's nutrition, diseases and emotional stress can also affect the development of the fetus.

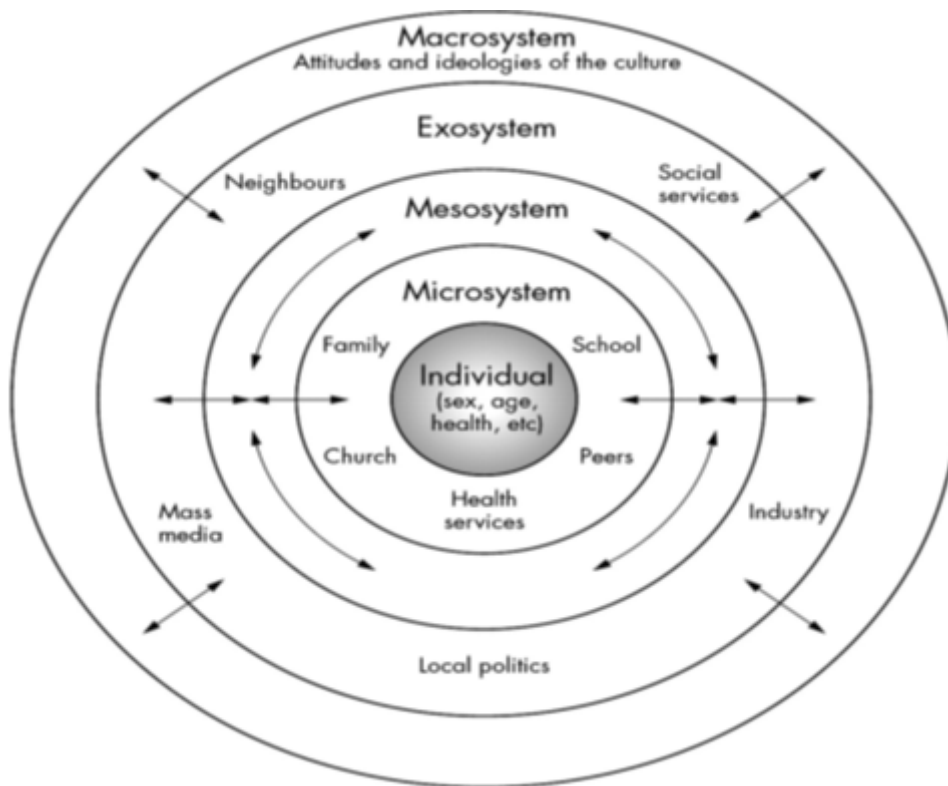


Fig. 4: Ecological systems

influencing human development

After birth, several types of environmental factors operate to influence the development of the child. The ecological systems theory of development views environmental factors organized as concentric circles of systems.

The micro-system is the immediate environment at home and interactions between them and the child's own characteristics. The meso-system consists of the relationships between family members and school and neighborhood. The exo-system refers to the influences of indirect agencies such as work-place of the parents or community services. The macro-system is the outermost layer which includes cultural values, laws and customs. The systems are ever-changing and dynamic. There is also a chronosystem

which refers to the time dimension. As the child grows, there are changes within each system, and also changes due to the child's interaction with the environment. On the whole, in ecological systems theory, children are both products and producers of their environments, in a network of interdependent effects.

The present view of development gives importance to both nature and nurture.

Heredity and environment are inseparably interwoven, each affecting the impact of the other on the child. Development therefore has certain universal features, as well as features unique to the individual. It is important to understand the role of heredity, but more fruitful to understand how the environment can be improved, so as to help the child develop in the best possible way within the limits set by heredity.

An understanding of the principles and determinants of development helps us in several ways:

1. It helps us to know what to expect of the individual's capability at a particular age.
2. It gives information on when to provide opportunities and stimulation for optimal development.
3. It helps to parents, teachers and others who work with children, to prepare them for the physical and psychological changes that are to take place.
4. It helps us to be prepared for changes in our bodies and personalities as we grow older.
5. It helps us to understand that it is possible to facilitate the process of development by providing an enriched environment.

INTEXT QUESTIONS 9.4

1. Fill in the blanks in the following statements:
 - a. Heredity is determined at the time of _____.
 - b. There are ____ pairs of chromosomes in a human cell.
 - c. Genes can be _____ or _____.
 - d. Characteristics that are carried in the genetic code but not displayed are called _____.
 - e. In the ecological systems theory, the layer which consists of cultural values,

laws and customs is called the _____.

f. External agents that can harm the unborn baby in the womb are called _____.

2. Write short answers to the following questions:

a. Explain the terms ‘genotype’ and ‘phenotype’.

b. Briefly explain the main features of the ecological systems theory of development.

WHAT YOU HAVE LEARNT

Development refers to the process by which the individual grows and changes during its life-span, from conception to death.

Development is systematic, orderly, progressive, multi-dimensional, multidirectional, plastic and contextual.

The major domains of development are physical, cognitive and socio-emotional.

Stages of development can be categorised for the sake of convenience, ranging from prenatal stage to late adulthood or old age.

Each stage is characterized by certain developmental tasks which are social expectations for a given age group.

Hereditary factors are determined at the time of conception and genetic information is carried by genes and chromosomes.

Genotypes refer to the characteristics which are carried genetically but not displayed.

Phenotypes refer to those characteristics which are displayed.

Environmental factors influence both prenatal and post-natal development.

External harmful agents can cause damage to the fetus in the mother's womb.

Maternal disease, nutrition and stress can influence fetal development.

The ecological systems theory proposes several sub-systems in the growing child's environment which can influence development.

Both nature and nurture jointly influence development.

TERMINAL QUESTIONS

1. Differentiate between the terms “development”, “growth”, “maturation” and “evolution”.
2. Describe the major domains of development.
3. Identify the major stages of human development.
4. Discuss the relative importance of nature and nurture in development.

ANSWERS TO INTEXT QUESTIONS

9.1

1. a. true
b. False
c. False
d. True
e. True
2. a. Refer to section 9.1.1
b. Refer to 9.1.3

9.2

- a. cognitive
b. socio-emotional
c. physical
d. three

9.3

1. a. False
b. False
c. True
d. True
e. False
f. True
2. a. Refer to Section 9.3.2
b. Refer to Section 9.3.3

9.4

1. a. conception

b. 23

c. dominant, recessive

d. genotypes

e. macrosystem

f. teratogens

2. a. Refer to section 9.4.2

b. Refer to section 9.4.3

Hints for Terminal Questions

1. Refer to section 9.1.2

2. Refer to section 9.2

3. Refer to section 9.3.1

4. Refer to section 9.4

10. CHILDHOOD

Childhood is a beautiful phase of life for most of us. It is a period associated with play, fantasy, and innocence. It is the crucial phase in life and involves important aspects of growth and development which are critical for the entire life span.

The childhood period can be further classified into four sub-stages:

- (a) Infancy (birth to 2 years),
- (b) Early childhood (2 to 6 years),
- (c) Middle childhood (6 to 11 years), and
- (d) Adolescence (12 to 19 years).

Infancy provides the foundation for all areas of development. It is a period marked by extremely rapid physical growth and development of sensori-motor skills. Infancy is also important for cognitive, language and socio-emotional development. This stage has already been discussed. Adolescence will be taken up in the next lesson.

In this lesson, we will try to understand how various aspects of development such as physical, motor, cognitive, social and personality development proceed during childhood. We will also study the socialization process during childhood in the cultural context.

OBJECTIVES

After studying this lesson, you will be able to:

describe physical and motor development in childhood;

understand cognitive development during childhood;

explain socio-emotional and personality development in childhood; and

understand the process of socialization within the cultural context.

10.1 PHYSICAL AND MOTOR DEVELOPMENT IN CHILDHOOD

While physical growth is extremely rapid during infancy, it is comparatively slower in childhood. As the child's body size, height and weight increase, there are also changes in muscular and skeletal structure. This leads to development of several motor skills in childhood. The studies have shown certain universal trends in both physical and motor development.

The sequence of physical and motor development is fairly uniform across different cultures. Yet there are large **individual differences**. One child may walk at 10 months, while another may walk only at 24 months. The major milestones of physical and motor development in infancy and childhood are given in Table 10.1

Table 10.1: Milestones of Physical and Motor Development

Lifting chin up while lying on stomach	1 month
Lifting head and chest	2 months,
Rolling over	4 months,
Picking up objects with palm without sitting with support	5 months,
Finer use of thumb and fingers	7 months,
Sitting alone (without support)	8 months
Standing with support (holding on to furniture)	8-9 months,
Crawling	9 months
Standing independently	9-10 months
Taking few steps without holding on and walking holding on	12-13 months

Grasping objects with thumb and forefingers	13-14months
Walking alone	15 months
Running and climbing stairs	2 years
Walking on tip toe	2½ years
Riding a tricycle	3 years
Throwing ball over head, walking down stairs with one foot to a step	4 years
Copying a complex figures	6 years

Understanding of physical and motor development is extremely important as it provides the basis for the development of child's personality, social and emotional development. It helps establish emotional bonding between the child and the mother figure. It also helps mastery over the child's own body and movements.

A. Gross and Fine Motor Development

Motor development is usually classified into ‘gross motor’ and ‘fine motor’ categories. **Gross motor development** refers to control over large muscle groups that enable the child to get around, e.g. crawling, standing and walking. They usually involve the limbs or the whole body. **Fine motor development** refers to control over smaller muscle movements e.g. grasping, pinching etc. Fine motor skills are necessary to engage in smaller, more precise movements, generally using the hands and fingers. Fine motor skills are different than gross motor skills which require less precision to perform the daily tasks.

Table 10.12: Gross and Fine Motor Development Milestones

Child’s age	Activities involving Fine Motor Skills:
0 and 3 months	Holding an object,
3 and 6 months	Trying to reach out for objects, putting things in mouth,
6 months and 1 year	Trying to hold food, using hand and fingers to playgames,
1 year and 1-1/2years	Scribble on paper, trying to play throw and catch withthe ball,
1-1/2 and 2 years	Drawing lines with pencil, using a spoon to eat food, with little help,
2 and 3 years	Brushing teeth and buttons cloths with help,
3 and 5 years	Builds using building blocks, uses a pencil to draw,turn pages of a book,
5 and 7 years	Draws various shapes very easily, brushes and combswithout support, cuts shapes very clearly

10.13 Physical and Motor Development in Early Childhood (2-6 years)

Early childhood covers the period from 2 to 6 years. It is also known as the preschool stage. The child who has become mobile is now able to widen the sphere of activities beyond the immediate family. Through interaction with the wider society and the environment the child learns the rules of appropriate social behavior and develops mental abilities which prepare him/her for formal education and schooling. Most infants appear to be quite clumsy in their physical and motor activities. But gradually their locomotion skills become refined and graceful. Body balance while walking and running improves noticeably. A 3-year-old can run in a straight line and jump smoothly without falling down. A 4-year-old can skip, jump on one foot and catch a large ball thrown from a distance.

Development of Fine Motor Skills during Childhood

2-3 years: Undresses self and assists with dressing, Copies a circle and a cross, Strings four large beads, Turns a single page, Snips with scissors on a line, Buttons/unbuttons large buttons.

3-5 years: Copies a square, Prints own name – tripod pencil grasp, Ties shoelaces, Cuts on line continuously, Prints some capital letters.

5-7 years: Copies a triangle, Cuts out simple shapes, Copies first name, Prints numerals 1 to 5, Color within lines, Pastes and glues appropriately

Between 2 and 3 years, young children stop “toddling,” and develop a smoother gait. They also develop the ability to run, jump, and hop. They can participate in throwing and catching games with larger balls.

Children who are 3 to 4 years old can climb up stairs using a method of bringing both feet together on each step before proceeding to the next step. However, they may still need some assistance to prevent fall as they are likely to be unsteady in this new skill. Children of this age can jump and hop higher as their leg muscles grow stronger.

During this period, children become better at catching and throwing, can hit a stationary ball with a bat, learn to ride a tricycle, and can kick a ball placed directly in front of their bodies. They can create things with their hands, such as building towers out of blocks, molding clay into rough shapes, and scribbling with a crayon. Children of this age often begin showing a preference for using one hand more often than the other, which is the beginning of becoming left or right-handed.

Between 3 and 4 years, children improve in eating food themselves and can use utensils like forks and spoons. They can now hold a crayon or pencil by the writing hand rather than just grasping it with the fist. They can also make twisting motion with their hands, useful for opening door knobs or twisting lids to open jars. Most children are toilet trained by 4 years of age.

As children reach the age of schooling, the rate of physical growth becomes slower until puberty when there is a rapid 'growth spurt'.

By the age of six, the child is physically capable of coordinated actions which require body balance. Small muscle coordination required for fine motor activities, such as putting on shirt buttons or copying a simple figure, improves quite dramatically during the early childhood years. Children can also complete other self-care tasks beyond dressing and undressing, such as brushing their teeth and combing their hair. Children of this age can also independently feed themselves without an adult's immediate supervision or help.

During 5 to 6 years of age, young children continue to refine the earlier skills. They can run faster and can start to ride bicycle with training wheels for added stability. In addition, they can step sideways. Children of this age begin mastering new forms of physical play such as the jungle gym, and begin to use the see-saw, slide, and swing on their own.

10.16 Development in Middle Childhood (6-11 years)

During this period physical growth becomes more gradual and rate of overt change becomes slower until puberty at about 11-13 years when there is again a rapid 'growth spurt'. There are changes in height, weight and muscular strength and swiftness. During this period, there are large gender differences. Girls are slightly shorter than boys during 6 to 8 years but then the trend reverses. Girls start putting on weight. A 10 year old girl may look taller and heavier than a boy of the same age. The growth spurt for boys comes later than for girls.

During this period, children achieve greater control over large and small muscle groups. They keep getting stronger, faster and attain better motor co-ordination. School children are energetic and enjoy all types of outdoor games. Increase in cognitive capacity also helps them learn the rules of new games.

The 6-7 year-old children can copy complex figures such as a diamond, color patterns and figures and assemble tools and model toys. They also become more skillful in games requiring skillful eye-hand coordination such as throwing, catching and hitting targets. They continue to refine fine motor skills and build upon earlier skills.

INTEXT QUESTIONS 10.1

Write short answers to the following questions:

1. Give names of any 3 gross motor skills developed in childhood.

2. Give names of any 3 fine motor skills developed in childhood.

10.2 COGNITIVE DEVELOPMENT

Cognitive development refers to the way children learn and process information. It includes improvement in attention, perception, language, thinking, memory and reasoning.

According to Piaget's cognitive developmental theory, our thoughts and reasoning are part of adaptation. Cognitive development follows a definite sequence of stages.

Piaget described four major stages of cognitive development:

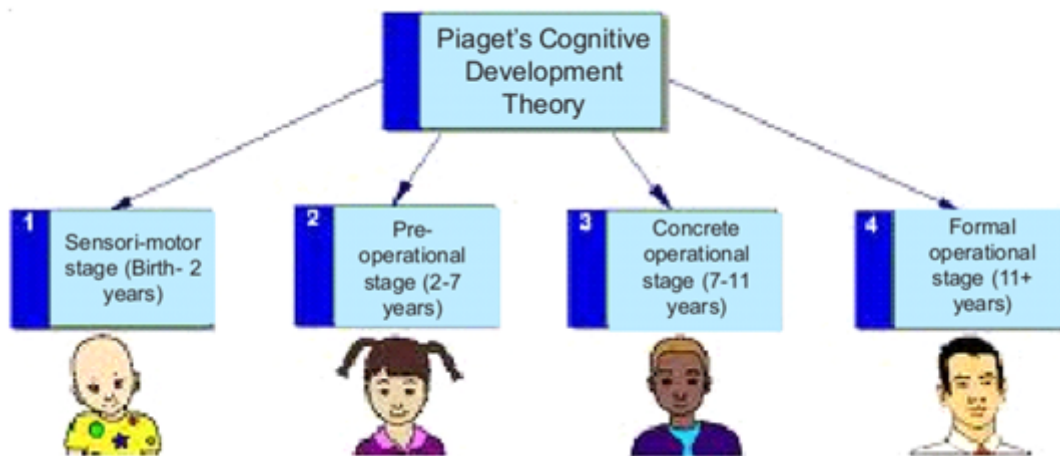
Sensori-motor stage (Birth- 2 years),

Pre-operational stage (2-7 years)

Concrete operational stage (7-11 years),

Formal operational stage (11+ years)

Fig 10.1: Piaget's Cognitive Development Theory



The following section takes a look at cognitive development through childhood as indicated by Piaget's theory.

Cognitive Development in Early Childhood (2 to 6 years)

During this period children become increasingly proficient in using symbols such as words and images to represent a variety of objects, situations and events. By the time children enter school they have a reasonably good vocabulary. In fact, children are very receptive at learning language when they are exposed to multiple languages in different contexts. They often grow up as bilingual or multilingual children. They have better understanding of language compared to monolingual children who use only one language.

Capacity for sustained attention improves during the early childhood. A 3-year-old child may persist on a task such as coloring with crayons, playing with toys or watching

television for not more than 15-20 minutes at a stretch. By contrast, a 6- year-old can be found to be working on an interesting task for an hour or more. Children also become more selective in their attention. As a result their perceptual skills also improve.

Thinking becomes more logical and capacity to remember and process information also improves. Through interaction with the environment the child learns the rules of appropriate social behavior which prepare him/her for schooling.

Early childhood, from 2 to 6 years, is the phase when the child progresses through the **Pre-operational stage**. There are 2 sub-stages of the Pre-operational Stage:

a) Symbolic Function (2 to 4 years)

b) Intuitive Thought (4 to 7 years)

During the Symbolic Function sub-stage, children can create mental images of objects and store them in their minds for later use. For example, such a child can draw a picture of or pretend to play with a puppy that is no longer present there. Children can talk about people who are traveling, or who live somewhere else. They can also talk about or draw places they visited, as well as create new scenes and creatures from their imagination. Children can also use their mental images of things to role-play in games.

Piaget also believed that pre operational children have a style of thinking characterized by **Egocentrism**, or the inability to see the world from someone else's point of view. According to Piaget, children with egocentrism explain situations from their own perspective and understanding.

The next sub-stage in Piaget's Preoperational cognitive development stage is the

Intuitive Thought sub-stage, which spans ages 4-7 years. Children in this substage of development learn by asking questions such as, "Why?" and "How come?" Piaget labeled this "intuitive thought" because he believed that children at this stage tend to be so certain of their knowledge and understanding that they are unaware of how they gained this knowledge in the first place. These children show "**Centration**". They focus on one characteristic of object and base their decisions or judgment on that only.

During early childhood, children's ability to understand, process, and to produce **language** also improves rapidly. There is a 'language explosion' between 3 and 6 years. At age 3, their spoken vocabularies consist of roughly 900 words. By age 6, spoken vocabularies expand dramatically to anywhere between 8,000 and 14,000 words.

As children move beyond using two word sentences, they start to learn and understand grammar rules. Beyond growing their vocabularies, young children start expanding their ability to use different forms of words (e.g., irregular verbs such as "She brought" rather than "She brang")? They also begin to form more complex sentences.

B. Cognitive Development in Middle Childhood

Children at the middle childhood level are full of curiosity and are keen to explore the environment. Memory and conceptual knowledge improve, facilitating logical

thinking beyond the immediate situation. Children can also engage in aesthetic activities such as music, art and dance and develop hobbies of their own.

In Piaget's theory, characteristics of **the Concrete Operational Stage** during middle childhood are as follows:

Understanding of logical principles.

Improvement in spatial reasoning.

Logical thinking limited to real and concrete situations.

In middle childhood years, **language development** proceeds in many ways. More than just learning new words, children are acquiring more adult definitions of the words they know. They create relationships among words, understand synonyms and antonyms, and understand how prefixes and suffixes affect word meaning.

INTEXT QUESTIONS 10.2

1. Write short answers to the following questions:

a. List any 3 major features of the pre-operational stage.

b. Describe any 3 major achievements of the concrete operational stage.

10.3 SOCIO-EMOTIONAL AND PERSONALITY DEVELOPMENT

Infancy is a period when basic emotions are established, and the child progresses in both emotional expression as well as recognition of emotions in others. During early childhood, children typically start to develop self-conscious emotions like shame and guilt as they start evaluating themselves, instead of simply reacting to caregivers' or other adults' evaluations.

As children become increasingly self-aware, more effective at communicating, and better at understanding the thoughts and feelings of others, their social skills improve. They become skilled at modifying and expressing their emotions to fit different social situations. For example, one child may feel angry, but he/she knows that having a tantrum at school is inappropriate. Another child learns that acting pleasant and happy, even though she is actually feeling shy, is a better way to meet people in a birthday party when she does not know many of the other children. Changing or

controlling one's emotions in social situations is an important skill that allows children to fit in with groups and helps in creating interpersonal relationships. Another emotional capacity that develops during early childhood is empathy, which is an important component of positive social behavior. As with other emotions, the development of empathy depends on cognitive and language development.

Learning how to appropriately express and deal with anger, aggression, and fear is a valuable life and social skill. Young children often need plenty of guidance and positive discipline in order to learn how to control their anger. Parents and caregivers not only directly teach ways of controlling emotions, but also indirectly influence children's behaviour by acting as role models.

As infants grow, they begin to mature in their ability to interact with others socially. A child's main developmental task is to create bonds and connect with primary caregivers. In contrast, young children branch out and begin to create other social relationships. When interacting with other children of their age, such as peers at day care or preschool, they engage in parallel play in which they play beside each other without truly interacting with each other.

Young children begin to play more cooperatively. In cooperative play, they engage in the common activity in a small group. Often, these first forms of cooperative play include pretend or symbolic play. As they continue to develop socially with peers, they often enter a stage of rough and tumble play which includes running, racing, climbing, or competitive games. This is the stage when social skills such as learning to take turns and follow simple group rules and norms are practiced.

B. Middle Childhood

During this period children show fast change in social relationships. They use social comparison to distinguish themselves from others. They start seeing things from other's perspective. A child begins to weigh self-worth and increasingly compares himself/herself with peers.

Children are now able to understand emotions of pride and shame and can experience more than one emotion in a given situation. They can suppress or conceal emotions more effectively and use self-acquired strategies to cope. In this period, children internalize or begin to control their values and feelings from within. They start making judgments at this age that impact moral development.

Gender differences are observed during middle childhood. Girls may attach greater importance to good interpersonal relationships and the family while boys may

emphasize the importance of social prestige.

A prevalent problem among children of this age group is of bullying. Researchers have found that bullies display certain characteristics, such as being intrusive, having

demanding but unresponsive parents. Victims are often depressed and have lower self-esteem.



Fig 10.2: *Girls at play*

In middle childhood, self-concept changes from a focus on observable characteristics and typical behavior and emotions to an emphasis on personality traits, positive and negative characteristics, and social comparisons. The child begins to see his or her own strengths and limitations. This helps the move towards achievement of an identity.

INTEXT QUESTIONS 10.3

1. Fill in the blanks in the following statements:

a. The development of empathy depends on _____ and _____ development.

b. Girls in middle childhood attach more importance to good _____ relations while boys give more importance to own social

_____.

- c. A child who is liked by some but disliked by other peers is called _____.
- d. When two children are playing without interaction, it is called _____ play.
- e. An internal determiner of emotional development is the child's _____.

10.4 SOCIALIZATION

The process of acquiring norms, values and beliefs, which are considered significant in a society, is called socialization. It is the process of transmitting cultural values, preferences and patterns of behavior to children. This is done by various processes,

institutions and people. Socialization involves regulating children's behavior and disciplining their undesirable or inappropriate behavioral tendencies.

Parents, families, children's peer groups, schools, religious institutions and mass media such as television are some important agents of socialization. They influence the process of child rearing directly as well as indirectly and reinforce culturally appropriate modes of thought and behaviour.

Early childhood is a crucial period of development because during this period children learn a great deal about the norms, conventions and practices in their families, society and culture. They acquire language and learn the fundamentals of culture. The primary socializing agents during this period are the family members.

Middle childhood is the time when influence of peers and school becomes prominent, though family context remains important. The effect of media such as television and computers is undeniable. This is the time when social stereotypes and prejudices are most likely to develop.

Much research has been done on the impact of **parenting styles** on the socialization of the child. There are 4 main types of parenting styles:

- 1. Authoritarian style:** demanding, controlling, insensitive parenting
- 2. Permissive style:** indulgent, non-demanding parenting
- 3. Authoritative style:** firm, consistent parenting, reasons given for discipline.

Neglectful or uninvolved style disinterested uncaring, low control and communication.

Besides family, peers, media and school, there are other factors which influence the socialization process. Ethnicity and socio-economic status of the parents makes an impact on the child's development directly and indirectly. Ethnicity is linked to family size structure, education, income, composition, extended networks.



Fig 10.3: Children from low SES home



Fig 10.4: *Child in high SES family*

INTEXT QUESTIONS 10.4

1. Give brief answers to the following questions.

a. What do you understand by the term ‘Socialization’?

b. State the major types of parenting styles.

WHAT YOU HAVE LEARNT

Physical development continues during early and middle childhood, but the rate is slower than in infancy.

There are gender differences in physical growth as girls tend to have a growth spurt earlier than boys during middle childhood.

Gross motor development is rapid during early childhood.

During middle childhood, children continue to develop gross and fine motor skills, while there is marked improvement in muscular control and eye-hand coordination.

In early childhood the child is in the pre-operational stage of cognitive development.

In middle childhood, the child reaches the concrete operational stage, when conservation, transformation, seriation and logical thinking are acquired, but the child still finds it difficult to deal with abstract concepts.

Emotional development in childhood involves the development of self-conscious emotions, and better self-regulation in emotional expression.

Social development progresses during childhood from parallel play to cooperative play. In middle childhood, the focus shifts gradually from family to peers and school.

Peer acceptance is an important determiner of a child's self-esteem and adjustment.

Socialization is the process of acquiring behaviors, norms, values and beliefs, which are considered significant in a society.

Parenting style influences the socialization process and personality development of the child.

Cultural and social factors influence parenting style and attitudes towards child rearing, and hence the socialization process.

TERMINAL QUESTIONS

- a. Discuss the major features of physical development in early childhood
- b. Describe the major features of motor development in middle childhood.
- c. Discuss the main differences between the pre-operational stage and concrete stage of Piaget's theory.
- d. Explain the major features of emotional development in early childhood.
- e. Discuss how cultural factors influence the process of socialization in childhood.

ANSWERS TO INTEXT QUESTIONS

10.1

1. a. Crawling, standing, walking
- b. grasping, pinching, holding pencil

10.2

1. a. Craete mental images, store it, talk about events, people create new scenes

from imagination

b. understanding logical principles improvement in spatial reasoning language development

10.3

1. a. cognitive, language

b. interpersonal, prestige

c. controversial

d. parallel

e. temperament

10.4

1. a. Acquiring norms values, beliefs, Regulation discipline of undesirable behaviour, agents of socialization.

b. Authoritarian, permissive, authoritative and neglectful.

Hints for Terminal Questions

a. Refer to unit 10.1.3

b. Refer to unit 10.1.4

c. Refer to unit 10.2.1 and unit10.2.2.

d. Refer to unit 10.3.1

e. Refer to unit 10.4

11. ADOLESCENCE AND ITS CHALLENGES

Human life completes its journey through various stages and one of the most vital stages is adolescence. Adolescence is the period of transition from childhood to adulthood and plays a decisive role in the formation of prosocial/antisocial adult. All of us undergo this stage which poses many challenges and is full of excitement. At the same time it demands adjustment on many fronts.

When we come to this world we are completely dependent upon others and learn gradually to be independent. In India, the adolescents do most of the work themselves but the final decision regarding various domains of life is taken by their parents. For example, an adolescent wishes to enjoy movies but parents may force him/her to complete study first. Parents claim that they have more practical knowledge and experience and tend to treat adolescents like children.

This lesson shall help you understand the nature of adolescence and its challenges and major tasks faced by the adolescents, the influences that shape their personality and some of the important problems faced by the adolescents.

OBJECTIVES

After studying this lesson, you will be able to:

- explain the nature of adolescence as a developmental stage;
- enlist the developmental tasks encountered during adolescence;
- explain the physical development during adolescence; and
- discuss gender roles and describe the phenomenon of generation gap.

11.1 CONCEPT OF ADOLESCENCE

Adolescence is a period of transition when the individual changes- physically and psychologically-from a child to an adult. It is a period when rapid physiological and psychological changes demand for new social roles to take place. The adolescents, due to these changes often face a number of crises and dilemmas. It is the period when the child moves from dependency to autonomy. It demands significant adjustment to the physical and social changes.

The adolescence period in the Indian social system comes under Brahmacharya

(apprenticeship). This is the first ashram (stage of life) of development stages. In this stage, the child learns the basic skills in relation to his future role as a responsible adult.

It is a fact that all living beings pass through specific stages or phases of development. Erikson believed that each stage of life is marked by a specific crisis or conflict between competing tendencies. Only if individuals negotiate each of these hurdles successfully they can develop in a normal and healthy manner. During this phase adolescents must integrate various roles into a consistent self- identity. If they fail to do so, they may experience confusion over who they are.

11.2 THE ADOLESCENT TRANSITION PERIOD

Indeed adolescence represents one of the greatest of these periods of crisis. In fact it is the stage of stress, strain and storm. It brings many ambiguities in life. During this phase one really does not know where he or she stands. It is believed that this uncertainty about one's role causes many conflicts.

It is a well-known fact that delinquency rates soar during the period of adolescence, that suicides become increasingly prevalent, that drug and alcohol addiction may have their beginning, and that much general happiness exists. Adolescence is also a period when satisfactory heterosexual adjustments are facilitated or hindered, when career is planned and philosophies of life are molded.

11.3 ADOLESCENCE : BIOLOGICAL OR SOCIAL?

Adolescence is both biological and social in nature. The beginning of adolescence is marked by biological changes in girls and boys. As a matter of fact, just before puberty there occurs what is known as pre-adolescent growth spurt. It takes place in girls mostly during the age of nine to twelve years, and in boys between eleven and fourteen years of age. During this period, and following shortly thereafter, the secondary sexual characteristics emerge. In girls there is typically rounding out of the hips, breast development, appearance of pubic hair, and menstruation. In boys, some of the secondary sexual characteristics that mark the beginning of adolescence

are appearance of pubic hair, facial hair, and change of voice. These changes are biologically induced.

Adolescence for both girls and boys is marked largely by social changes. Such factors as when an adolescent leaves home, gets a job and can vote determine when his or her transition from childhood to adulthood is accomplished. The length of this period is thus primarily a social phenomenon.

The problems adolescents face during the long period of growing up have both biological and social roots. Physical changes and deviation can create many problems. Society also creates problems for adolescents. Adolescents in western societies behave differently than adolescents in eastern societies owing to varying social norms expectations and family structure.

Activity 1 Learning by doing

List some issues on which you and your friends feel uneasy because you are not free to do your job independently due to your parents' interference. Discuss with elders whether we can manage some negotiations to minimize the conflict with parents.

11.4 FACTORS DETERMINING THE PATTERN OF TRANSITION

Optimal development in adolescence depends on successful accomplishment of the developmental tasks in infancy and childhood. How easy it will be for the adolescent to make the transition into adulthood will depend partly upon the individual, partly on environmental aids or obstructions, and partly on their experiences. Let us examine some of these factors:

Speed of transition: The changes during adolescence take place at very fast pace. During no other period does the individual undergo such a sudden and drastic change in such a short time period and at no other age is he less prepared to cope with the problems that this change brings.

Length of transition: Those who mature rapidly (in term of physical growth), find adjustment especially difficult. They are expected to behave like adults simply because they look like adults. On the other hand, a prolonged adolescence also brings problems. The adolescent gets into the habit of being dependent, and this, is difficult to overcome later.

Discontinuities in training: Much of the stress and strain during adolescence is due to the discontinuities in training. For example, the assumption of responsibility during adolescence is difficult because the child has so far been trained to be dependent and submissive.

Degree of dependency: How dependent the young adolescent will be is determined mainly by the kind of training he/she received during childhood. Parents often foster dependency because they feel that adolescents are not ready to assume responsibility for their own behaviours.

Ambiguous status: In the societies like India, a child is expected to follow the footsteps of his/her parents. This gives him/her a pattern of behaviour to imitate. In open society, by contrast, it is assumed that every individual should be free to choose his/her own course of self- development.

Conflicting demands: The adolescent is often confronted with conflicting demands from parents, teachers, peers and the community.

Degree of realism: When the adolescent begins to look like an adult, (s)he is permitted an added degree of freedom. If (s)he feels is not ready, either physically or psychologically, to play the adult role, (s)he feels dissatisfied.

Motivation: The adolescent goes through a period of wondering how he or she will meet the new problems the life presents. He or she would like to grow up but being unsure of the ability to cope with the challenges of adulthood. So long as this feeling of insecurity exists, there will be little motivation to make the transition in adulthood.

As the barriers to growing up are lowered or removed by parents, teachers, and society, the adolescent can move smoothly in the direction of reaching the goal of adulthood, and motivation to make the transition is normally increased.

INTEXT QUESTIONS 11.1

1. Explain the factors that determine pattern of adolescent transition.

2. Define adolescence. Identify the various challenges faced by an adolescent.

11.5 DEVELOPMENTAL TASKS

The term “developmental task”, refers to those problems that individuals typically face at different periods during their life. An infant or a small child must master the complexities of learning to walk, learning to talk, and controlling the elimination of waste products of the body. In middle childhood such skills as learning to play games and learning to read become of major importance.

So far as adolescents are concerned, the developmental tasks present the vital

problems which must be met and solved during the transition from childhood to adulthood. These problems are not entirely unique to the adolescent period, but they are ones upon which the adolescent must work if he eventually expects to achieve a successful adult role.

Developmental Tasks for Adolescents

Having Hurst has listed the following tasks.

- Achieving new and more mature relations with age mates of both sexes.
- Achieving a masculine or feminine social role.
- Accepting one's physique and using the body effectively.
- Achieving emotional independence of parents and other adults.
- Achieving assurance of economic independence.
- Selecting and preparing for an occupation.
- Preparing for marriage and family life.
- Developing intellectual skills and concepts necessary for civic competence.
- Desiring and achieving socially responsible behaviour.
- Acquiring a set of values and an ethical system as a guide to behaviour.

INTEXT QUESTIONS 11.2

1. Developmental tasks of adolescent's include:

- a. preparing for marriage and family
- b. crawling and creeping
- c. reflecting childish behaviours
- d. None of the above

2. How many developmental tasks were described by Robert Havinghurst?

- a. 10 b. 6

c. 9 d. 8

Adolescence is a long period, and many young adolescents have little motivation to master the developmental tasks for their age. In the later period, however, they realize that adulthood is rapidly approaching. This provides them the necessary motivation

to prepare for their new status. As a result, they make greater strides toward the goal of maturity than they did during early adolescence.

Successful achievement of the developmental tasks for one period in life leads to success with later tasks, while failure leads not only to personal unhappiness and disappointment but also to difficulties with later tasks.

Regardless of whether the adolescent has successfully mastered the developmental tasks of adolescence, he/she is, in most “advanced” cultures, automatically given the status of adult when he/she reaches the age of legal maturity. This is in direct contrast to more simple cultures where young people must demonstrate adult status to their elders, in “puberty rites,” privileges, and the responsibilities.

11.6 PHYSICAL DEVELOPMENT

Adolescence is fundamentally a period of physical and physiological changes. The young person's reactions to physical changes may range from pride and pleasant expectation to bewilderment and fear. When adolescents evaluate each other, especially when they consider qualities they like in a manner of the opposite sex, "good looks" are often mentioned. In particular they dominate in descriptions of girls than of boys but other physical characteristics such as body build, height and the like are often noted when boys are being "sized up".

One of the most characteristic features of adolescence is the increase in size i.e. in height and weight. However, the beginning of adolescence, known as pubescence, brings with it abrupt and psychologically significant physical changes. While many of these physical changes, such as growth in height and deepening of the voices are obvious and commonly known, others are hidden or less obvious but nonetheless significant.

Surge in Growth

Toward the end of childhood, usually girls at about ten or eleven and boys at twelve or thirteen, show a period of rapid growth in height and weight (general bodily growth). This acceleration of growth is closely linked to the increased hormonal output of the pituitary gland, which serves not only as the catalyst to produce growth but also as the controller of other glands (i.e. adrenals, gonads, and thyroid) that determine both tissue growth and function.

The rapid growth continues for a period of about three or four years, with the greatest increment in growth coming at an average age of 12.6 years for girls and 14.8 years for boys. During this period it is not uncommon for a child to grow as much as six to eight inches in height and to gain 18 to 22 kilograms in a

year's time.

For a variety of reasons, physical development in this period is characterized by asynchronization that brings concerns to adolescents and their parents and teachers. Skeletal and muscular development is more rapid than learning required for making use of the new muscle mass and motor habits. The body requires new learning. Bodily proportions also undergo changes. The facial features alter, because the growth of the lower part of the face lags behind the growth of the upper part. Legs, proportionally, usually grow more rapidly than the body stem, and hands and feet anticipate by several years the total body size of their owners.

General bodily growth is paralleled by physiological changes such as cardiovascular and respiratory growth and changes in metabolism and in general movement toward the adult states.

Skeletal Muscles: Muscles attached to bones, which bring about various types of body movement like the limb movements.

Puberty and Sexual Development

Following closely on the heels of accelerated growth, the pituitary gland directs the adrenal cortex and the gonads into more activity. Prior to this time, virtually equal amount of androgenic (male) and estrogenic (female) hormones are produced for both sexes by the adrenal cortex on the direction of anterior pituitary gland. Now an increased amount of hormonal production differentiates the sexes; the males producing more androgens, the females more estrogens.

Sex Hormones are substances secreted by the gonads for reproductive functions

and determination of secondary sex characteristics; e.g. estrogen in the female and testosterone in male. Testosterone is the male sex hormone responsible for the development of the male secondary sexual characteristics as they become sexually mature, while in the female body, this role is played by estrogen.

Moreover, the genital and sex-appropriate tissues become more sensitive to these sex specific hormones. The dawn of these changes is reflected in boys by beginning of the enlargement of the testes. There follows in each sex a series of physical changes whose appearance is highly predictable and whose sequence is unalterable. In girls the enlargement of breast is followed by the growth of pubic hair, the menarche and axillary hair. In boys, after the initial enlargement of the testes, there occurs pubic hair, axillary hair, voice changes, and beard, in that order.

Thus, by observing adolescents, it is fairly easy to determine the point in pubescence that they have reached. The observable secondary sex characteristics, such as development of axillary hair and voice change are secondary to the primary sex changes. Primary sex characteristics include reproductive organs whose maturity is signaled by the menarche in the girl and by the first ejaculation in the boy.

Strength, Skill and Fitness:

Physical development as mentioned above presents only the gross picture of the

changes that occur in adolescence. Even more significant, perhaps, are the growth patterns of strength and skills. The post-pubescent boy, even though he may be of the same chronological age as his friends who have not yet entered pubescence, will almost certainly be stronger and will likely have greater agility, motor coordination, and bodily skills. He will, of course, rapidly overtake girls, whose strength already has increased about a year earlier and who briefly challenged him. You would like to know what forces account for this development. First, it is clear that the accelerated production of male hormone (androgens) brings with it added muscular strength. Second, the nature of skeletal growth, increased shoulder breadth, bigger chest cavity, and finally, the greater lung size, heart size, and increased blood pressure are all favourable conditions for greater physical strength.

While many of these same changes occur among girls, their physical strength increases at a slower rate, and, of course, never equals that of males. There are two reasons for this superiority: (a) biologically, the male is favoured by larger shoulder breadth, a bigger chest cavity, and better leg leverage; (b) culturally, girls receive little encouragement for the development of strength. In fact they are encouraged in the opposite direction-to be weak and dependent, or at least to pretend that they are.

Along with increases in strength come the development of motor coordination, reaction speed, and perceptual-motor skills. In an appraisal of physical development, one important consideration has to do with what is optimal. Height-weight charts are based on the average, but may, when a large segment of youth is overweight, lead to faulty conclusions. What is average is certainly not optimal.

INTEXT QUESTIONS 11.3

1. Explain how adolescence is a period of rapid physical growth and sexual development.

11.7 GENDER ROLES

Gender is a much more complex term. It relates to a given society's beliefs about the traits and behaviours supposedly characteristic of males and females. **'Gender roles'** refer to the expectations concerning the roles males and females should perform and the ways they should behave. Such expectations come into play as soon as a child is born.

The beliefs about gender roles continue to influence us and our behaviour throughout life. Children's recognition that they belong to one sex or the other called '**gender identity**' is established quite early in one's life by the age of three or four. At this time, however, they are uncertain as to whether they will always be a boy or a girl. It is not until they are about six or seven that children acquire '**gender consistency**'— the understanding that even if they adopted the clothing, hair style, and behavior associated with the other sex they would still retain their current sexual identity.

INTEXT QUESTIONS 11.4

1. Explain the meaning of gender role.
-

WHAT YOU HAVE LEARNT

- Adolescence is the period of transition between childhood to adulthood and it plays a decisive role in relation to the formation of prosocial/antisocial adult.
- Optimal development in adolescence depends on successful accomplishment of the developmental tasks in infancy and childhood.
- Adolescence is fundamentally a period of physical and physiological change.

The most important single feature of adolescent development consists of the changes that take place in the young person's body.

- Toward the end of childhood, girls at about ten or eleven and boys at twelve or thirteen begin a period of rapid growth in height and weight.
- Gender role refers to expectations concerning the roles male and females should adopt and the ways they should behave.

TERMINAL QUESTIONS

1. What are the primary and secondary characteristics during adolescence?
2. What are the developmental tasks of adolescents?
3. Give a detailed account of major challenges faced during adolescence.

ANSWERS TO INTEXT QUESTIONS

11.1

1. Period of transition, rapid physical and psychological changes, New social roles, independence, faces conflicts.

11.2

1. d

2. a

11.3

1. Speed and length of transition, training, dependency, status, demands, degree of realism, motivation.

2. Surge in growth, height and weight, skeletal and muscular growth, sex hormones, sexual development, sex hormones, strength and skill.

11.4

1. Society's beliefs about behaviours characteristic of males and females. Gender consistency and behaviour of males and females.

Hints for Terminal Questions

1. Refer to section 11.3

2. Refer to section 11.6

3. Refer to section 11.5

12. ADULTHOOD AND AGEING

What is it like to grow old? What changes occur as people grow older? What does it mean to say that some one is old? These are some of the questions which come to our mind when we see or meet an aged person. We find that our parents do not behave as we do. The kind of images which normally come to our mind when we hear the word 'adult and aging' are responsibility, stability, forgetfulness and some times poor health. We view the older years as a stage in life when people become more dependent on others. Are these stereotypes or myths?

When we look into our society today we notice that the segment of older population is growing. It has been estimated that by the year 2021 a major portion of the world's population will be in the category of the aged people. Becoming adult and growing old is a biological process, but many other developments also take place along with it. You will study about these changes in this lesson. This lesson will also help you understand the value of the aged people in the society and relating to them in a better way.

OBJECTIVES

After studying this lesson you will be able to:

- explain the characteristics of the period of adulthood and old age;
- describe the key developmental changes during adulthood;
- describe the problems of ageing and coping with it; and
- appreciate the concerns of ageing and aged people.

12.1 THE ADULTHOOD STAGE

We are often told by our elders to act according to our age. Have you ever wondered what this means? There seem to be certain normative expectations about how

individuals of a certain age should act. The period of adulthood is marked by society's expectations of performing one's duty as an adult. Some special roles and relationships like family and career are the major milestones of adulthood. The spurt of growth during adolescence stabilizes by adulthood and the individual looks forward to more concrete roles in the society. Our capabilities and behaviours continue to change as we grow older not only because of the biological growth but also because of our knowledge and the way we interact with environment. In the Indian tradition the period of adulthood refers to Grihastha Ashram. During this stage an individual enters the social life and accepts responsibilities of family, marriage and entering into one's career. The stage of adulthood has the following sub periods: (1) Young adulthood (20s to 30s), (2) Middle adulthood (40s to 50s) and (3) Later adulthood (60s and above.). Let us understand the various characteristics of each of these sub periods.

Young Adulthood

During this period one sees himself or herself as an autonomous and independent adult. The individual becomes financially less dependent on the parental family. A major developmental task during this period is the choice of a career. Young men and women tend to settle down in a career of their choice. In addition a young adult tries to settle down and start the family life. They are also concerned about various social issues and forming close relationships with one's professional peers and members of community. The duties of a person demand two-fold responsibilities which are towards oneself as well as the society.

The stage of young adulthood is characterized by new tasks and challenges in life such as establishing financial and emotional independence and entering into marital relationship. Unemployment and marital discord are two typical crisis conditions during early adulthood. According to Erikson this period is characterized by a crisis of 'intimacy versus isolation'. The young adults must develop the ability to form deep intimate relationships with others, particularly in marriage. Otherwise, they can become socially and emotionally isolated. Most young adults develop a dream of future accomplishments or a vision of what they want to achieve as a life goal. Such dreams motivate young adults towards goal directed efforts. Occupational and marital choices and gaining social and economic independence are some major tasks of early adulthood. Very often, during the early thirties individuals make a reappraisal of their choices and seek to make specific changes in their career choice and social relations. These experiences are described as "age thirty transition".

Characteristics of Young Adulthood

Choice of a career,
starting a family life,

forming close relationships with peers,
concerns about society.

Do It Yourself

Meet two adults, a male and a female. Ask them to describe their roles and responsibilities. Record their descriptions and your own description. Compare and discuss with your peers and tutor.

INTEXT QUESTIONS 12.1

1. Identify the characteristics of young adulthood.

12.2 ADULTHOOD OR MIDDLE AGE

From the period of twenties and thirties, the individual arrives at middle age in the forties and fifties. Middle age is characterized by competence, maturity, responsibility and stability. This is the time when one wants to enjoy the success of job, satisfaction derived from the family and social life. People look forward to the success of their children. Attention gets more focused on health, the future of children, aging parents, the use of leisure time and plans for old age. For women, menopause occurs between the age of forty-five and fifty. Menopause is supposed to be accompanied by some distressing physical and psychological symptoms in women. During this period men show greater concern towards health, strength, power and sexual potency.

Middle age is also viewed as a period of creativity and significant contribution to society. It is found that the best works of scientists, writers and artists are produced during the late forties and early fifties. Erikson characterizes middle adulthood as a phase of crisis involving generativity versus self-absorption. Unless a person makes this a period of meaningful contribution he/she may become preoccupied with selfish needs and desires. The middle age transition is also a period of turbulence. People start viewing themselves as older and become conscious of their mortality which often leads to emotional instability. They also engage in taking stock of their lives to assess the extent to which they have realized their dreams. Such re-assessments often lead to occupational changes and changes in marital relationships such as divorce.

For most people who are employed in regular jobs, the end of the middle adulthood years involves the task of coming to terms with possible retirement from work life and the associated readjustments in life.

Characteristics of Middle Adulthood

- Competence, maturity, responsibility and stability.
- Attention focused on children.

Plans for old age.

- Occurrence of menopause in women.

Old Age

Later adulthood or the period of old age begins around the age of sixty. During this stage most individuals lose their jobs as they retire from active service. They begin to fear about their physical and psychological health. In our society, the elderly are typically perceived as not so active, deteriorating intellectually, narrow-minded and attaching significance to religion and spirituality. Many of the elderly people lose their spouses and suffer from emotional insecurity. However, this may not be true of everybody. Many people at the age of sixty or above remain very healthy and active in life. The life style including exercise, diet, and regular health check up helps people enjoy a meaningful and active life during old age.

Aging, however, is also a psychological phenomenon. There are numerous examples of writers, statesmen, professors and yogis who lead a productive and active life during old age. So it is important to note that old age can be a much happier and fuller time. Mental or physical decline does not necessarily have to occur. Persons can remain vigorous, active and dignified until their eighties or even nineties. This is quite possible if they take adequate care of themselves right from their middle age years. In fact, the older persons have vast reservoir of knowledge, experience and wisdom on which the community can draw upon.

INTEXT QUESTIONS 12.2

1. Describe any two characteristics of middle adulthood.

12.3 PHYSICAL AND COGNITIVE CHANGES DURING ADULTHOOD AND OLD AGE

In the present day society, with increasing health consciousness, both men and women want to be physically and mentally fit. With advancing age certain inevitable and universal changes such as chemical changes in cells or gradual loss of adaptive reserve capacity take place. There are also certain cognitive changes taking place from middle adulthood onwards. These changes are slow and gradual. Normally people see aging as a period of physical and mental health decline. These changes

are mainly physical and cognitive in nature. Let us try to understand these changes.

(a) Physical Changes: It has been found that the body organs of most persons show a 0.8 to 1 percent decline per year in the functional ability after the age of 30 years. Part of this decline is normal, some are disease-related and some caused by factors such as stress, occupational status, nutritional status and many other environmental factors. The major physical changes associated with aging are described as external changes, internal changes and changes in the sensory capacities and cognitive abilities. Let us understand these changes.

(1) External Changes: This refers to the symptoms of growing old that can be seen in an individual. With many individuals the symptoms of aging can be seen as graying hair, aging skin shift in posture and appearance of wrinkles and other changes.

Key to Longer and Healthier Life

- (a) Eat healthy diet.
- (b) Exercise regularly.
- (c) Reduce stress level.
- (d) Don't smoke or drink alcohol and other drugs.
- (e) Cultivate certain positive qualities.
- (f) Engage with activities of social welfare.
- (g) Develop spirituality and faith.

(2) Internal Changes: These changes refer to the symptoms of growing old that are not visible. These include changes in the respiratory system, gastrointestinal system, cardiovascular system, and central nervous system.

Some Great Achievers of late adulthood

- Mahatma Gandhi led India's struggle of freedom at the age of 77.
- Baba Amte
- Ramakrishna Paramhans
- Mother Teresa

(3) Changes in Sensory Capacities: With advancing age, there is gradual slow down in the sensory abilities. As you know we communicate with the outer world through our senses. Losing in any of the senses can have profound psychological and social consequences.

Cognitive Changes: Cognition refers to the process by which information is acquired, stored, and used. During adulthood and aging various aspects of cognition like memory, learning, attention and speed of performance get adversely affected. The aged people often report forgetfulness, difficulty in learning new techniques and acquiring skills. Their concentration and responsiveness to environmental stimulus decreases. Dementia and Alzheimer and important diseases of old age. In general people become more vulnerable during old age.

12.4 COPING WITH THE PROBLEMS DURING OLD AGE

You must be wondering whether the process of ageing can be stopped or can anyone remain young throughout? We have noted that ageing is a biological process. This process gets affected by many other things such as stress, tension, habit, undisciplined life, ill health work habits etc. Leading a disciplined life means giving the body enough time to rest, proper work habits, less tension, proper nutrition and spirituality. It seems that aging can be delayed but no one can escape it. One must not deny and accept one's age and act in a realistic but optimistic manner. Different people adopt different coping strategies to meet their life challenges. Some of the effective coping strategies are as follows:

- (1) Develop an attitude of flexibility so that one may adapt to the pressures and problems of old age.
- (2) Recognize that one has to explore new ways of coping with life events.
- (3) Greater use of information seeking and problem solving strategies rather than withdrawing or isolating.
- (4) Increasing self-confidence, self-reliance, developing healthy attitudes about one's strengths and weaknesses.
- (5) Learning and maintaining effective coping skills and adopting an active approach towards the environment.

Use of the above styles of coping can be effective in making healthy adjustments during old age.

Aging and Social Support

Enlarging social network is another means of coping with life problems. Participating in various group activities such as joining clubs. Certain organizations for informal social interaction are found very helpful for the aged. Building a social network of people of their own age group in neighbourhood or elsewhere provides them with greater opportunity to share their life circumstances and find emotional expressions

for their existing problems. Through such social networks, one can get an unconditional expression of approval, share secrets, provide new experiences to each other, and develop trusting relationships.

Involvement in grand parenting helps elderly satisfy many of their personal and emotional needs. Grand parents can serve as important role models. Older people find these roles emotionally self-fulfilling and tend to derive self-satisfaction through achievement of their grand children.

12.5 IMPORTANCE OF THE AGED PEOPLE IN THE SOCIETY

“Those who greet elders and do service to them, they grow in all four dimensions such as longevity, knowledge, fame and strength”.

One of the reasons for the growth of the Indian society is its multigenerational structure which means staying of many generations together in one family. This phenomenon is expressed in the form of joint family system. Aged people have been always considered as individuals with a vast repertoire of knowledge and experience. Staying with them and respecting them helps not only in personal growth but facilitates multidimensional growth of a person.

Today many of the problems seen in the society are due to breakdown in the family system. Young couples prefer to stay alone to seek independence and doing away with the restrictions. But the repercussion of this system is largely being felt on the younger generation. The elders in the family have always been a medium of cultural transmission. Can you forget the stories told to you by your grand parents and their morals? The cultural traditions followed in our country have been passed from generation to generation. With the emerging nuclear family system, the younger generation is becoming deprived of this rich cultural transmission. It has been found that emotional expressions and values in young generation have also been affected by the absence of elders in the family.

INTEXT QUESTIONS 12.3

State whether the following statements are true or false:

1. The body organs show a decline of 0.8 to 1 percent after the age of 30. T/F
2. Gradual changes in sensory abilities take place with ageing. T/F

3. One must not accept one's age and become younger. T/F
4. Developing healthy attitudes do not help in coping with age. T/F
5. Aged people should be considered as treasure of the society. T/F

WHAT YOU HAVE LEARNT

- Some special roles and relationships like family and career are the major milestones of adulthood.
- Our capabilities and behaviours continue to change as we grow older not only because of the biological growth process but also because of our understanding and the way we interact with the environment.
- The stage of young adulthood is characterized by new tasks and challenges in life such as establishing financial and emotional independence and entering into marital relationship.
- Middle age is also viewed as a period of creativity and significant contribution to society.
- In our society, the elderly are typically perceived as not so active, deteriorating intellectually, narrow-minded and attaching significance to religion and spirituality; however it may not be true.
- The major physical changes associated with ageing are described as external, and internal changes and changes in sensory capacities.
- Leading a disciplined life means giving the body enough time to rest, proper work habit, less tensions, proper nutrition and spirituality. This may help in coping with the problems of old age.
- Aged people have been always considered as individuals with a vast repertoire of knowledge and experience.

TERMINAL QUESTIONS

1. Describe the key to longer and healthier life.
2. What attitude is helpful to meet out the pressures and problems of old age?
3. What is the main cause of today's problems in the society?
4. How can one remain healthy during old age?

ANSWERS TO INTEXT QUESTIONS

12.2

1. Autonomous, independent, choice of career, new tasks and challenges, intimate relationships.

12.2

1. (a) Competence

(b) Attention on children

(c) Plans for old age

(d) Occurrence of menopause

2. By exercise, proper diet and regular health checkup

12.3

1. True

2. True

3. False

4. False

5. True

Hints for Terminal Questions

1. Refer to section 12.3

2. Refer to section 12.4

3. Refer to section 12.5

4. Refer to section 12.3

CURRICULAM

PSYCHOLOGY:Secondary Level Course (222)

RATIONALE

The study of psychology deals with important aspects of human behaviour and mental processes. The study of psychology is significant to one's personal growth and development and for facilitating the process of social development.

The way we think, feel and behave with the people and world around us influence our adjustment. A course in psychology at the Secondary level will help the learners understand themselves and their place in society and realize their potentials.

OBJECTIVES

The course is designed to enable the learner:

Understand the basic concepts of psychology and their application.

Relate oneself positively to family, neighborhood and society.

Behave responsibly and in a value based manner.

Learn to live a purposeful life of health and happiness.

MODULES

Module I Foundations of Psychology 15

Module II Basic Psychological Processes 15

Module III Human Development 15

Module IV Social Psychological Processes 15

Module V Health and Well Being 15

Module VI Work Life and Environmental Concerns 15

Module VII Yoga : A Way of life 10

Maximum Marks 100

Curriculum

Module I: Foundations of Psychology

Marks: 15

Approach

This module helps students understand the nature of psychology and the range of behaviours and mental processes which constitute its subject matter. It would also help them to understand the methods and some important fields of psychological applications .

Contents

Lesson 1 – An Introduction to Psychology

The nature of psychology and its definitions

Scope of psychology

Basic psychological processes

Fields of psychology

Lesson 2 – Methods of Psychology

Approaches to the study of psychological processes

Methods to understand psychological processes

Psychological Tools

Lesson 3 – Individual Differences

Nature of individual differences

Assessing individual differences

Nature of Intelligence and its assessment

Aptitudes and interests, and their assessment

Nature of personality

Techniques of personality assessment

Module II: Basic Psychological Processes

Marks: 15

Approach

This module familiarises the learners with the concepts and processes involved in knowing the world. It helps to understand the basic ideas involved in acquiring knowledge and retaining it for further use. This module will also describe the processes of perception and the way we become aware of the world and think. It also tries to provide an understanding of the ways we feel and appreciate our inner world

Contents

Lesson 4 – Learning

Learning

Reinforcement and observational learning

Some other types of learning

Transfer of training

Lesson 5 – Sensory Processes: Attention and

Perception

Sensation

Perception

Perceptual organisation

Attention

Extra sensory perception

Applications of perception in everyday life

Lesson 6 – Memory

Memory and Forgetting

Stages of Memory

Types of memory

Forgetting

Strategies for enhancing memory

Repression: Forgetting painful events

Lesson 7 – Thinking and Problem Solving

The nature of thinking

Problem solving

Creating and its role in life

Decision making

Lesson 8 – Motivation and Emotion

Nature of motivation

Nature of emotion

Role of emotion and motivation in human behaviour

Module III: Human Development

Marks: 15

Approach

This module aims at describing the processes, characteristics and stages of human development and explaining the influence of biological factors and socio-cultural context on human development.

Contents

Lesson 9 – Nature Determinants of Development

Nature of development

Domains of development

Stages of development

Influence on development

Lesson 10 – Childhood

Physical and motor development in Childhood

Cognitive development

Socio-emotional and personality development

Socialisation

Lesson 11 – Adolescence and its Challenges

Concept of Adolescence

The adolescent transition period

Adolescence: Biological or social?

Factors determining the pattern of transition

Development tasks

Physical development

Gender roles

Lesson 12 – Adulthood and Aging

The adulthood stage

Adulthood or middle age

Physical and cognitive changes during adulthood and old ages

Importance of the aged people in the society

Module IV: Social Psychological Processes

Marks : 15

Approach

This module aims at describing the influence of social institutions. It helps to understand how people behave in groups and also explains the formation of human beliefs, attitudes and values. The module will also explain the process of

communication and familiarise the learners with some major social problems.

Contents

Lesson 13 – Groups and Leadership

Importance of groups in our life

Characteristics of group

Group dynamics

Difference between ‘Us’ and ‘Them’

Formal and informal groups

Group development

Leadership

Qualities of leader

Theories of a leadership

Lesson 14 – Communication

Understanding the meaning of communication

Elements of communication

Types of communication

Effective communication

Features of effective communication

Non-verbal communication in Indian context

Role and impact of media

Lesson 15 – Attitudes, Beliefs and Social Cognition

Nature of attitude

Formation and change of attitudes

Relationship between attitudes and behaviour

Beliefs and behavior

Beliefs and behaviour

Social cognition

Lesson 16 – Social and Educational Problems

Poverty

Some other social problems

School dropouts, stagnation and wastage

Module V: Health and Well Being

Marks: 15

Approach

This module aims at developing an understanding about the concepts of happiness and well being. It describes the obstacles in the pursuit of happiness and psychological processes to overcome them. It also explains the concepts of health, hygiene and describe practices for positive mental health.

Contents

Lesson 17 – Happiness and Wellbeing

Concepts of happiness and well being

Self actualization and mature personality

Recognizing strengths of self Actualized individuals

Developing positive emotions

Self efficacy and compassion

Lesson 18 – Mental Disorders and their Treatment

Stress as a source of disturbed mental health

Types of conflicts and frustations

Types of mental disorders

Psychotherapeutic process

Process of coping with stress

Lesson 19 – Mental Health and Hygiene

Concept of mental health and hygiene

Mental health, mental ill-health and mental illness

Indicators/signs of poor mental health

Behaviours which sustain good health

Strategies for time management

Module VI: Work Life and Environmental Concerns

Marks : 15

Approach

This module aims at developing an appreciation of the concerns for vocational life and understanding of the challenges of world of work. It also explains the relevance of individual differences for job requirements, notions of work motivation. The relationship between human being and environment, causes and

remedies of environmental stress are also described.

Contents

Lesson 20 – Aptitude, Interest and Job Requirements

Aptitude

Interest

Ability

Mental age and I.Q.

Importance and relationship among aptitude, interest and ability

Analysing and communicating job requirements

Lesson 21 – Preparation for the Vocational Role

Meaning of vocational role

Need for preparation for right vocation

Special training

Concepts of career, vocations and occupation

General requirements for a vocation

Vocational choice emerging perspective

Lesson 22 – The World of Organization

Organization as a system

Organizational climate

Interpersonal relationship at work

Productivity and excellence at work

Strategies for developing productivity at work

Lesson 23 – Environmental Stress

Environmental stress

Environmental stress

Pollution issues

Sustainable development

Module VII: Yoga: A Way of Living

Marks: 10

Approach

This module tries to develop an appreciation among the learners about Yoga as a way of life being practised from ancient times. The module also explains the relevance of Yoga in the various stages of our life. It would also provide guidelines to the learners for facing various stresses and tensions in life and for self development.

Contents

Lesson 24 – Healthy Mind in Healthy Body

What is Yoga

Different types of yoga

Yoga to keep fit

Pranayam

Meditation

Lesson 25 – Self-development and Yoga

Self-development through Yoga

Applications of Yoga in different walks of life

How Yoga shapes our attitudes, thinking and behaviour?

Yoga for motivation and excellence: (Competence) in action

Patanjali on 'Mind': Its nature, functions and states

The eight-fold path of Yoga

Lesson 26 – Nurturing the Whole Being: An Indian perspective

Wholistic Personality: An ideal view

Theory of Trigunas

Theory of panch koshas (Five Sheaths of existence)

Development of Koshas

Sri Aurobindo on levels of consciousness

Lesson 27 – Controlling and Disciplining the Mind

Importance of controlling and regulating the mind

Jeevan Vigyan

Preksha Dhyan

Art of Living

Transcendental meditation

Vipassana

